End of line (EOL) test description

Customer:	**General**
System:	**ABS 9.x** **ESP® 9.x** **ESP®plus 9.x,**
Brake split distribution:	**Diagonal (X); front/rear (II)**
Transmission:	**Rear wheel drive / front wheel drive / four wheel drive**
Number of HU channel:	**3, 4**
Test rig:	**2 axle brake dynamometer respectively Roller test bench with 4 independent and detachable rolls**
Diagnostic protocol:	**KWP2000, UDS (customer specific)**
Tester requests via:	**K-Line (opt.), CAN, FlexRay (customer specific)**
"D"	
Distribution list:	**Customer**

Department:	Date:	Signature:
CC/ESU1	11.08.09	sgd. Kühner

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Modifications:

No.	Date	Page	Description	Signature
00	10.08.2009	All	New document	sgd. A. Horvath

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1 Attachments and abbreviations

Attachments:

No.	BOSCH – No.:	Name
1	---	Hydraulic circuit diagram ABS 9
2	---	Hydraulic circuit diagram ESP® 9i
3	Y 265 B32 347/E	Electrical circuit diagram ESP® 9i

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Abbreviations:

ABS	antilock braking system
ABS-SILA	ABS warning light
AC	Accumulator chamber
AV	outlet valve
ALS (Ax)	longitudinal acceleration sensor
AQS (Ay)	lateral acceleration sensor
BLS	brake light switch
BS	Brake switch
DTC	diagnostic trouble code
EBD	electronic brake distribution
ECU	electronic control unit
ESP®	electronic stability program
ESP®-MFL	ESP multifunction (warning) light
EV	inlet valve
FA	front axle
FCM	fault code memory
FL	front left
FR	front right
HAS	handbrake switch (park brake switch)
HSV	high pressure switching valve (priming valve)
MCC	master cylinder circuit
PATA	passive taster (TCS/ ESP® passive taster)
PB	process byte
PS (DG, DS)	pressure sensor
RA	rear axle
RFP	recirculation pump
RL	rear left
RR	rear right
SAS (LWS)	steering angle sensor
SV	solenoid valve
TCS (ASR)	Traction control system
UDS	Unified diagnostic services
UMR	pump motor feedback
USV	pilot valve
UVR	Valve relay feedback
WSS	wheel speed sensor
YRS	yaw rate sensor

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2 Introduction

This document describes test steps which are recommended to ensure the correct function of the Bosch ABS / ESP® system and its interfaces after the vehicle assembly. The tests consist of static- and dynamic function tests.

The tests are based on the customer specific diagnosis. The dynamic tests are carried out on a brake dynamometer or selectively on a roller test bench with 4 independent and detachable rolls.

The appropriate version of the customer/vehicle specific ”customer specific descriptions of diagnosis” provides information on the handling and function of the diagnosis.

Requirements for the test facilities, which are provided by the customer:

- Communication with the ABS/ESP® ECU based on the corresponding diagnostics
- Control of the test sequence
- Control of the dynamometer
- Evaluation of the test data
- Issue test protocol

Product of Bosch:

- Functional description of diagnosis (customer specific)
- Bosch specific EOL diagnostic routines

The test steps and the test timing is optimized in order to meet a short cycle time and to avoid double checks of the system.

Note:

If the dynamic ABS / ESP® EOL tests are performed in another way as described in this end of line test description, the ABS / ESP® control functions could be activated or the system disabled due to error detection. Bosch cannot be liable for such of problems respectively critical driving situations caused by a not specified test procedure.
To avoid damages at the valve coils by overheating after a previous executed evac/fill or repair bleed process a defined time has to be waited before further valve activivations are executed at the EOL process.

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3 Test scope

All the Bosch ABS / ESP® components, such as ECU, hydraulic modulator, wheel speed sensors are checked individually in the Bosch assembly line prior to shipment to the customer's assembly plants.

The entire system including harnesses, dashboard lights and power supply of the vehicle is assembled the first time at the customer plant and must be checked for proper connections and functions.

The ECU is capable to run a number of complexity self monitoring software features, such as continuity and plausibility checks and in case of errors to identify these faults and to store fault code numbers.

This feature is used at the end of line test by reading out the fault codes as error identification as well as information for repair works.

However, some of the failures can not be detected by the ECU itself (such as cross connections of hydraulic pipes or wheel speed sensors and correct functioning of some input signals, i.e. brake light switch). In order to find these failures static or dynamic tests should be performed while the vehicle is at rest or during a dynamic test on a dynamometer.

<u>The major items of this test steps is to check for:</u>

- correct functioning of all input and output signals of the ECU
- correct functioning of all wheel speed sensors (signal quality and correlation)
- correct functioning of all valves and hydraulic components (functioning and correlation)

The ABS / ESP® hydraulic modulator is shipped to the customer in a dry state (no brake fluid). Therefore it needs to be evacuated and filled on line by using the customer specific diagnostics for valve and pump motor activation's. To ensure a proper process, the process information (e.g. ok or nok) can be stored in the EEPROM of the ECU. The value of the process bytes are stored by the evac/fill or bleed equipment and can be checked at the end of line test to verify the previous processes. The delivery state of the process bytes contains a value which should be used for nok process.

The entire test procedure is designed to cover all the important items of the system and to minimize the testing time.

<u>Important notes to the ABS/ESP® EOL tests:</u>

a) Variant coding:
Preconditions for a faultless ABS/ESP® system are, that all required variant data are coded.

b) Calibrations:
Preconditions for a faultless ABS/ESP® system are, that all integrated and connected sensors (e.g. longitudinal- (AXS), lateral- (AQS) acceleration and steering angle) are calibrated.

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4 Test step description

Test step		Description	
4.1		Initial start up actions & standstill tests:	
	.1	Start of the customer specific diagnostic mode	
	.2	ECU identification	
	.3	Check of the process bytes (optional)	
	.4	Coding (e.g. engine and gear box variant, vehicle identification number (VIN), etc.) - optional	
	.5	Calibration of the in the ABS/ESP® ECU integrated Ax/Ay sensors (optional)	
	.6	Calibration of the steering angle sensor (only ESP® systems – optional)	
	.7	Check of the input and output signals	
	.8	Execution of the static test – optional	
	.9	Execution of the pump motor slow-down time test (only ESP® systems) – optional	
	.10	Read and clear fault code memory	
4.2		Dynamic tests on a 2 axle brake dynamometer (low speed):	
	.1	Wheel speed sensor tests	WSS correlation test WSS signal quality test Rotation direction and mounting position tests (optional e.g. Bosch WSS 11i sensors used)
	.2	Dynamic tests	Base brake- and ABS test
	.3		ESP® test
4.3		Alternate: Dynamic tests on a 4 roller test bench with 4 independent rolls (high speed):	
	.1	Wheel speed sensor tests	WSS signal quality test Rotation direction and mounting position tests (optional e.g. Bosch WSS 11i sensors used)

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	.2	Dynamic tests	Base brake- and ABS test (incl. WSS correlation test)
	.3		ESP® test (incl. WSS correlation test)
4.4		Finishing & standstill tests:	
	.1	Read fault code memory	
	.2	ABS/ESP® warning and safety lamp tests / change into normal mode (end of the EOL test procedure)	
4.5		Driving EOL test (only ESP® systems - optional)	

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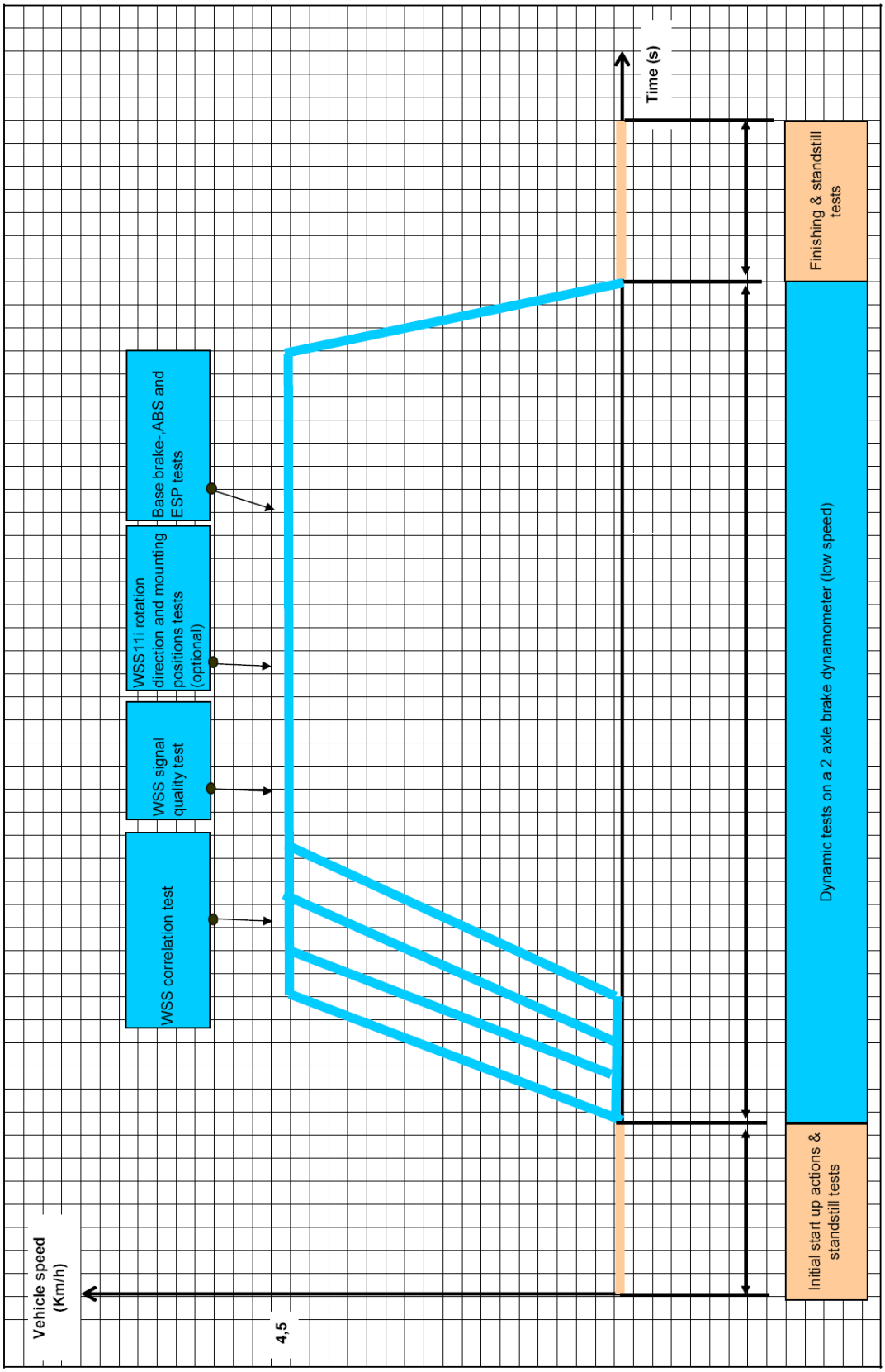
Graphical presentation of the complete EOL test procedure on a 2 axle brake dynamometer (low speed):



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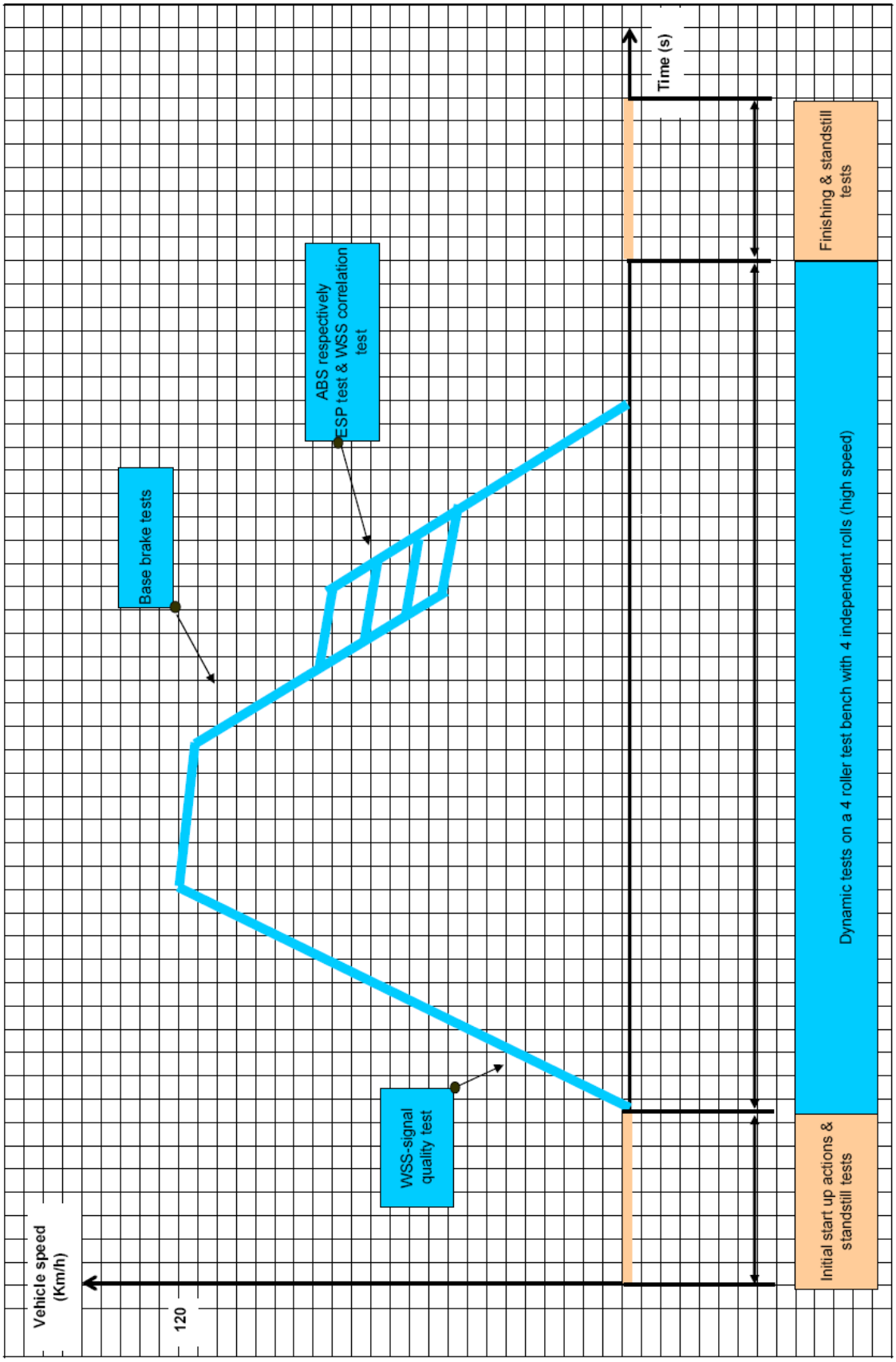
Graphical presentation of the complete EOL test procedure on a 4 roller test bench (high speed):



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4.1 Initial start up actions & standstill tests

4.1.1 Start of the customer specific diagnostic mode

Start of the customer specific diagnostic mode	
Operator:	Tester / ECU:
→ Ignition off. → Connect diagnostic interface (OBD2 connector). → Ignition on. → Wait till the ABS/ESP® system check (initial tests after ignition on) is finished: **Time: t_{ABS/ESP_system_check}**	
	Start of customer specific diagnostic mode.

Remarks:

Before the EOL tests are executed the ignition should be switched off to be sure that the ABS/ESP® system is active and has not been switched off by a previous assembling process which set the ABS/ESP® system into an unwanted faulty mode.

After every switching on of the ignition, the ABS/ESP® system check is started. To avoid to interrupt this system self tests (ECU self diagnosis), the customer specific diagnostic mode should be started after the defined **time t_{ABS/ESP_system_check}**. A premature change into the customer specific diagnostic mode might stop this system self test. In this case the static test should be executed.

The communication between the tester and the ABS/ESP® ECU is based on the customer specific diagnostic interface.

Before e.g. the pump motor slow-down time test (only available at ESP® systems - optional), the WSS tests, the dynamic ABS/ESP® test can be executed, the ABS/ESP® SG must be set into the customer specific diagnostic mode. Only in this special diagnostic mode such of EOL functions are available respectively executable.

In case of a not successful execution of the function „start of communication“, it should be repeated minimum 3 times before the process is interrupted.

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The customer specific diagnostic mode must be kept alive with sending a valid diagnostic service (e.g. tester present service) < P3 Timeout, otherwise this diagnostic mode will become invalid.

Conditions for a change into the customer specific diagnostic mode (EOL mode):

- No system control
- **Speed of the vehicle (km/h) ≤ V_1.**

Parameter:

Kind of parameter:	Parameter name:	Unit:	Physical value:
Condition	Vehicle speed V_1	km/h	tbd
Diagnostic	P3 Timeout	Milliseconds (ms)	5000
Time	$t_{_ABS/ESP_system_check}$		5000

More details to the start of communication can be read in the customer specific documentations.

All with tbd signed parameters are customer specific.

Start of communication successful ⇒ **Continue the EOL test procedure**

Start of communication not successful ⇒ **Abort, print out fault message and send vehicle to repair area**

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4.1.2 ECU identification

ECU identification	
Tester:	ECU:
Read and evaluate the ECU identification data Selection of the right valve activation respectively EOL test sequence.	Output of the requested ECU data.

Notes:

The ECU identification which has to be evaluated data must be confirmed by the customer and its assembly plants.

The ECU identification data must be compared with the in the tester software stored desired ECU data. This is required to select the right valve activation respectively EOL test sequence. The ECU identification data might be changed in case of an ABS/ESP® software update respectively modification, therefore it must be easily possible with editor menu to update the data base of the tester respectively EOL equipment.

More details to the chapter “ECU identification” can be read in the customer specific documentations.

The evaluation of the ECU respectively vehicle identification has following reason:

1) To ensure that the expected ABS/ESP® system is installed
2) Selection of the required test sequence

Known system ⇒ **Continue**

Unknown system ⇒ **Abort, print out fault message and send vehicle to repair area**

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4.1.3 Check of the process bytes (optional)

Check of the process bytes (optional)	
Tester:	ECU:
Requests and evaluates the process bytes.	Reads the process bytes from the E^2PROM and responses the requested data.

Notes:

The process byte must be equal to the demanded data which are stored in the tester software data base. The evaluation of the process bytes is optional and customer specific.

The process bytes contain the information if the previous vacuum and filling (online) or the previous repair bleed (offline) process was successfully executed.

Precondition to evaluate these data is, that as well as the vacuum and filling and the repair bleed facility supports the process byte writing.

The ABS/ESP® ECU is delivered with predefined value (customer specific). This value is written into the ABS/ESP® EEPROM by the Bosch plant. In case of none successful communication with the ECU at the customer assembly plant, the delivery value is still stored. These conditions must be considered as a not successful respectively faulty evacuation and filling or repair bleed process.

In case of a successful evacuation and filling or repair bleed process these data are modified by the facility tester. The used data must be agreed and confirmed between the facility supplier and customer.

More details to the chapter “Check of the process bytes – optional” can be read in the customer specific documentations.

The evaluation of the process byte data has to ensure the following process:

1) ABS/ESP® system respectively base brake system was successfully filled by the online evacuation and filling or offline repair bleed process

Written process bytes ok ⇒ **Continue**

Written process bytes nok ⇒ **Abort, print out fault message and send vehicle to repair area**

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4.1.4 Coding (e.g. engine- and gear box variant, vehicle identification number, etc.) - optional

Coding (e.g. engine- and gear box variant, vehicle identification number, etc.)- optional	
Tester:	ECU:
1. Start of the login procedure for the ABS/ESP® ECU. 2. Start of the coding procedure for the ABS/ESP® ECU.	1. ECU sends the seed, checks the received login key and allows access. 2. ECU stores the coding data in the E^2PROM.

Notes:

To have writing access to the ABS/ESP® E^2PROM, a login procedure has to be executed within the same diagnostic session (otherwise the ECU answers with a negative response „access denied“).

At first the ECU checks the received key. In case of the key is valid the ECU is unlocked till the customer specific diagnostic session is left (e.g. finished by a received stop communication command, a software reset or a hard key off).

After the ECU is unlocked the tester can send the coding data. The ECU validates the data (depending on the data) and stores them in the E^2PROM of the ABS/ESP® ECU. After the next ignition cycle this data will be verified (vehicle must be fully assembled) and will become valid.

The login procedure and used algorithm is customer specific.

This step is only required, if the coding data are written during the EOL tests.

More details to the chapter “Coding” can be read in the customer specific documentations.

Coding successful ⇒ **Continue**

Coding not successful ⇒ **Abort, print out fault message and send vehicle to repair area**

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4.1.5 Lateral (AQS) and longitudinal (AXS) acceleration sensor offset value calibration - optional

Lateral (AQS) and longitudinal (AXS) acceleration sensor offset value calibration - optional		
Operator:	Test rig / Tester:	Remarks:
☞ Turn steering wheel to straight ahead position Car should be positioned straight ahead on the equipment	Calibration of the lateral (AQS) and longitudinal acceleration sensor (AXS) offset values	The requirement of this calibration depends on the mounting tolerance of the ESP system and the used system functions (hill hold control (HHC), etc.). Details are customer specific.

Conditions during sensor offset calibration:

- Maximum inclination angle of the test rig within the allowed limits (both alignment directions, for details look in the chart below)
- Test rig must be stationary (i.e. not on production line)
- Steering wheel in straight ahead position
- Correct tire pressure
- Normal loading condition
- Low capacity of the vehicle tank allowed (production fuelling)
- Car standing on its wheels
- Only the driver should be sitting in the car
- No additional disturbances such as closing the door or closing the engine hood are allowed
- Maximum inclination angle of the ESP® system / bracket within the allowed mounting tolerance

The calibration may be done together with the End-of-line test on the rolling road tester, or at a previous assembly station with a suitable flat area (e.g. alignment test rig).

Tolerance chain of all possible single tolerances:

Tolerance chain of all single tolerances			
No.	Kind of the single tolerance	Max. value (°)	Description
1	Tolerances of the ESP ECU and the AXS/AQS sensors	≤ 2,00	Inaccuracy during the offset calibration in the Bosch assembly plant (ECU level), hydraulic tolerance (tolerance ECU/HU), fixing rubbers (tolerance HU/bracket), signal

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			accuracy of the acceleration sensors (e.g. temperature drift, signal interference)
2	Mounting tolerance of the ESP unit in the vehicle	≤ tbd (customer specific)	Design of the bracket, accuracy of the mounting position (e.g. position of the welded mounting bolts)
3	Accuracy of the chassis	≤ tbd (customer specific)	Accuracy of the chassis which influence the inclination angle of the vehicle (e.g. different tire pressure, asymmetric suspension FA/RA)
4	Maximum inclination angle of the test rig respectively ground during calibration	≤ 0,57	Inclination angle of the vehicle caused by unevenness ground respectively alignment station or test rig during execution of the calibration.
---	---	tbd	Sum of all tolerances at a statistic reflection (radical of the square sum); $S_{_statistic_reflection} = \sqrt{s_1^2 + s_2^2 + s_n^2}$
---	---	tbd	Sum of all single tolerances „Worst Case“; $S_{_all_single_tolerances} = s_1 + s_2 + s_n$)
---	---	tbd	Average value (= max. value) of the sum of all tolerances at a statistic reflection and the sum of all single tolerances; $s_{_max} = S_{_statistic_reflection} + S_{_all_single_tolerances} / 2$)

Remarks:

To guarantee correct ABS / ESP® functions the AXS/AQS sensor inclination angle might not exceed a defined maximal value during execution of the calibration procedure.

This maximal value is the average value of the sum of all tolerances at a statistic reflection and the sum of all single tolerances. In addition the resolution of the customer specific diagnostic function has to be considered.

Especially the tolerance regarding the flatness of the ground where the calibration is executed should be as small as possible to have a bigger range for the assembling, mounting and chassis (tire pressure, suspension, etc.) tolerances.

The ESP® ECU is delivered to the customer with predefined AXS and AQS offset values in the E^2PROM. The values show that the sensor is not yet calibrated. As long these values are stored a fault code is set in the fault code memory. This fault code indicates a missing calibration of the AXS and AQS. During the calibration procedure, the ECU writes the measured values of the AXS and AQS sensor offsets into the E^2PROM. From now on, the ECU uses the stored offsets in the E^2PROM to correct the measured acceleration values.

It is not possible for the ECU to check whether the calibration conditions described above have been met, so the customer must guarantee the correct conditions during calibration of the sensor in the plant.

If the measured **AXS or AQS signal offset value A_1** is too large, the ECU will reject the calibration. The ECU will also reject the calibration if during the calibration the spread (difference) of measured

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values over the **calibration time $t_{cal_AXS_AQS}$** is too large. In this case the fault code is still set in the fault code memory.

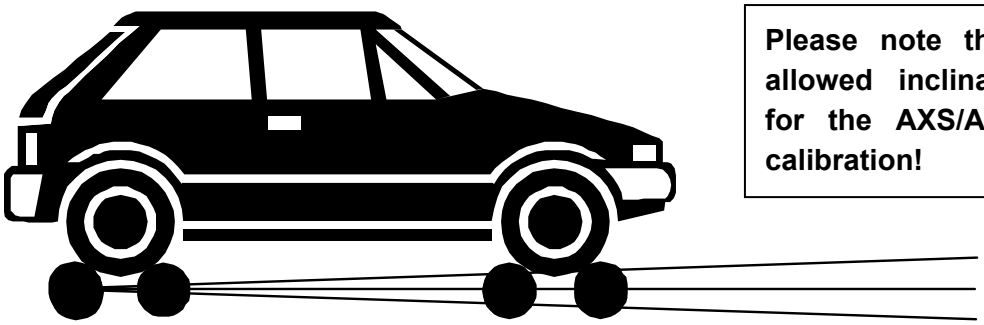
If the ESP® inclusive hydraulic unit is changed in the rework area, a new calibration must be executed. This procedure is also valid for service and workshops at the dealer.

More details to the chapter "Lateral (AQS) and longitudinal (AXS) acceleration sensor offset value calibration - optional" can be read in the customer specific documentations.

Parameter:

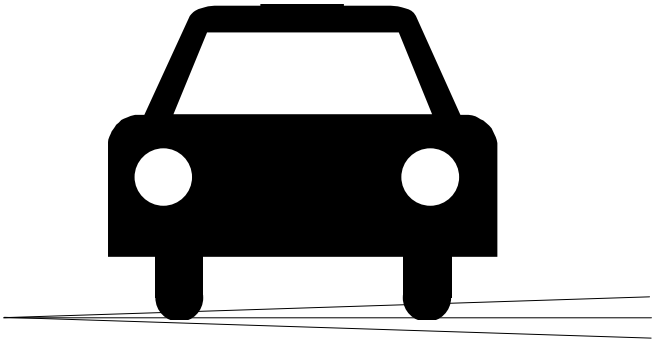
Kind of parameters:	Parameter name:	Unit:	Physical value:
Fault condition	AXS or AQS signal offset value A_1	Degrees (°)	> 10
Time	Calibration time $t_{cal_AXS_AQS}$	Milliseconds (ms)	≤ 1500

Side view of the vehicle (AXS/AQS sensor element calibration):



Please note the maximal allowed inclination angle for the AXS/AQS sensor calibration!

Front view of the vehicle (AXS/AQS sensor element calibration):



Please note the maximal allowed inclination angle for the AXS/AQS sensor calibration!

AXS/AQS calibration successful ⇒ **Continue**

AXS/AQS calibration not successful ⇒ **Abort, print out fault message and send vehicle to repair area**

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4.1.6 Calibration of the steering angle sensor (SAS) - optional

Calibration of the steering angle sensor (SAS) - optional	
Operator:	Test rig / Tester:
☞ Turn the steering wheel to straight ahead position SAS calibration tolerance measured at the steering wheel: $\leq \pm A_2$	Calibration (zeroing of the offset) of the steering angle sensor

Remarks:

The calibration may be done together with the End-of-line test on the rolling road tester, or at a previous assembly station (e.g. alignment test rig).

Following steps have to be done in the described sequence:

1. Initialisation of the SAS by turning of the steering wheel after ignition on (hardware respectively customer specific).

2. Turning of the steering wheel to straight ahead position.

The maximal **SAS calibration tolerance W_2** measured at the steering wheel: $\leq \pm A_2$. These conditions have to be guaranteed for the total **calibration time t_{Cal_SAS}**.

3. Calibration of the steering wheel sensor with the customer specific diagnostic function.

4. Verification of the calibration by evaluation of the fault code memory respectively the customer specific SAS status information. In case of a not successful calibration of the SAS, the calibration procedure has to be executed once again before the process is aborted.

In case of a wrong executed or missing calibration an actual customer specific fault code is stored in the fault code memory of the ESP® ECU (ESP® warning lamp on).

Parameter:

Kind of parameters:	Parameter name:	Unit:	Physical value:
Condition	SAS calibration tolerance W_2	Degrees (°)	5
Time	Calibration time t_{Cal_SAS}	Milliseconds(ms)	≤ 1500

The requirement to calibrate the steering angle sensor depends on the used hardware (customer specific)!

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More details to the chapter “Calibration of the steering angle sensor (SAS) - optional” can be read in the customer specific documentations.

SAS calibration successful ⇒ **Continue**

SAS calibration not successful ⇒ **Abort, print out fault message and send vehicle to repair area**

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4.1.7 Check of the input / output signals

Check of the input / output signals		
Tester / ECU		Notes:
Checks respectively activate the input / output signals (channels).		The tester checks the input and output signals for plausibility. Some outputs / channels must be activated before they can be tested.
ECU data:	Demanded state of the ECU output:	→ Operator / Tester / Notes:
Example of I/O check for an ABS system:		
Hand brake switch:	on	→ Actuate the hand brake (switch)!
	off	→ Release the hand brake (switch)!
Brake light switch:	on	→ Apply the brake pedal!
	off	→ Release the brake pedal!
ABS lamp:	on	Switch on the ABS lamp!
		→ Check, if the ABS lamp is switched on!
	off	Switch off the ABS lamp!
		→ Check, if the ABS lamp is switched off!
Example of the additional part of a I/O check for a ESP® system (ABS I/O check must also be executed):		
ESP® passive-switch:	on	→ Actuate the ESP® passive-switch!
	off	→ Release the ESP® passive-switch!
ESP® passive-lamp:	on	Switch on the ESP® passive-lamp!
		→ Check, if the ESP® passive- lamp switched on!
	off	Switch off the ESP® passive-lamp!
		→ Check, if the ESP® passive- lamp switched off!
Yaw rate sensor (YRS):	- 1,2…+ 1,2 °/s	---
Longitudinal acceleration sensor (AXS):	- 0,384…+ 0,384 m/s^2	Sensors (ALS and AQS) are calibrated; inclination of the ground respectively test rig within the allowed tolerance
Lateral acceleration sensor(AQS):	- 0,384…+ 0,384 m/s^2	

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Pressure sensor (PS):	≥ 35 bar	→ Apply the brake pedal!
	- 7,7 ≤ PS ≤ 7,7 bar	→ Release the brake pedal!
Steering angle sensor (SAS):	-5°... + 5°	→ Turn the steering wheel into straight ahead position!
	SAS ≥ 30°	→ Turn the steering wheel counter clockwise ≥ 30°!
	SAS ≤ -30°	→ Turn the steering wheel clockwise ≤ -30°!

Notes:

The check of the components respectively inputs and output signals have to be done step by step depending on the operator (e.g. 1. step: operator applied brake pedal; 2. step: operator released brake pedal). To be checked components respectively inputs at the ABS/ESP® ECU are customer specific (all mentioned tests are only examples).

The actual ECU data must be compared with the demanded data which are stored in tester data base.

The operator prompts (”release brake pedal”, ”apply brake pedal”, etc.) must be identified on display.

Precondition for the described steering angle sensor test is that the SAS calibration was executed at a previous assembly process.

More details to the chapter “Check of the inputs and output signals” can be read in the customer specific documentations.

The evaluation of the input / output signals has to ensure the following functions and assembly processes:

1) To ensure all electrical connections to the ABS/ESP® ECU
2) To ensure the mechanical connection between the SAS and steering column
3) To ensure the function of the pressure sensors
4) To ensure the function of the brake light switch, etc.

Data correct ⇒ **Continue**

Data not correct ⇒ **Abort, print out fault message and send vehicle to repair area**

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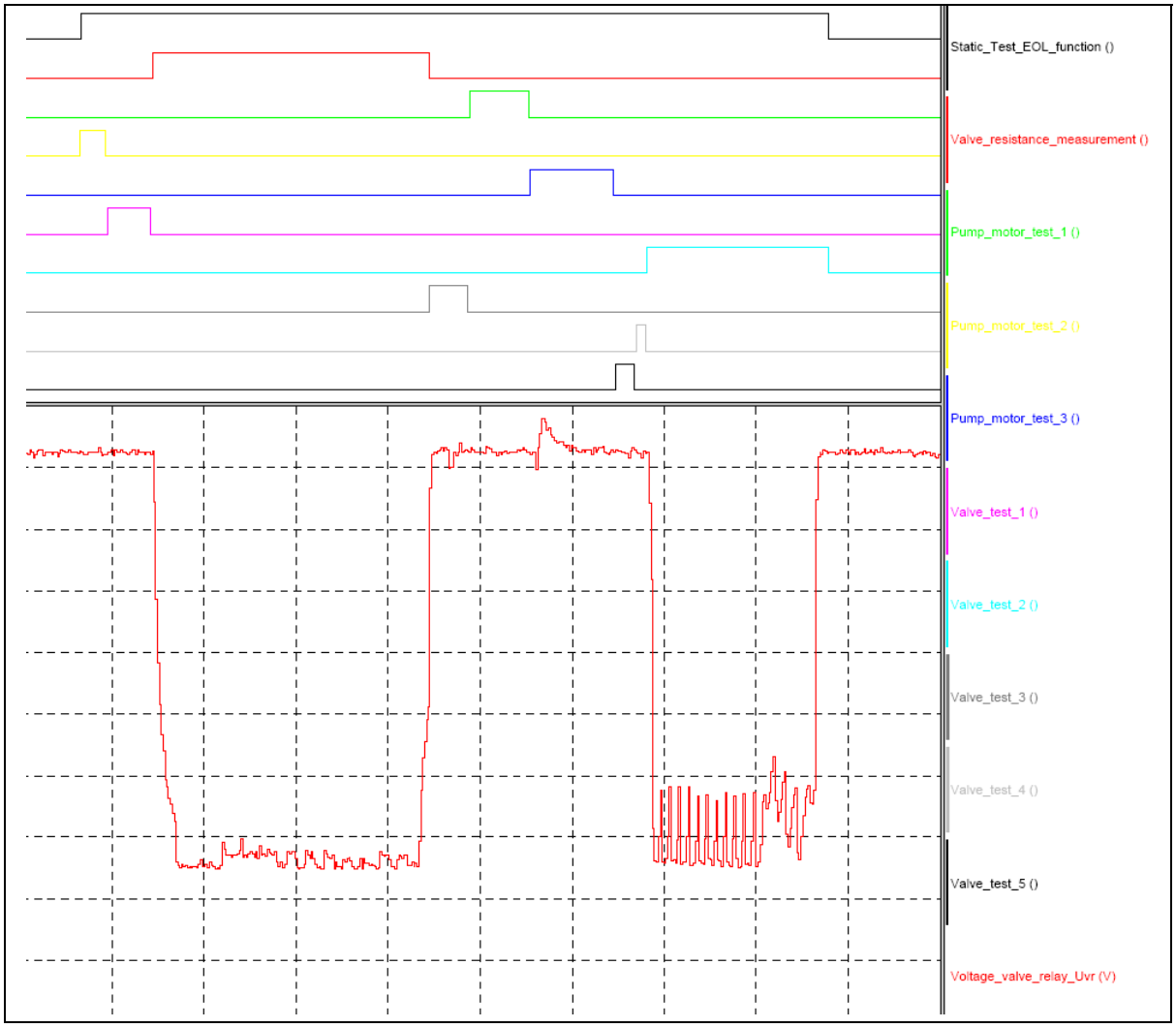
4.1.8 Static Test - optional

Static Test		
Operator:	Tester:	Remarks:
→ Don't apply the brake pedal!	1. Start of the static test 2. Evaluation of the fault code memory.	The static test doesn't provide any specific results. In case of any detected failure a fault code is set. That's the reason why the fault code memory has to be evaluated after the test is finished.

Graphical presentation of the static test:



Remarks:

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The static test starts all valve and pump motor tests, which are normally triggered by the ABS/ESP® ECU after ignition on (preconditions: e.g. no special diagnostic mode active, no under voltage, etc.). All these valve and pump motor tests are identical with the ECU self tests in standstill conditions after power on. In addition valve and pump motor tests are executed, which are normally triggered by the ABS / ESP® ECU only at a defined vehicle speed.
The static test doesn't provide any specific results. After execution of this test the fault code memory has to be evaluated, because in case of a detected fault the corresponding fault code is set in the fault code memory by the static test. For more **details about the used diagnostic function please refer chapter 5.1.**
The **test time t_{ABS} respectively test time t_{ESP}** depends on the used system respectively of the possible test repetitions of each suspicious valve or pump motor test.

In case of an occurred **timeout $t_{_Timeout_St}$** the static test has to be repeated, before the vehicle is indicated as faulty. If this is not possible caused by a limited cycle time within the assembly line, the vehicle has to be marked as faulty car and sent to the rework area. At this location the static test has to be repeated. In case of the timeout occurs again the vehicle has to be investigated for under voltage conditions or wiring harness faults (e.g. loosen ground connection, etc.). This is also valid in case of no fault code is stored.

Test conditions:

- Vehicle engine is running, to guarantee a **minimum voltage at the ABS/ESP® ECU ≥ U_1**.
- Vehicle stands still.

Parameter:

Kind of parameters:	Parameter name:	Unit:	Physical value:
Condition	Minimum voltage U_1	Volt (V)	≥ 12
Time	Timeout $t_{_Timeout_St}$	Milliseconds (ms)	> 10000
	Test time t_{ABS}		1500 to 4500
	Test time t_{ESP}		2000 to 6000

More details to the chapter "Static Test - optional" can be read in the customer specific documentations.

The execution of the static test has to ensure the following functions and assembly processes:

1) To ensure the electrical function of all valves and the pump motor.
2) Start of the valve coil resistance measurement

Static Test ok ⇒ Continue

Static Test nok ⇒ Abort, print out fault message and send vehicle to repair area

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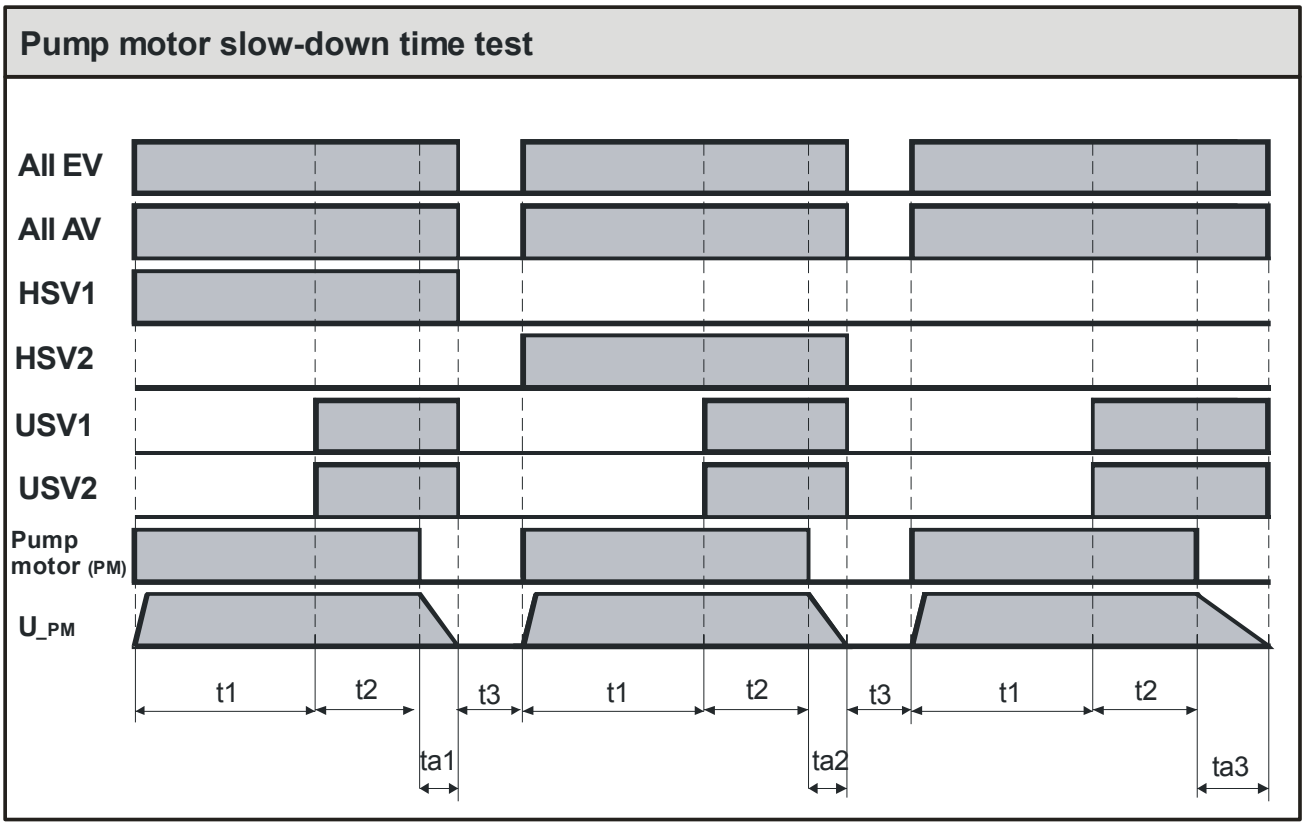
4.1.9 Pump motor slow-down time test (only ESP® systems) – optional

Pump motor slow-down time test		
Operator:	Tester:	Remarks:
→ Don't apply the brake pedal!	Start of the pump motor slow-down time test and evaluation of the test results.	The diagnostic tester evaluates the ECU test results and indicates if the test is passed successfully.

Graphical presentation of the pump motor slow-down time test:



Remarks:

The test will be used to check the feed rate of the pumps respectively the active pressure increase function as well as the tightness of different valves. This test should be executed before the dynamic ESP® test to detect a not successful filled hydraulic unit. In case of failed test the vehicle has to repair bleed.

After the tester has been started the pump motor slow-down time test , the ESP® hydraulic unit creates in each master cylinder circuit step by step a internal **system pressure of p_1 bar** between the closed inlet and pilot valves as a result of the running pump motor and the opened priming valves. After switching off of the pump motor at the test step 1 (MCC1) and test step 2 (MCC2) the ECU records the **post run times t_{a1} and t_{a2}** of the pump motor. Finally the **post run time t_{a3}** of the pump

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motor is measured in unpressurized conditions (test step 3). The post run times are measured and calculated from the generated post run voltage after switching off of the pump motor. The post run times t_{a1}, t_{a2} and t_{a3} have to be read by the tester after the pump motor slow-down time test is finished. For more details about the used diagnostic function please refer chapter 5.2.

Conditions:

- Vehicle engine is running, to guarantee a **minimum voltage at the ESP® ECU ≥ U_1**.
- Vehicle stands still.
- The brake pedal might not applied during the whole running test! The tester has to ensure this by reading and evaluating the BLS status.

Details to the test step 1:

After the start of the routine respectively the start of the test step 1 all outlet (AV) and inlet valves (EV) as well as the prime valve 1 (HSV1) and the pump motor are switched on.

After the **start time t_1 ($t_1 \geq t_{1min}$)** is expired both USV valves are switched on to create an internal system pressure in the master cylinder circuit 1 (MCC1). After the **start time t_2 ($t_2 \geq t_{2min}$)** is expired the pump motor is switched off and the post run time t_{a1} is measured.

At the end of the measurement all valves are switched off and after the **waiting time t_3** (time constant in the software) test step 2 is started.

Details to the test step 2:

After the start of the routine respectively the start of the test step 2 all outlet (AV) and inlet valves (EV) as well as the prime valve 2 (HSV2) and the pump motor are switched on.

After the **start time t_1 ($t_1 \geq t_{1min}$)** is expired both USV valves are switched on to create an internal system pressure in the master cylinder circuit 2 (MCC2). After the **start time t_2 ($t_2 \geq t_{2min}$)** is expired the pump motor is switched off and the post run time t_{a2} is measured.

At the end of the measurement all valves are switched off and after the **waiting time t_3** (time constant in the software) test step 3 is started.

Details to the test step 3:

After the start of the routine respectively the start of the test step 3 all outlet (AV) and inlet valves (EV) and the pump motor are switched on. At this test step no prime valve is switched on.

After the **start time t_1 ($t_1 \geq t_{1min}$)** is expired both USV valves are switched on (no system pressure is created). After the **start time t_2 ($t_2 \geq t_{2min}$)** is expired the pump motor is switched off and the post run time t_{a3} (unpressured conditions) is measured.

At the end of the measurement all valves are switched off.
After test step 3 is finished the tester has to request the routine results of all 3 test steps with a separate service.

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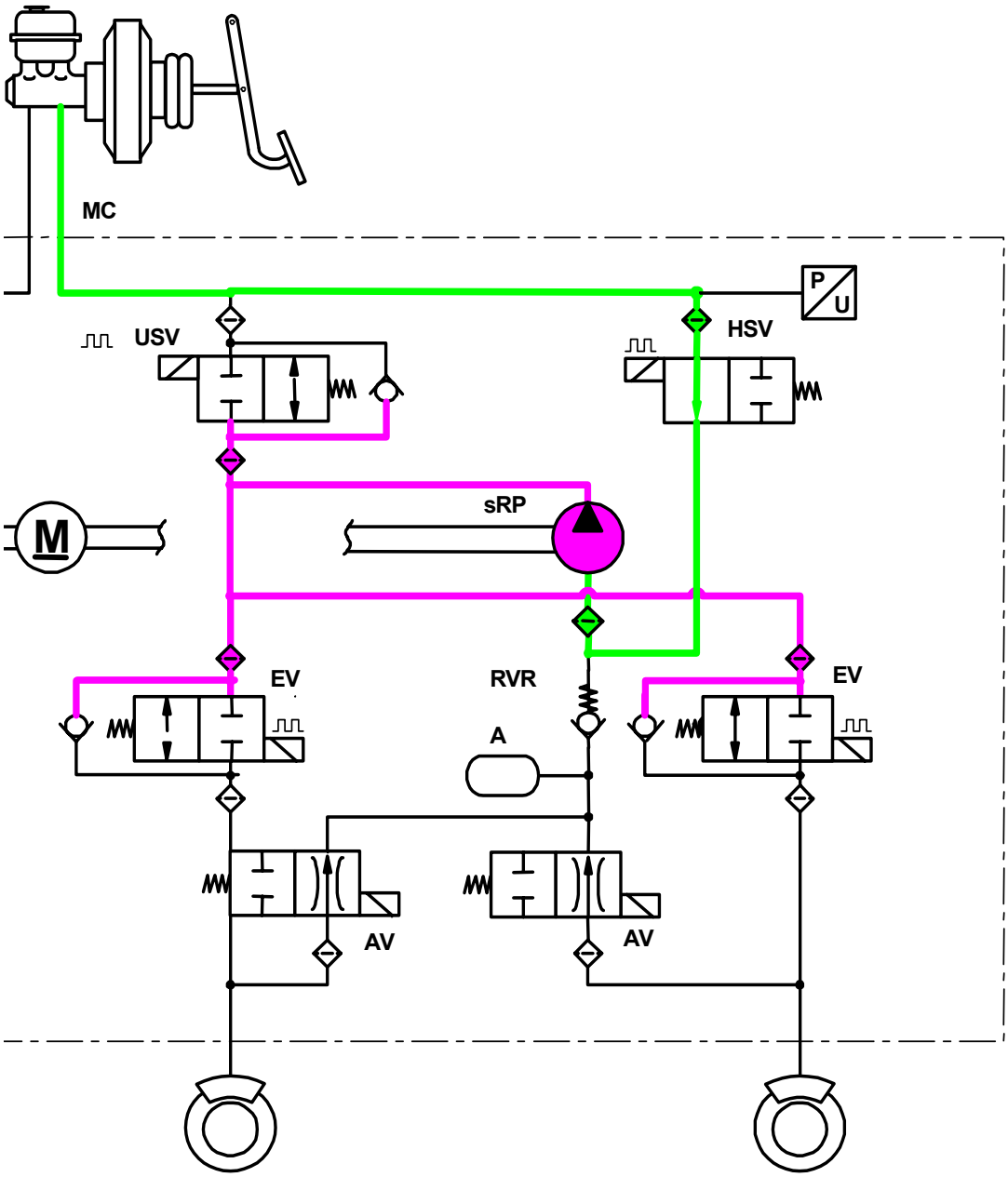
Details to the internal system pressure of the test step 1 & 2:



By closing of the EV and USV valve, opening of the HSV valve and switching on of the pump motor a system internal pressure is created. This system pressure (p_1 bar) is limited by the maximal allowed activation current for the USV valve (Details to the valve protection measurements can be read in the chapter 6). To avoid an influence of a possible EV leakage on the wheel circuits, all AV valves are switched on.

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Parameter:

Kind of parameters:	Parameter name:	Unit:	Physical value:
Condition	Minimum voltage U_1	Volt (V)	12
	System pressure p_1	Bar	110 ± 20
Time constant in the software	Waiting time t_3	Milliseconds (ms)	1500
Start of routine	Start time t_1		$\geq t_{1min}$
	Start time t_2		$\geq t_{2min}$
Minimum requirement for the start of routine	Start time t_{1min}		400
	Start time t_{2min}		200
Routine results	Post run time t_{a1}		xx
	Post run time t_{a2}		xx
	Post run time t_{a3}		xx
Evaluation parameter	Difference time $t_{a_delta_1}$		tbd
	Difference time $t_{a_delta_2}$		tbd
	Post run time $t_{a1/a2_min}$		tbd
	Post run time $t_{a1/a2_max}$		tbd
	Post run time t_{a3_min}		tbd
	Post run time t_{a3_max}		tbd
Time	Total test time $t_{total_test_time}$		6500

Remarks:

The given total test time $t_{total_test_time}$ of the pump motor slow-down time test depends on the selected start times (t_1 & t_2) and the test conditions. For the given time following was assumed:

$t_{total_test_time} = 3^* t_{1min} + 3^* t_{2min} + 2 * t_3 + 3 * t_{a3_max}$;
for t_{a3_max} = 3 * 600 ms e.g. if the ESP® hydraulic unit is completely empty (not filled).

Different causes e.g. the ESP® ECU supply voltage, the state of filling (filled / not filled) of the ESP® hydraulic unit or the allowed valve leakages influence the post rum times (t_{a1} & t_{a2} & t_{a3}), therefore the customer has to define the evaluation parameter (**difference time $t_{a_delta_1}$, difference time $t_{a_delta_2}$, post run time $t_{a1/a2_min}$, post run time $t_{a1/a2_max}$, post run time t_{a3_min} and post run time t_{a3_max}**) by doing a statistic judgement.

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Recommended evaluation logic:
$t_{a1/a2_min} \leq t_{a1} \leq t_{a1/a2_max}$
$t_{a1/a2_min} \leq t_{a2} \leq t_{a1/a2_max}$
$t_{a3_min} \leq t_{a3} \leq t_{a3_max}$
$t_{a3} - t_{a1} \geq t_{delta_1}$
$t_{a3} - t_{a2} \geq t_{delta_1}$
$t_{a1} - t_{a2} \leq \pm t_{delta_2}$

More details to the chapter “pump motor slow-down time test (only ESP® systems)” can be read in the customer specific documentations.

<u>The execution of the pump motor slow-down time test has to ensure the following functions and assembly processes:</u>

1) To ensure the pump flow rate respectively a active ESP® pressure increase function (system internal pressure increase) at vehicle standstill conditions
2) To ensure a sufficient filled secondary and priming circuit of the ESP® system

Pump motor slow-down time test ok ⇒ Continue

Pump motor slow-down time test nok ⇒ Abort, print out fault message and send vehicle to repair area

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4.1.10 Evaluation and erasing of the fault code memory

Evaluation and erasing of the fault code memory
Tester / ECU
Erase the fault code memory. Immediately after execution of the erasing command the ECU executes the so-called „good check“. In case of actual faults are existing they will be stored again.
Recommend **waiting time t_{good_check}** till the fault code memory can be evaluated again. Alternate an ECU reset function can be executed.
Evaluation of the fault code memory. Following strategy is recommended: • If system free of faults (see 1. example), continue with the EOL tests. • If system not free of faults (see 2. example), stop the EOL tests, print out fault message and send vehicle to repair area.

Example:	1. diagnostic trouble code (DTC):	Status of 1. DTC:	1.+n. DTC:	Status of 1+n. DTC:	Evaluation:
1)	---	---	---	---	ok
2)	WSS front left, line fault	actual	---	---	nok

Remarks:

The ECU outputs the entire contents of the fault code memory. These data include different information. The number of the maximal storable fault codes is customer specific.
Following data of the received fault information have to be evaluated:

- Number of DTC
- Status of DTC (actual/historic)

The tester has to evaluate the fault data and to transfer the number of DTC into a failure text and to print out a failure protocol. The status information of a DTC shows if a fault is actual or historic.

The waiting time t_{good_check} after execution of the good check is required to ensure, that all fault paths are tested by the ECU and in case of they are still active (actual), they will be stored again. The time which is required for the renewed fault storing depends on the monitoring (filter) time of each individual fault.

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Parameter:

Kind of parameters:	Parameter name:	Unit:	Physical value:
Time	Waiting time t_{good_check}	Milliseconds (ms)	5000

More details to the chapter “Evaluation and erasing of the fault code memory” can be read in the customer specific documentations.

The evaluation of the fault code memory has to ensure the following functions and assembly processes:

1) To ensure that the ABS/ESP® system is free of faults before the dynamic Tests are executed on a 2-axle brake dynamometer (low speed) respectively on a roller test bench with 4 independent and detachable rolls (high speed)

ABS/ESP® system free of faults ⇒ **Continue**

ABS/ESP® system not free of faults ⇒ **Abort, print out fault message and send vehicle to repair area**

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4.2 Dynamic tests on a 2-axle brake dynamometer (low speed)

4.2.1 Wheel speed sensor correlation, signal quality and WSS11i tests (optional)

Wheel speed sensor tests (correlation and signal quality)									
Test rig speed $V_{_Test}$:			ECU data to be evaluated:					Proposal for evaluation:	Test time:
1. test step: wheel speed sensor correlation test									
Roll FL	On	x km/h	wheel FL		xx	km/h		$V_{FL} \geq V_{roll_on}$	$t_{_WSS_correlation}$
Roll FR	Off	y km/h	wheel FR		xx	km/h		$V_{FR} \leq V_{roll_off}$	
Roll RL	Off	y km/h	wheel RL		xx	km/h		$V_{RL} \leq V_{roll_off}$	
Roll RR	Off	y km/h	wheel RR		xx	km/h		$V_{RR} \leq V_{roll_off}$	
Roll FL	On	x km/h	wheel FL		xx	km/h		$V_{FL} \geq V_{roll_on}$	
Roll FR	On	x km/h	wheel FR		xx	km/h		$V_{FR} \geq V_{roll_on}$	
Roll RL	Off	y km/h	wheel RL		xx	km/h		$V_{RL} \leq V_{roll_off}$	
Roll RR	Off	y km/h	wheel RR		xx	km/h		$V_{RR} \leq V_{roll_off}$	
Roll FL	On	x km/h	wheel FL		xx	km/h		$V_{FL} \geq V_{roll_on}$	
Roll FR	On	x km/h	wheel FR		xx	km/h		$V_{FR} \geq V_{roll_on}$	
Roll RL	On	x km/h	wheel RL		xx	km/h		$V_{RL} \geq V_{roll_on}$	
Roll RR	Off	y km/h	wheel RR		xx	km/h		$V_{RR} \leq V_{roll_off}$	
2. test step: wheel speed sensor signal quality test									
Roll FL	On	x km/h	wheel FL	max:	xx	km/h		$V_{FLmax} \leq V_{FLØ} + V_{WSS_tolerance}$	$t_{_WSS_signal_quality}$
				min:	xx	km/h		$V_{FLmin} \geq V_{FLØ} - V_{WSS_tolerance}$	
				Ø	xx	km/h		$V_{_Test} - V_{tolerance} \leq V_{FLØ} \leq V_{_Test} + V_{tolerance}$	
Roll FR	On	x km/h	wheel FR	max:	xx	km/h		$V_{FRmax} \leq V_{FRØ} + V_{WSS_tolerance}$	
				min:	xx	km/h		$V_{FRmin} \geq V_{FRØ} - V_{WSS_tolerance}$	
				Ø	xx	km/h		$V_{_Test} - V_{tolerance} \leq V_{FRØ} \leq V_{_Test} + V_{tolerance}$	
Roll RL	On	x km/h	wheel RL	max:	xx	km/h		$V_{RLmax} \leq V_{RLØ} + V_{WSS_tolerance}$	
				min:	xx	km/h		$V_{RLmin} \geq V_{RLØ} - V_{WSS_tolerance}$	
				Ø	xx	km/h		$V_{_Test} - V_{tolerance} \leq FR_{LØ} \leq V_{_Test} + V_{tolerance}$	
Roll RR	On	x km/h	wheel RR	max:	xx	km/h		$V_{RRmax} \leq V_{RRØ} + T_{WSS_tolerance}$	
				min:	xx	km/h		$V_{RRmin} \geq V_{RRØ} - V_{tolerance}$ WSS_tolerance	
				Ø	xx	km/h		$V_{_Test} - V_{tolerance} \leq FR_{RØ} \leq V_{_Test} + V_{tolerance}$	
3. test step (optional): WSSi additional information „rotation direction”									
Roll FL	On	x km/h	rotation direction FL					$RD_{FL} = RD_{_demand}$	$t_{_WSS_rotation_direction}$
Roll FR	On	x km/h	rotation direction FR					$RD_{FR} = RD_{_demand}$	
Roll RL	On	x km/h	rotation direction RL					$RD_{RL} = RD_{_demand}$	

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Roll RR	On	x km/h	rotation direction RR	$RD_{RR} = RD_{_demand}$	
4. test step (optional): WSSi additional information „mounting position”					
Roll FL	On	x km/h	mounting position FL	$MP_{FL} = MP_{_demand}$	$t_{_WSS_mounting_position}$
Roll FR	On	x km/h	mounting position FR	$MP_{FR} = MP_{_demand}$	
Roll RL	On	x km/h	mounting position RL	$MP_{RL} = MP_{_demand}$	
Roll RR	On	X km/h	mounting position RR	$MP_{RR} = MP_{_demand}$	

Remarks:

1. test step: wheel speed sensor correlation test:

To check the correlation of the wheel speed sensors (connection of the WSS with the related wheel) the test rig switches on the rolls step by step with the **test speed $V_{_Test}$**. In parallel the tester takes the wheel speed data of the ECU and compares them with the **speed $V_{_roll_on}$** und **$V_{_roll_off}$** during the **test time $t_{_WSS_correlation}$**.

2. test step: wheel speed sensor signal quality test:

During the signal quality test all rolls are switched on (this test step will be executed immediately after the correlation test). The tester starts the in the ECU software included WSS test routine for the **test time $t_{_WSS_signal_quality}$** (time depends on the test speed and wheel circumference), which picks up with the **sampling time $t_{_sampling}$** all sensor speed data. The selected test time has to cover a complete revolution of a wheel.
The WSS test routine stores the **minimum, maximum and average speed data (V_{FLmin}, V_{FLmax}, $V_{FLØ}$, etc.)** of each wheel, which were measured during the selected test time. After the end of the test is reached, the routine outputs these data on request of the tester.
The test equipment evaluates these data and compares them with the calculated limit values from the average speed data and the **tolerance $V_{WSS_tolerance}$**. Additional the tester has to check if the measured average speed of each wheel is within a defined speed range. This is done by comparing of the individual average wheel speed and the test rig test speed ± **tolerance $V_{tolerance}$**. **For more details about the evaluation please refer the chapter 4.2.1.4. Details about the used diagnostic function can be read in the chapter 5.3.**

3. & 4. test step (optional): WSSi rotation direction and mounting position test:

These additional and optional test steps are only required for vehicles with intelligent wheel speed sensors (e.g. Bosch WSS11i). These tests are directly executed after the standard correlation and signal quality tests. To ensure that the WSS weren’t wrongly mounted the tester reads and checks the information “rotation direction” and “mounting position”. These data have to be read with the corresponding diagnostic function during one revolution of the wheel. At first the information “rotation direction of all wheels” within the **test time $t_{_WSS_rotation_direction}$** is read and compared with the expected value **rotation direction demand RD_{demand}**. During the following **test time $t_{_WSS_mounintg_position}$** (time depends on the test speed and wheel circumference) the information “mounting position of all wheels” is read 3 times, at the beginning, in the middle and at the end of the test time. The read ECU data are compared with the expected **mounting position demand MP_{demand}**. The requirement to read this data only 3 times is sufficient, because sporadic occurred fault information is stored in the ECU for the **storing time t_{St}**. During this time the stored data can be read via the corresponding diagnostic service. **For more details about the evaluation please refer the chapter 4.2.1.5 and 4.2.1.6.**

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Parameter:

Kind of parameters:	Parameter name:	Unit:	Physical value:
Time constant in the software	Sampling time $t_{_sampling}$	Milliseconds (ms)	5
	Storing time t_{St}		3000
Time	Test time $t_{_WSS_correlation}$		tbd
	Test time $t_{_WSS_signal\ quality}$		tbd
	Test time $t_{_WSS_rotation_direction}$		tbd
	Test time $t_{_WSS_mounting_position}$		tbd
Condition	Lower test speed $V_{_min}$	Kilometre per hour Km/h)	≥ 3
	Upper test speed $V_{_max}$		≤ 15
Evaluation parameter	Test speed $V_{_roll_on}$		$= V_{Test} - T_{Vvalid}$
	Test speed $V_{_roll_off}$		1
	tolerance $V_{WSS_tolerance}$	Percent (%)	tbd
	tolerance $V_{tolerance}$		10
	Rotation direction RD_{demand}	---	forward
	Mounting position MP_{demand}		tbd

Notes to the given parameters:

The test times depends on the test rig speed, the times to accelerate the rolls and the used tire sizes. These times are to be defined and specified by the customer.
For the definition of the test times $t_{_WSS_signal_quality}$ and $t_{_WSS_mounitng_position}$ following formula is valid:

Test time= wheel circumference / test rig speed

Example:

Test rig speed: 4.5 km/h (1,25 m/s)

Wheel circumference: 2 Meter

Test time = 1,6 seconds (2m / 1,25 m/s)

→ recommended test time with some safety: 2 seconds

To ensure that all important wheel speed information (wheel speed signals respectively mounting position (air gap) information – only vehicles with WSS11i) are available, the **test speed $V_{_Test}$** might not below the **lower test speed $V_{_min}$**.

To ensure the highest sampling rate per tooth respectively speed resolution, the **test speed $V_{_Test}$** might not exceed the **upper test speed $V_{_max}$**.

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As low as possible the test speed is selected, even more is the sampling rate per tooth. Generally following reference values are valid:

Speed (km/h)	Samplings per tooth
2,5	12
5	6
10	3
15	**2**

Details to all 4 test steps and evaluations can be seen in the graphics on the following pages.

<u>The execution of the wheel speed sensor tests have to ensure the following functions and assembly processes:</u>

1) To ensure the correct correlation between the wheels speed sensors and the wheels
2) To ensure the signal quality regarding missing tooth respectively single division error
3) Correct installation (evaluation of the additional WSSi information “rotation direction”)
4) To ensure the correct air gap respectively magnetic field intensity (evaluation of the additional WSSi information “mounting position”)

No WSS failure detected ⇒ **Continue**

WSS failure detected ⇒ **Failure message, abort, prints out fault message and send vehicle to repair area**

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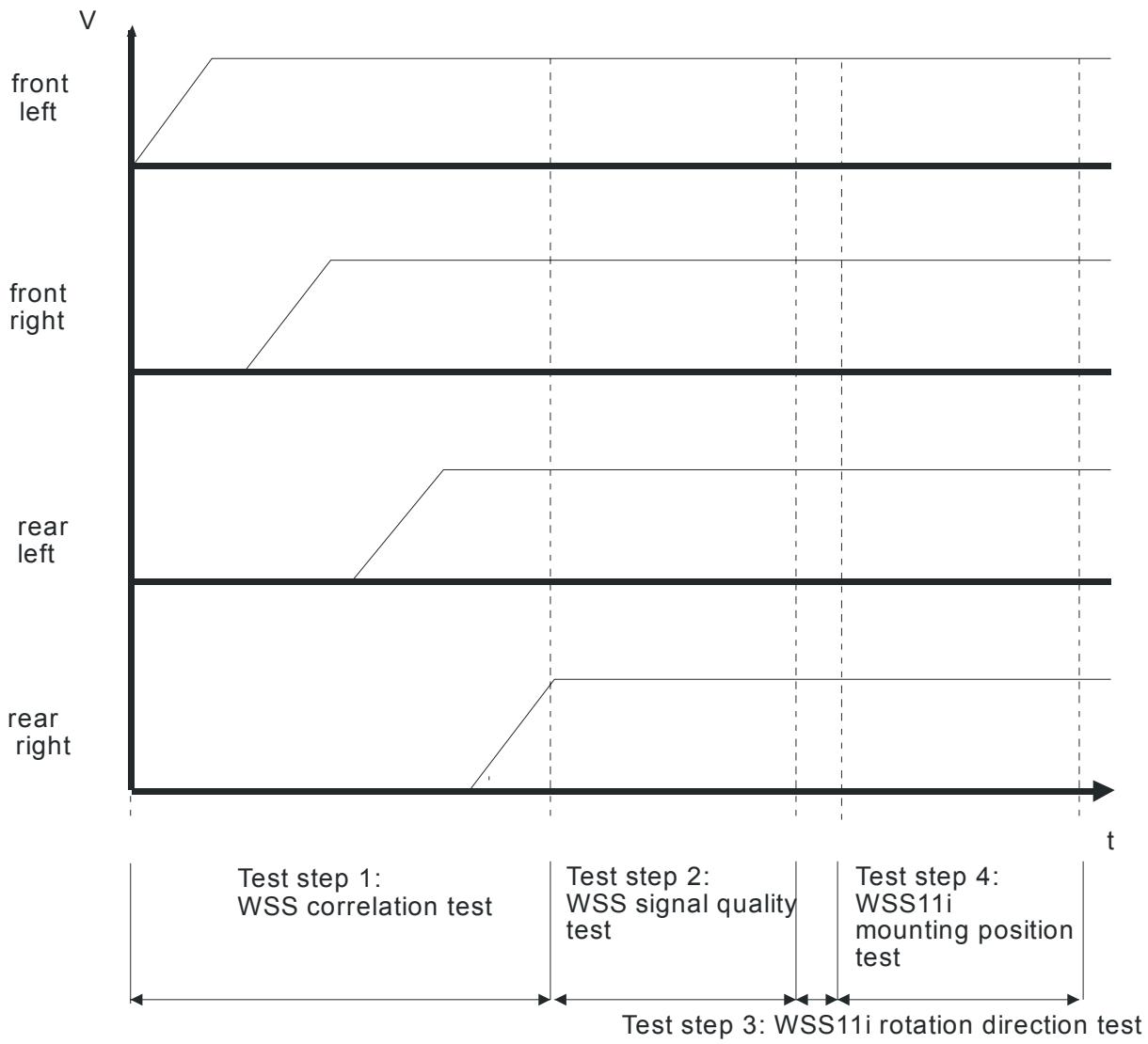
4.2.1.1 Graphical presentation of the entire wheel speed sensor tests (test step 1-4)

Wheel speed sensor test on a 2-axle brake dynamometer



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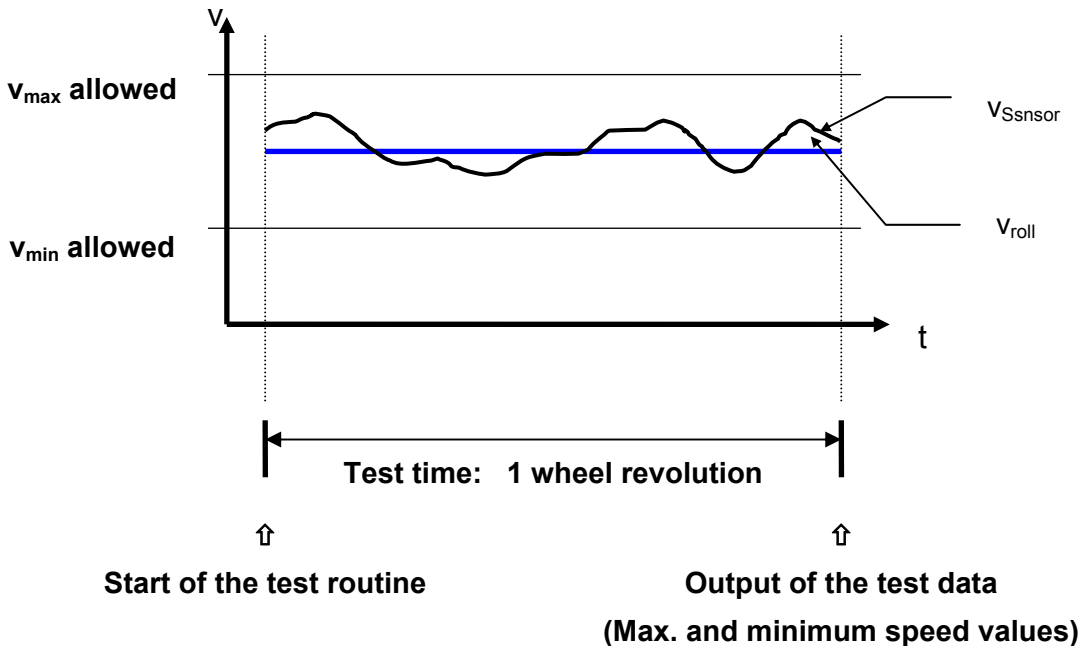
4.2.1.2 Evaluation diagrams of the WSS signal quality test

Wheel speed sensor test (signal quality test)

Test diagram (Example at one wheel):



Remarks:

The test rig turns on the individual rolls by external drives step by step while the tester requests the speed data of the sensors and checks the correct correlation.
The tester then runs a function (routine) that checks the sensor quality over a minimum of one revolution for all wheels. The ECU internally picks up each sensor speed and stores the minimum, maximum and average (optional) speed data within the selected test time. In order to identify faults of the sensor, toothed wheels, air gaps or eccentricities the tester has to evaluate whether the speeds are within a certain limit. These limits have to be defined to cover all WSS signal tolerances. The tolerance of test bench must be hold as small as possible. The selected test rig speed has to be within the function speed of the system.

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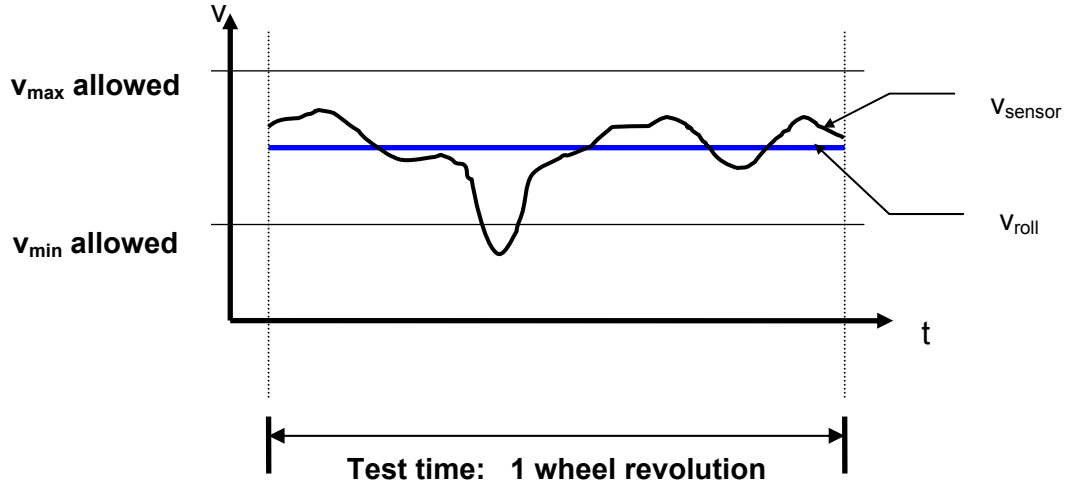
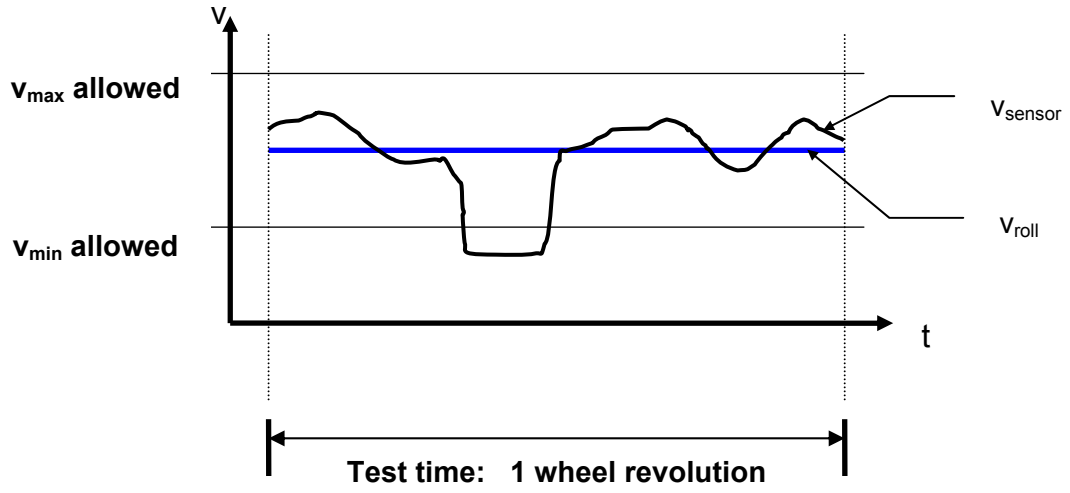
Fault simulation:

Fault simulation: faulty tooth wheel



Fault simulation: Air gap to big or eccentricity



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4.2.1.3 Exemplary determination of the allowed WSS signal tolerance

Influences on the WSS signal quality test:	Nominal tolerance in %:	Real tolerance in %:	Remarks:
Constant velocity respectively repetition accuracy of the test rig speed	3,00	2,00	The nominal test rig tolerance has to be taken out of the test rig documentation. The real test rig tolerance should be measured and then used as test tolerance. The definition of this tolerance is up the customer responsibility.
Signal tolerance (WSS and tooth wheel)	5,00	2,50	The real signal tolerance is normally better than the specified nominal value by Bosch. Consequently this real signal tolerance has to be defined by a statistic evaluation and then be used as test tolerance. The definition of this tolerance is up the customer responsibility.
Total tolerance of the sum of all single tolerances (SST)	---	4,50	Maximal tolerance of the sum of all single tolerances.
Total tolerance of the radical of the square sum of all single tolerances (RSS)	---	3,20	Maximal tolerance of the statistic evaluation.
Tolerance $V_{WSS_tolerance}$:	---	**3,85**	To guarantee the most exact test of the WSS signals, not the nominal tolerance should be used, but the average value of total tolerance of the sum of all single tolerances (SST) and total tolerance of the radical of the square sum of all single tolerances (RSS).

4.2.1.4 Evaluation proposal for the WSS test (signal quality regarding single pitch deviation, eccentricity, etc.):

1. step: Comparison of the maximal measured wheel speeds (V_{FLmax}, etc.) with the average wheel speeds ($V_{ØFL}$, etc.) plus the **tolerance $V_{WSS_tolerance}$**:

step 1.1: $V_{FLmax} \leq V_{ØFL} + V_{WSS_tolerance}$
step 1.2: $V_{FRmax} \leq V_{ØFR} + V_{WSS_tolerance}$
step 1.3: $V_{RLmax} \leq V_{ØRL} + V_{WSS_tolerance}$
step 1.4: $V_{RRmax} \leq V_{ØRR} + V_{WSS_tolerance}$

2. step: Comparison of the minimal measured wheel speeds (V_{FLmin}, etc.) with the average wheel speeds ($V_{ØFL}$, etc.) minus the **tolerance $V_{WSS_tolerance}$**:

step 2.1: $V_{FLmin} \geq V_{ØFL} - V_{WSS_tolerance}$
step 2.2: $V_{FRmin} \geq V_{ØFR} - V_{WSS_tolerance}$
step 2.3: $V_{RLmin} \geq V_{ØRL} - V_{WSS_tolerance}$
step 2.4: $V_{RRmin} \geq V_{ØRR} - V_{WSS_tolerance}$

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3. step: To ensure, that all wheels are within a plausible speed during this test, the measured average speed of each wheel has additional to be compared with the test rig speed V_{test} plus respectively minus the **tolerance $V_{tolerance}$**:

step 3.1 $V_{test} – V_{tolerance} \leq V_{ØFL} \leq V_{test} + V_{tolerance}$
step 3.2: $V_{test} - V_{tolerance} \leq V_{ØFR} \leq V_{test} + V_{tolerance}$
step 3.3: $V_{test} - V_{tolerance} \leq V_{ØRL} \leq V_{test} + V_{tolerance}$
step 3.4: $V_{test} - V_{tolerance} \leq V_{ØRR} \leq V_{test} + V_{tolerance}$

Example:

Measured minimal, maximal and average speeds (km/h) of each wheel:

V_{FLmin}	V_{FLmax}	V_{FRmin}	V_{FRmax}	V_{RLmin}	V_{RLmax}	V_{RRmin}	V_{RRmax}
4,4	4,6	4,5	4,6	4,2	4,5	4,4	4,65
$V_{ØFL}$	$V_{ØFR}$	$V_{ØRL}$	$V_{ØRR}$				
4,48	4,48	4,48	4,48				

Test rig speed (V_{test}): 4,5 km/h

Judgement:

1.1 step: V_{FLmax} (4,6 ≤ 4,652) = ok
1.2 step: V_{FRmax} (4,6 ≤ 4,652) = ok
1.3 step: V_{RLmax} (4,3 ≤ 4,652) = ok
1.4 step: V_{RRmax} (4,65 ≤ 4,652) = ok
2.1 step: V_{FLmin} (4,4 ≥ 4,308) = ok
2.2 step: V_{FRmin} (4,5 ≥ 4,308) = ok
2.3 step: V_{RLmin} (4,2 ≥ 4,308) = nok
2.4 step: V_{RRmin} (4,4 ≥ 4,308) = ok

3.1 step: $V_{ØFL}$ (4,032 ≤ 4,48 ≤ 4,928) = ok
3.2 step: $V_{ØFR}$ (4,032 ≤ 4,48 ≤ 4,928) = ok
3.3 step: $V_{ØRL}$ (4,032 ≤ 4,48 ≤ 4,928) = ok
3.4 step: $V_{ØRR}$ (4,032 ≤ 4,48 ≤ 4,928) = ok

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4.2.1.5 Evaluation of the additional information "rotation direction" of a intelligent WSS (e.g. Bosch WSS11i) - optional

Description „RD_{Demand}“:	Judgement:
Condition not correct (measured value not valid)	nok
Rotation direction forward	**ok**
Rotation direction backward	nok

Remarks:

This diagnostic service is sent by the tester during the wheel is turned by the test rig. The ECU outputs the actual identified rotation direction. An additional rearward drive test is not required, but it can be executed instead of the forward drive test.

4.2.1.6 Evaluation of the additional information "mounting position" of a intelligent WSS (e.g. Bosch WSS11i) - optional

Description „MP_{demand}“:	Judgement:	Possible fault causes:
Mounting position of the WSS ok	**ok**	**---**
Mounting position of the WSS near of the limits	Conditional failure*	Magnetic field intensity (mT) ≤ 5,2 till 9,6 (typical 7,2) caused by: • Sensor neck or bracket damaged or burred • Dirt between sensor neck and the bracket • Mounting screw loosen • Axle or WSS bracket have not the right dimension stability • Weak magnetic field intensity of the tooth wheel
Faulty mounting position of the WSS	Failure	Magnetic field intensity (mT) ≤ 0,9 till 2,6 (typical 1,4) caused by: • Sensor neck or bracket damaged or burred • Dirt between sensor neck and the bracket • Mounting screw loosen • Axle or WSS bracket have not the right dimension stability • Weak magnetic field intensity of the tooth wheel
Conditions not fulfilled (measured value not valid)	Failure	• Electrical defect (e.g. WSS line interrupted, etc.) • Info not available, because $\mathbf{V_{_min}} > V_{_Test} > \mathbf{V_{_max}}$

Remarks:

During the **test time $t_{_WSS_mounting_position}$** the tester has to evaluate the mounting position information 3 times. The first time directly after the start of the test time, the second at the middle and the third time at the end of the test. The evaluation of 3 times is enough because a sporadic occurring mounting position fault information is stored for the **storing time t_{St}**, so that this information can be dependably read via the diagnostics. Please not that the test time must be defined to cover one wheel revolution.
* Depending on the design of the air gap (between WSS and tooth wheel) the information "mounting position of the WSS near of the limits" have to be judged as good or ok case. In case of the range of

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all possible air gap tolerances allows judging the information “mounting position of the WSS near of the limits as not good or nok case; Bosch recommends strongly doing that, because of more safeness.

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4.2.2 Base brake- and ABS test

Base brake test (inclusive the optional leakage test) and ABS test (function "pressure decrease and increase" and brake pipe correlation test)				
Operator:	Test rig:	To be evaluated brake forces:	Demanded brake forces:	Measure sample:
→ Don't apply Brake pedal!	Measuring of the freewheeling forces	Wheel FL: xx N Wheel FR: xx N Wheel RL: xx N Wheel RR: xx N	$F_{FL} \leq F_1$ $F_{FR} \leq F_1$ $F_{RL} \leq F_2$ $F_{RR} \leq F_2$	M1
→ Apply the brake pedal till the expected brake forces are reached!	Pressure increase at all wheels Measuring of the brake forces	Wheel FL: xx N Wheel FR: xx N Wheel RL: xx N Wheel RR: xx N	$F_3 \leq F_{FL} \leq F_4$ $F_{FL} - F_{brake\ tolerance} \leq F_{FR} \leq F_{FL} + F_{Brake\ tolerance}$ $F_5 \leq F_{RL} \leq F_6$ $F_{RL} - F_{Brake\ tolerance} \leq F_{RR} \leq F_{RL} + F_{Brake\ tolerance}$	M2
→ Keep the brake pedal constant!	Measuring of the brake forces before and after the **measure time t_9** → leakage test (optional)	Wheel FL: xx N Wheel FR: xx N Wheel RL: xx N Wheel RR: xx N	$F_{FL} \leq F_9$ $F_{FR} \leq F_9$ $F_{RL} \leq F_9$ $F_{RR} \leq F_9$	M3.1 M3.2
→ Keep the brake pedal constant!	Pressure decrease FL Measure brake forces	Wheel FL: xx N Wheel FR: xx N Wheel RL: xx N Wheel RR: xx N	$F_{FL} \leq F_7$ $F_3 \leq F_{FR} \leq F_4$ $F_5 \leq F_{RL} \leq F_6$ $F_5 \leq F_{RR} \leq F_6$	M4
→ Keep the brake pedal constant!	Pressure increase FL Pressure decrease FR Measure brake forces	Wheel FL: xx N Wheel FR: xx N Wheel RL: xx N Wheel RR: xx N	$F_{FL} \geq F_3$ $F_{FR} \leq F_7$ $F_5 \leq F_{RL} \leq F_6$ $F_5 \leq F_{RR} \leq F_6$	M5
→ Keep the brake pedal constant!	Pressure increase FR Measure brake forces	Wheel FL: xx N Wheel FR: xx N Wheel RL: xx N Wheel RR: xx N	$F_{FL} \geq F_3$ $F_{FR} \geq F_3$ $F_5 \leq F_{RL} \leq F_6$ $F_5 \leq F_{RR} \leq F_6$	M6
→ Keep the brake pedal constant!	Pressure decrease RL Measure brake forces	Wheel FL: xx N Wheel FR: xx N Wheel RL: xx N Wheel RR: xx N	$F_{FL} \geq F_3$ $F_{FR} \geq F_3$ $F_{RL} \leq F_8$ $F_5 \leq F_{RR} \leq F_6$	M7
→ Keep the brake pedal constant!	Pressure increase RL Pressure decrease RR	Wheel FL: xx N Wheel FR: xx N Wheel RL: xx N	$F_{FL} \geq F_3$ $F_{FR} \geq F_3$ $F_{RL} \geq F_5$	M8

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	Measure brake forces	Wheel RR: xx N	$F_{RR} \leq F_8$	
→ Keep the brake pedal constant!	Pressure increase RR Measure brake forces	Wheel FL: xx N Wheel FR: xx N Wheel FL: xx N Wheel FR: xx N	$F_{FL} \geq F_3$ $F_{FR} \geq F_3$ $F_{RL} \geq F_5$ $F_{RR} \geq F_5$	M9
→ Release the brake pedal!	Measuring of the freewheeling forces	Wheel FL: xx N Wheel FR: xx N Wheel RL: xx N Wheel RR: xx N	$F_{FL} \leq F_1$ $F_{FR} \leq F_1$ $F_{RL} \leq F_2$ $F_{RR} \leq F_2$	M10

Remarks:

The base brake and ABS tests are executed directly after the WSS tests. As first step during the base brake test (inclusive the optional leakage test) the freewheeling and base brake forces are measured. In addition the optional leakage of the entire brake system inclusive the tightness of the AV valves of the ABS / ESP® system are tested.
After the test rig rolls are switched on the freewheeling forces are measured, afterwards the operator is asked to apply the brake pedal immediately. As soon as the expected brake forces are reached, the operator has to keep the brake pedal constant and shouldn't release it till all brake forces of the base brake and the optional leakage test are measured. At the optional leakage test all EV and USV valves are switched on for a short time, to reduce the influence of the operator on this test.
The ABS test is executed immediately after the base brake and optional leakage test. The operator has to keep the brake pedal still constant. The tester activates the corresponding valves and the pump motor step by step for the ABS test. In addition the tester measures the corresponding brake forces at each step and wheel.

All mentioned force values and tolerances can be read in table in the chapter 4.2.3.4.

The operator has to apply, keep constant and release the brake pedal after he was requested by the tester!

The respective brake forces differences are calculated by the tester and compared with the fixed limits.

To reduce the entire cycle time of the ABS test, the test cycle respectively corresponding vale activation for each wheel can overlap.

The base brake test has to be executed at all vehicles independent of the installed control system. The test is required to ensure the function and tightness of base brake.
The ABS test can be used as well as for vehicles with an ABS or ESP® system. In case of a 3 channel ABS system is used, the described test procedure for the wheel RL is valid for the total rear axle. The described test step for the wheel RR can be removed, but only in case of a 3 channel system.

The base brake test, the optional leakage and ABS test have to ensure the following functions and assembly processes:

1) To ensure the function respectively tightness of the base brake system inclusive ABS / ESP® tightness (to exclude AV leakage)
2) To ensure the correlation of the brake pipes between the ABS / ESP® hydraulic ports and each wheel

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3) To ensure the function of the EV and AV valves after brake filling process (e.g. to exclude an impairment of function caused by contamination, etc.)
4) To ensure delivery rate of both pumps

ABS test ok ⇒ **Continue**

ABS Test nok ⇒ **Failure message, abort, print out fault message and send vehicle to repair area**

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4.2.2.1 Test sequence of the ABS test

Start mes-sage:	Operator prompt:	Activation before the time:	Time :	Activation after the time:	Measure sample:
---	Don't apply Brake pedal!	---	---	---	M1
---	Apply the brake pedal till the ex-pected brake forces are reached!	---	---	---	M2
1	Keep the brake pedal constant!	Pump motor on	---	---	---
2		Inlet valves of the front axle on	---	Inlet valves of the rear axle on	M3.1
3		Both USV (only ESP® systems) on	t9	---	M3.2
4		Both USV (only ESP® systems) off	---	---	---
5		AVFL on	t5	AVFL off	M4
6		EVFL off	t1.3	EVFL on	---
7		AVFR on	t5	AVFR off	M5
8		EVFR off	t1.3	EVFR on	M6
9		AVRL on	t5	AVRL off	M7
10		EVRL off	t1.4	EVRL on	---
11		AVRR on	t5	AVRR off	M8
12		EVRR off	t1.4	EVRR on	M9
13	Release the brake pedal!	Inlet valves of the front axle off	---	Inlet valves of the rear axle off	---
14		---	t8	Pump motor off	M10

Remarks:

As first step all test rig rolls are switched on and then the freewheeling forces are measured (measure sample M0) at all wheels. These values are subtracted from the afterwards measured brake forces. As next step the operator is prompt to apply the brake pedal till the expected brake force range is reached (M2). Afterwards the operator has to keep the brake pedal constant.
Then the tester switches on the pump motor (start message 1) and all inlet valves (start message 2). At ESP® systems both USV valves are additionally switched on (start message 3), to reduce the influence of the operator on this test. After this step the brake forces are measured before and after the time t9 at all wheels. With this step the tightness of the base brake system is tested respectively ensured (M3.1 & M3.2). At the end of this step both USV valves are switched off (start message 4).

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Afterwards the pressure in the wheel FL is released by switching on of the outlet valve for the time t5 (start message 5). After this test step the brake forces are measured at all wheels to ensure the pressure decrease at the wheel FL (M4).
As next step the inlet valve is immediately opened for the time t1.3 to increase the pressure in the wheel FL. In parallel the pressure in the wheel FR is released by switching on of the outlet valve for the time t5 (start messages 6 & 7). After this test step the brake forces are measured at all wheels again to ensure the pressure increase at the wheel FL and the pressure decrease at the wheel FR (M5).
Thereupon the pressure of the wheel FR is increased again by opening of the inlet valve for the time t1.3 (start message 8). In addition the brake forces are measured (M6).
The test at the rear axle is done in the same way (Start messages 9 till 12; measure samples M7 till M9).
At the end of the ABS test the operator is prompt to release the brake pedal. Afterwards all inlet valves are immediately switched off (start message 13). The pump motor is switched off after the time t8 (start message 14). This is required to discharge the accumulator chambers. In parallel the brake forces of all wheels are measured again and then they are compared with the freewheeling forces which were measured at the beginning of the base brake test (M10).

Details about the used diagnostic function can be read in the chapter 5.5.

Parameter:

See chapter 4.2.3.4.

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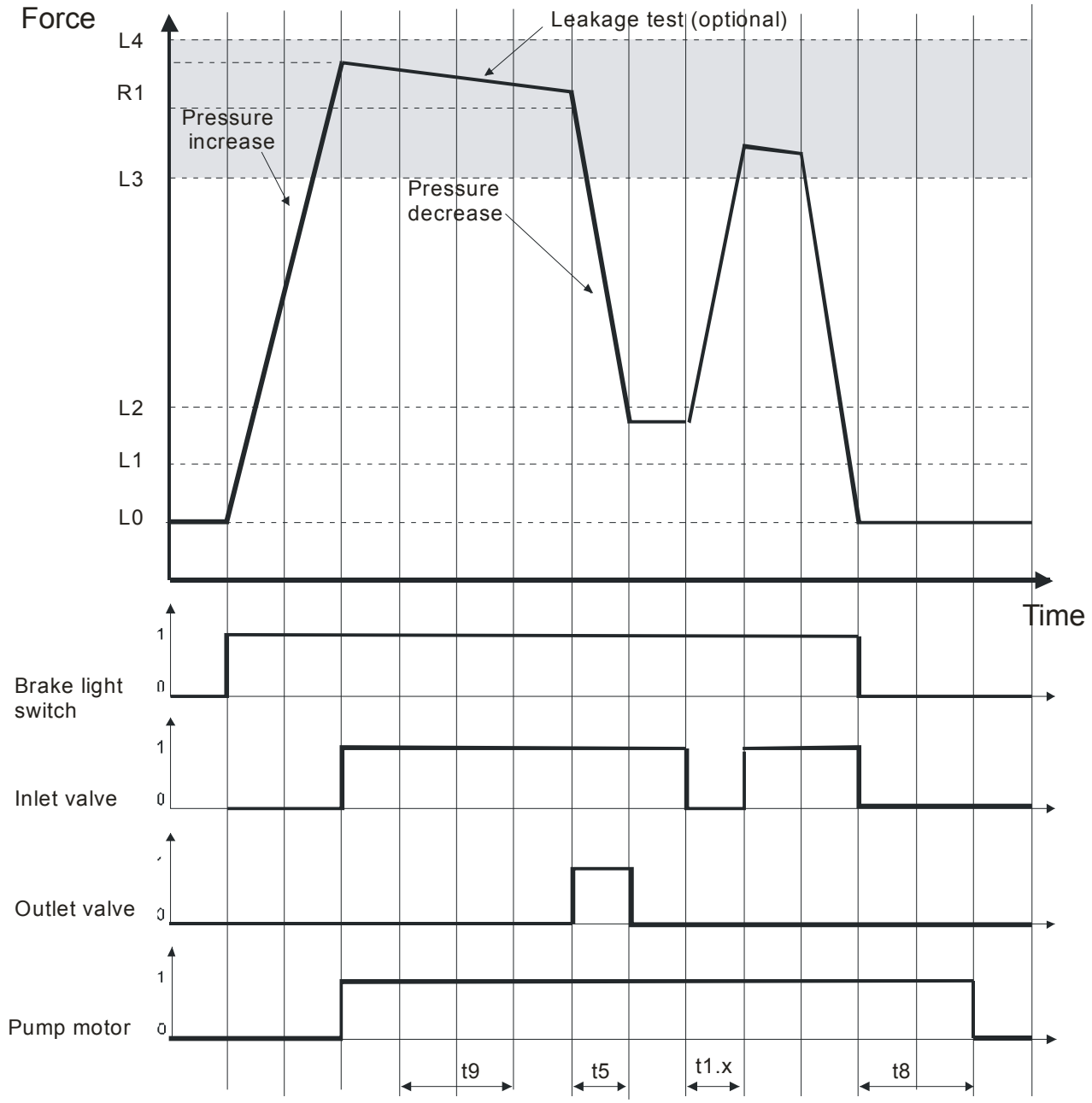
4.2.2.2 Diagram of the ABS test (example at one wheel)

ABS test

Test diagram (example at one wheel):



Remarks:

- At first step the test rig (tester) measures the freewheeling forces. These values are compared with the maximal allowed limit (L1).
- Afterwards the operator is requested to apply the brake pedal and keep constant the entire test cycle.

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- With applying of the brake pedal the brake forces are increased by the operator. The brake forces which have to be reached have to be within the defined limits (L3/L4). Afterwards the pump motor and all inlet valves are switched on.
- Now the optional leakage test is executed for the time t9. The measured brake forces before and after this time might not exceed the maximal allowed brake force difference (R1).
- As next step the tester switches on the corresponding outlet valve for the time (t2) to create a pressure decrease at the corresponding wheel. In addition the brake forces are measured and compared with the defined maximal allowed limit (L2).
- Immediately after the previous test step is finished, the tester switches off the corresponding inlet valve for the time (t1.x) to create a pressure increase at the corresponding wheel. In addition the brake forces are measured and compared with the defined maximal allowed limit (L3).
- As soon as the complete test is finished the operator is requested to release the brake pedal. Afterwards all inlet valves are opened respectively switched off. To discharge the accumulator chamber, the pump motor is switched off first after the time t8.

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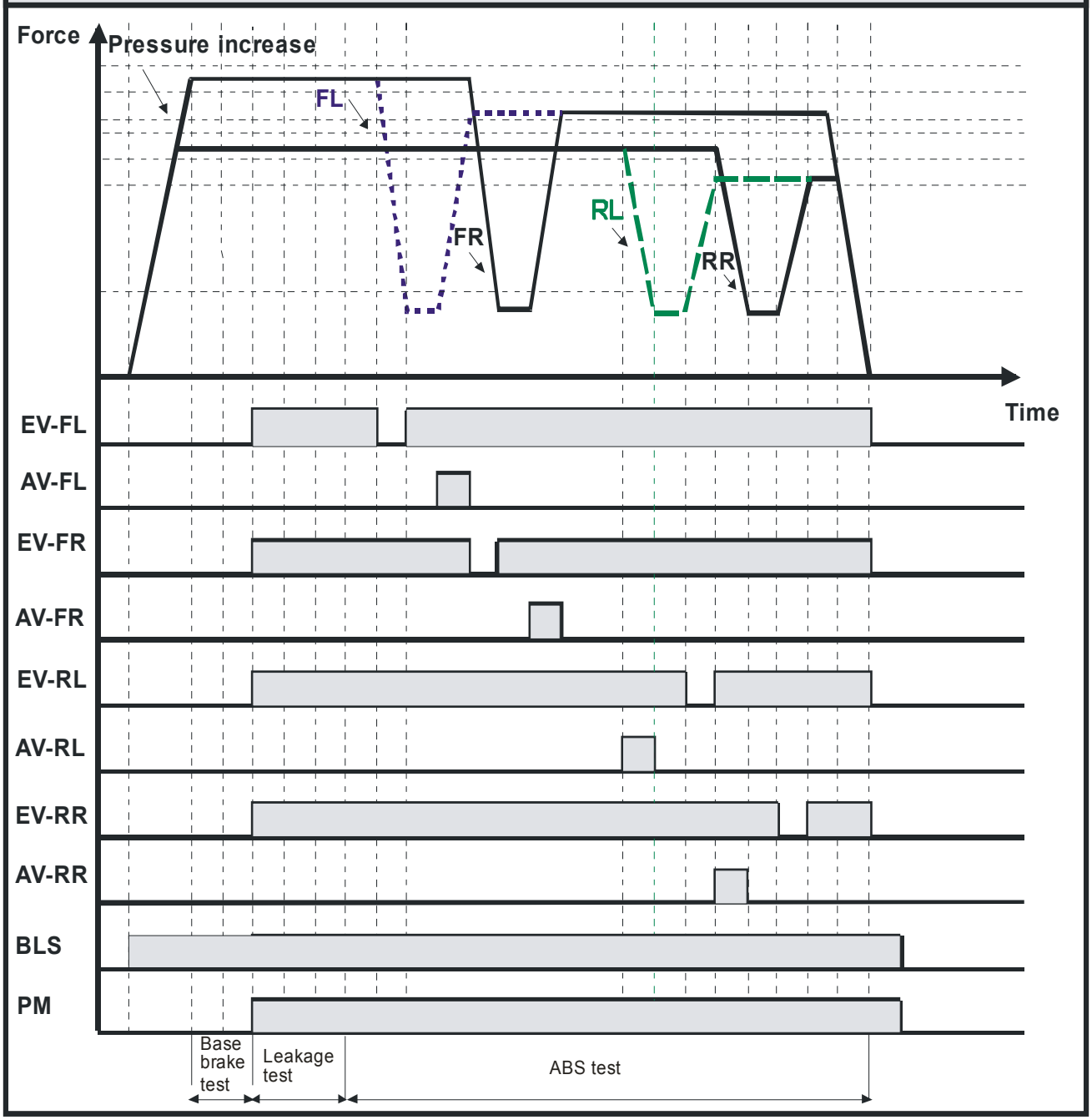
4.2.2.3 Diagram of the ABS test at all wheels

ABS test

Test diagram:



Remarks:

Please note the maximal allowed valve activation times. More information about these times can be read in the chapter 6.

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4.2.2.4 Proposal of a evaluation logic for the optional leakage and ABS test

a) Evaluation of the freewheeling force before the wheels are braked (exemplary for the wheel FL):

If
F_FL ≤ F1

then F_FL = F_FL_freewheel and execute test step b)

else Stop process and output “Freewheel force at the wheel FL nok”

b) Evaluation base brake test (exemplary for the front axle):

F_FL_corr = F_FL – F_FL_freewheel
F_FR_corr = F_FR – F_FR_freewheel

If
F3 ≤ F_FL_corr ≤ F4
and
F_FL_corr – F_brake_tolerance ≤ F_FR_corr ≤ F_FL_corr + F_brake_tolerance

then execute test step c)

else Stop process and output “Base brake test at the front axle nok”

c) Evaluation of the optional leakage test (exemplary for the wheel FL):

If
F_FL_1 (before the measure time) – F_FL_2 (after the measure time) ≤ F9

then execute test step d)

else Stop process and output “Leakage test at the wheel FL nok”

d) Evaluation ABS test „pressure decrease“ (exemplary for the wheel FL):

If
F_FL_corr ≤ F7

then execute test step e)

else Stop process and output “ABS pressure decrease at the wheel FL nok”

e) Evaluation ABS test „pressure increase“ (exemplary for the wheel FL):

If
F_FL_corr ≥ F3

then execute test step f)

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else Stop process and output “ABS pressure increase at the wheel FL nok”

f)) Evaluation of the freewheeling force after the wheels are released (exemplary for the wheel FL):

If

F_FL ≤ F1

else output “Base brake test FA and ABS test at the wheel FL ok”

else output “Freewheel force at the wheel FL after the ABS test nok”

Example:

Variables for evaluation*:
F1 = 250N
F3 = 800N
F4 = 2000N
F7 = 350N
F9 = 200N
F_brake_tolerance = 25%
Measure time t9: 1000ms

*All described variables for evaluation are examples and must be finally defined by the customer!

a) Measured brake forces (N) of each wheel at the test step „freewheeling force before the wheels are braked“:

F_FL	F_FR	F_RL	F_RR
150	155	105	100

b) Measured brake forces (N) of each wheel at the test step „base brake test“:

F_FL	F_FR	F_RL	F_RR
1550	1575	650	655

c) Measured brake forces (N) of each wheel at the test step „optional leakage test“:

F1_FL	F1_FR	F1_RL	F1_RR
1500	1570	645	645
	Measure time t9		
F2_FL	F2_FR	F2_RL	F2_RR
1450	1560	640	645

d) Measured brake forces (N) of each wheel at the test step „pressure decrease“:

F_FL	F_FR	F_RL	F_RR
350	1560	640	645

e) Measured brake forces (N) of each wheel at the test step „pressure increase“:

F_FL	F_FR	F_RL	F_RR
1350	1560	640	645

f) Measured brake forces (N) of each wheel at the test step „freewheeling force after the wheels are released“:

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F_FL	F_FR	F_RL	F_RR
150	155	105	100

Judgement of each test step for the wheel FL respectively FA:

a) Freewheeling at the wheel FL before the base brake test: 150 ≤ 250 **= ok**

b) Base brake test front axle: 800 ≤ 1400 ≤ 2000 && 1050 ≤ 1425 ≤ 1750 **= ok**

c) Optional leakage test at the wheel FL: 50 ≤ 200 **= ok**

d) ABS test „pressure decrease FL“: 200 ≤ 350 **= ok**

e) ABS test „pressure increase FL“: 1200 ≥ 800 **= ok**

f) Freewheeling at the wheel FL after the ABS test: 150 ≤ 250 **= ok**

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4.2.3 ESP® test

ESP® test (function test „active pressure increase“)				
Operator:	Test rig:	To be evaluated brake forces:	Demanded brake forces:	Measure sample:
→ Don’t apply Brake pedal!	Measurement of the free-wheeling forces	Wheel FL: xx N Wheel FR: xx N Wheel RL: xx N Wheel RR: xx N	$F_{FL} \leq F_1$ $F_{FR} \leq F_1$ $F_{RL} \leq F_2$ $F_{RR} \leq F_2$	M0
→ Don’t apply Brake pedal!	Pressure increase FL Measure brake forces	Wheel FL: xx N Wheel FR: xx N Wheel RL: xx N Wheel RR: xx N	$F_{FL} \geq F_3$ $F_{FR} \leq F_7$ $F_{RL} \leq F_8$ $F_{RR} \leq F_8$	M1+n
→ Don’t apply Brake pedal!	Pressure decrease FL Measure brake forces	Wheel FL: xx N Wheel FR: xx N Wheel RL: xx N Wheel RR: xx N	$F_{FL} \leq F_7$ $F_{FR} \leq F_7$ $F_{RL} \leq F_8$ $F_{RR} \leq F_8$	M2
→ Don’t apply Brake pedal!	Pressure increase FR Measure brake forces	Wheel FL: xx N Wheel FR: xx N Wheel RL: xx N Wheel RR: xx N	$F_{FL} \leq F_7$ $F_{FR} \geq F_3$ $F_{RL} \leq F_8$ $F_{RR} \leq F_8$	M3+n
→ Don’t apply Brake pedal!	Pressure decrease FR Measure brake forces	Wheel FL: xx N Wheel FR: xx N Wheel RL: xx N Wheel RR: xx N	$F_{FL} \leq F_7$ $F_{FL} \leq F_7$ $F_{RL} \leq F_8$ $F_{RR} \leq F_8$	M4
→ Don’t apply Brake pedal!	Pressure increase RL Measure brake forces	Wheel FL: xx N Wheel FR: xx N Wheel RL: xx N Wheel RR: xx N	$F_{FL} \leq F_7$ $F_{FL} \leq F_7$ $F_{RL} \geq F_5$ $F_{RR} \leq F_8$	M5+n
→ Don’t apply Brake pedal!	Pressure decrease RL Measure brake forces	Wheel FL: xx N Wheel FR: xx N Wheel RL: xx N Wheel RR: xx N	$F_{FL} \leq F_7$ $F_{FL} \leq F_7$ $F_{RL} \leq F_8$ $F_{RR} \leq F_8$	M6
→ Don’t apply Brake pedal!	Pressure increase RR Measure brake forces	Wheel FL: xx N Wheel FR: xx N Wheel RL: xx N Wheel RR: xx N	$F_{FL} \leq F_7$ $F_{FL} \leq F_7$ $F_{RL} \leq F_8$ $F_{RR} \geq F_5$	M7+n
→ Don’t apply Brake pedal!	Pressure decrease RR Measure brake forces	Wheel FL: xx N Wheel FR: xx N Wheel RL: xx N Wheel RR: xx N	$F_{FL} \leq F_7$ $F_{FL} \leq F_7$ $F_{RL} \leq F_8$ $F_{RR} \leq F_8$	M8

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Remarks:

The ESP® test is executed directly after the WSS tests. Due to the fact that all rolls are running, the freewheeling forces can be measured immediately (measure sample M0). Afterwards the tester executes step by step the required valve and pump motor activations and measures the brake forces at all wheels (M1 till M8).
During the entire ESP® test the brake pedal might not applied by the operator. To ensure that the tester should cyclic evaluate the brake light switch status between the individual pressure decrease cycles.

All mentioned force values and tolerances can be read in the table in the chapter 4.2.3.4.

Details to ensure the tightness and the function of base brake system can be read in the chapter 4.2.2.

The ESP® Test has to ensure the following functions and assembly processes:

1) To ensure the correlation of the brake pipes between the ESP® hydraulic ports and each wheel
2) To ensure the tightness respectively function of the EV and AV valves after brake filling process (e.g. to exclude an impairment of function caused by contamination, etc.)
3) To ensure delivery rate of both pumps respectively the active ESP® pressure increase function at all wheels

ESP® test ok ⇒ **Continue**

ESP® test nok ⇒ **Failure message, abort, print out fault message and send vehicle to repair area**

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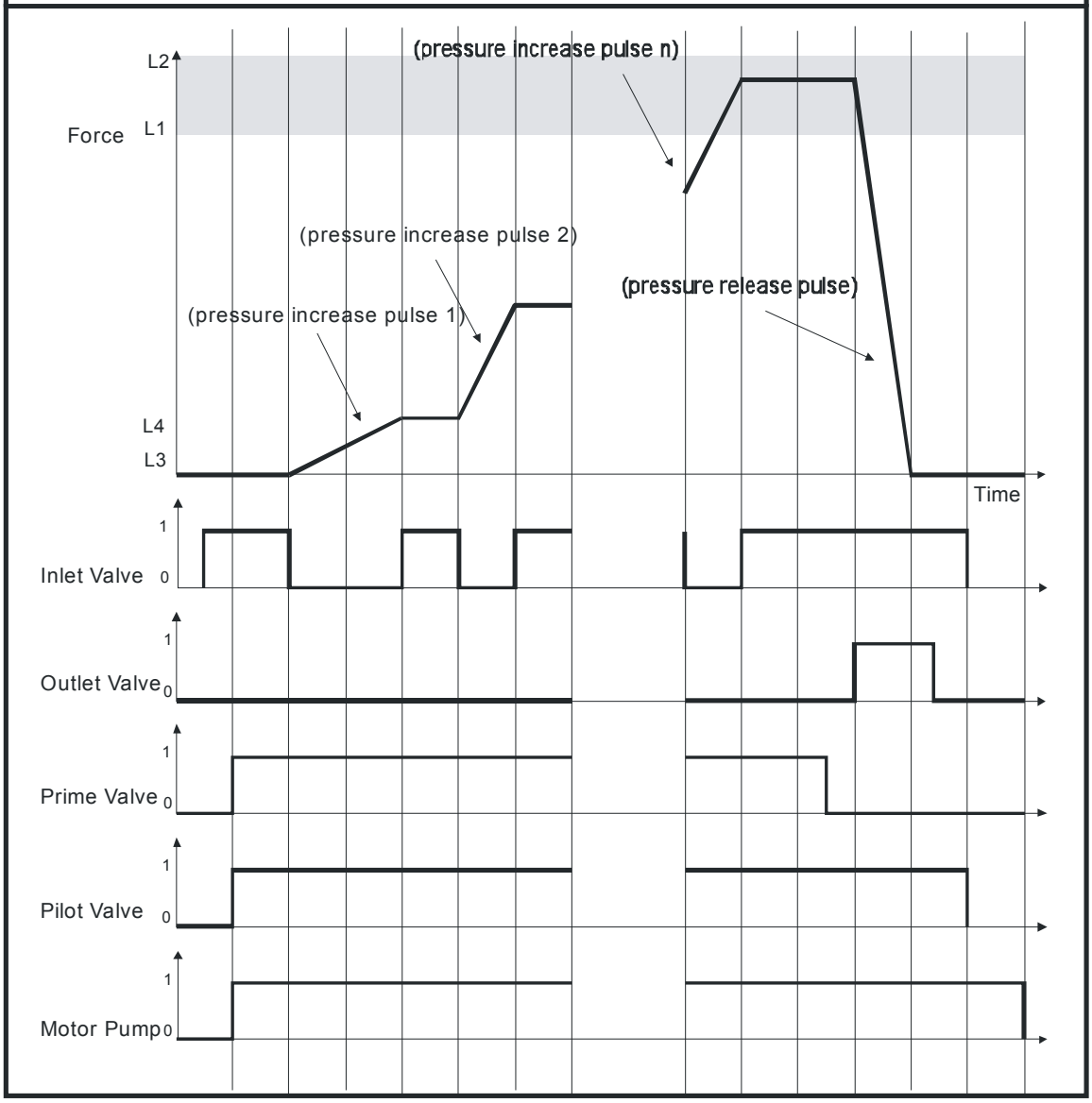
4.2.3.1 Schematically sequence exemplary at one wheel

Dynamic ESP Test

Test Diagram (Example of one wheel):



Remarks:

The hydraulic unit creates a pressure by running of a loop of pressure increase pulses. This loop is executed as long as the defined limit (L1) is reached. The limit (L2) should not be exceeded.The first pressure increase pulse is longer than the following. This is required to reduce the air gap between the brake disk and pads.

After the tester executed a pressure release function, the tester measures the brake forces and compares them with maximal allowed limits (L3/L4).

The diagram shows the test cycle at one wheel. The total test cycle includes this test at each wheel.

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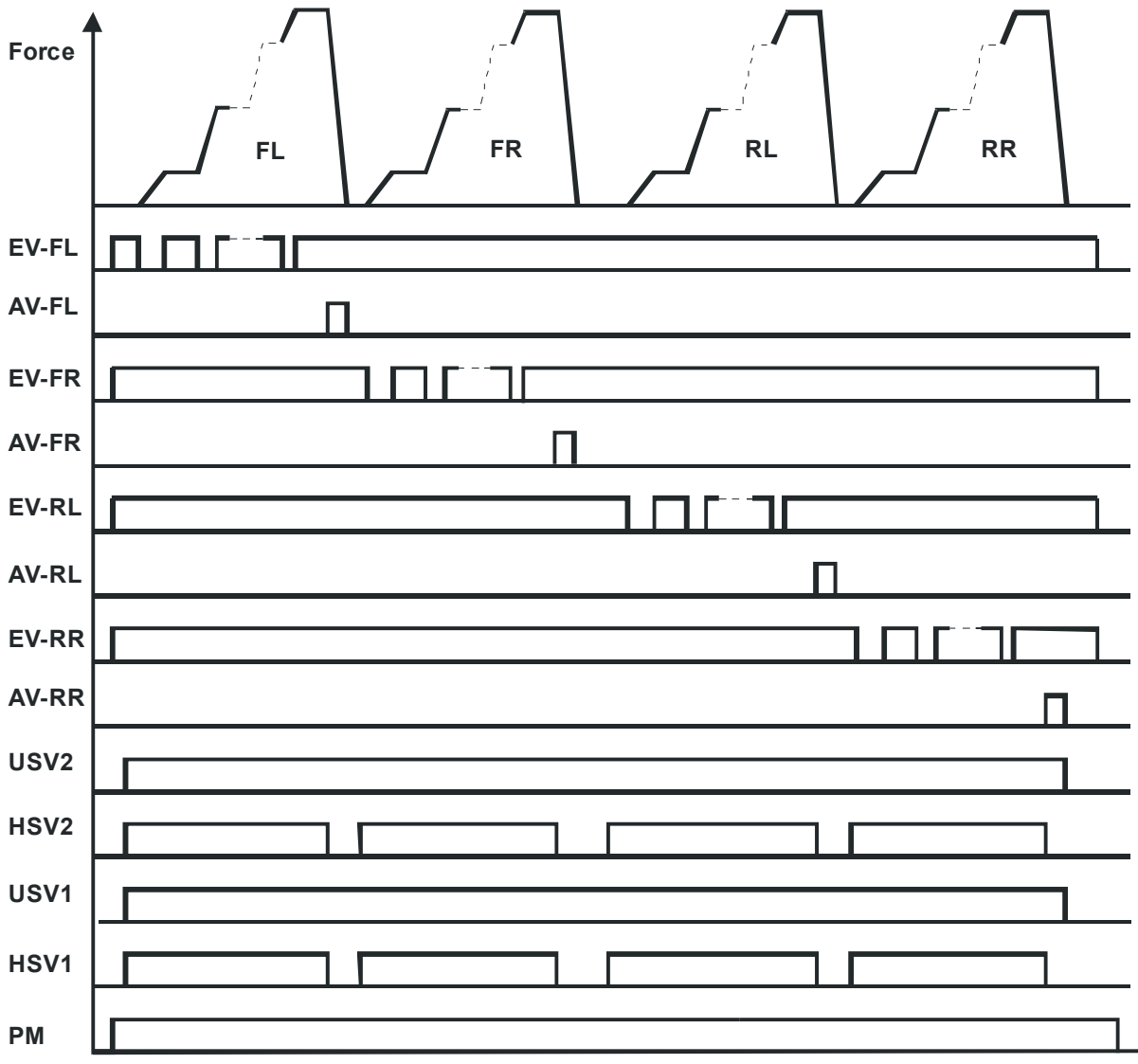
4.2.3.2 Graphical presentation of the entire ESP® tests on a 2 axle brake dynamometer

ESP test

Test diagram:



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4.2.3.3 Test sequence for the ESP® test

Start mes-sage:	Operator prompt:	Activation before the time:	Time:	Activation after the time:	Measure sample:
--	Don't apply Brake pedal!	---	---	---	M0
1		Pump motor on	---	---	---
2		Inlet valves front axle on	---	Inlet valves rear axle on	---
3		USV1 and USV2 on	---	HSV1 and HSV2 on	---
4		EVFL off	t1.1	EVFL off	M1
5.1 – 5.n		EVFL off	t2	EVFL off	M1+n
6		HSV1 & HSV2 off	---	---	---
7		AVFL on	t5	AVFL off	M2
8		HSV1 & HSV2 on	---	---	---
9		EVFR off	t1.1	EVFR on	M3
10.1 – 10.n		EVFR off	t2	EVFR on	M3+n
11		HSV1 & HSV2 off	---	---	---
12		AVFL on	t5	AVFL off	M4
13		HSV1 & HSV2 on	---	---	---
14		EVRL off	t1.2	EVRL on	M5
15.1 – 15.n		EVRL off	t2	EVRL on	M5+n
16		HSV1 & HSV2 off	---	---	---
17		AVRL on	t5	AVRL off	M6
18		HSV1 & HSV2 on	---	---	---
19		EVRR off	t1.2	EVRR on	M7
20.1 – 20.3		EVRR off	t2	EVRR on	M7+n
21		HSV1 & HSV2 off	---	---	---
22		AVRR on	t5	AVRR off	M8
23		USV1 & USV2 off	---	---	---
24		Inlet valves front axle off	---	Inlet valves rear axle off	---
25		---	t8	Pump motor off	---

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Remarks:

The operator might not apply the brake pedal during the entire ESP® Test, because the brake (system) pressure is active created by the ESP® hydraulic unit.
After the freewheeling forces are measured (measure sample M0) at all wheels, the tester switches on the pump motor (start message 1) at first, then all inlet valves (start message 2) and all USV respectively HSV valves (start message 3). To get a sufficient brake force increase at the wheel FL, the tester runs a number of pressure increase pulses. The first pulse should be longer than the following. This is required to reduce the air gap between the brake pads and disk (start message 4, time t1.1). Afterwards the tester runs a number of loops with a smaller pressure increase time (start message 5.1 till 5.n, time t2). This is required to control the brake force in very small steps. After the expected brake force is reached, the tester switches off both HSV valves (start message 6) at first, then the pressure of the wheel FL is released by switching on of the outlet valve (start message 7, time t5).
Between and after each individual test step all brake forces are measured to ensure the active pressure increase respectively pressure release function at the wheel FL (M1 - M1+n, M2).
To be able to execute the same procedure at the wheel FR both HSV valves have to be switched on again (start message 8). Afterwards the pressure is increased at first, then released at the wheel FR (start messages 9-12).
Between and after each individual test step all brake forces are measured to ensure the active pressure increase respectively pressure release function at the wheel FR (M3 – M3+n, M4).
The test at the rear axle is done in the same way (start messages 13 till 22). To ensure the active pressure increase respectively pressure release function at hte raer axle all brake forces are measured between and after each individual test step (M5 – M5+n, M6 respectively. M7 – M7+n, M8).

At the end of the ESP® tests all activated valves are switched off (start messages 23-24). Then the pump motor is switched off after the time t8 (start message 25).

Details about the used diagnostic function can be read in the chapter 5.5.

Parameter:

See chapter 4.2.3.4.

Important remark:

At ESP® systems pilot valves without pressure limitation are used. To protect these systems against over pressure, the pressure is limited by controlling of the pilot valve current. This current corresponds to the maximal system pressure (fluctuation by the allowed tolerances possible). More details can be read in the chapter 6.

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4.2.3.4 Parameter (Base brake, ABS and ESP® test)

Kind of parameter:	Parameter name:		Unit:	Physical value:
Time	1. Pressure increase pulse FA (ESP® test)	t1.1	Milliseconds (ms)	120
	1. Pressure increase pulse RA (ESP® Test)	t1.2		80
	General pressure increase pulse FA (ABS test)	t1.3		60
	General pressure increase pulse RA (ABS test)	t1.4		40
	Pressure increase pulse for the loop (general)	t2		20
	Pressure decrease pulse (general)	t5		300
	Post run time of the pump motor	t8		2000
	Measure time EV leakage test	t9		1000
Number	Number of loops for the pressure increase	x1	---	5
Evaluation parameter	Freewheeling force FA	F1	Newton (N)	250
	Freewheeling force RA	F2		150
	Force after pressure increase min. value FA	F3		800
	Force after pressure increase max. value FA	F4		2000
	Force after pressure increase min. value RA	F5		400
	Force after pressure increase max. value RA	F6		1200
	Force after pressure decrease value FA	F7		350
	Force after pressure decrease value RA	F8		250
	Force after EV leakage test max. value	F9 (R1)		200
	Tolerance $F_{_brake_tolerance}$	---	Percent (%)	25

Remarks:

The mentioned times are proposed times such as the activation times for the pressure increase and release cycles, the stabilisation and measure times. All these times depend on the vehicle (e.g. size of brake, orifices in the hydraulic unit, etc.) respectively the test rig, therefore these times must be confirmed respectively modified during trials at the customer plant.

Also the mentioned evaluation parameter have to be defined by a statistic evaluation with the first tested vehicles .The definition of all these data is up the customer responsibility.

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4.2.3.5 Proposal of a evaluation logic for the ESP® test

a) Evaluation of the freewheeling force before the wheels are braked (exemplary for the wheel FL):

If
F_FL ≤ F1

then F_FL = F_FL_freewheel and execute test step b)

else Stop process and output "Freewheel force at the wheel FL nok"

b) Evaluation ESP® test „pressure increase" (exemplary for the wheel FL):

F_FL_corr = F_FL – F_FL_freewheel

If
F_FL_corr ≥ F3

then execute test step c)

else Stop process and output "ESP® pressure increase at the wheel FL nok"

c) Evaluation ESP® test „pressure decrease" (exemplary for the wheel FL):

If
F_FL_corr ≤ F7

then output "ESP® test at the wheel FL ok"

else output "ESP® pressure decrease at the wheel FL nok"

Example:

Variables for evaluation*:
F1 = 250N
F3 = 800N
F7 = 350N

*All described variables for evaluation are examples and must be finally defined by the customer!

a) Measured brake forces (N) of each wheel at the test step „freewheeling force before the wheels are braked":

F_FL	F_FR	F_RL	F_RR
150	155	105	100

b) Measured brake forces (N) of each wheel at the n. test step „ESP® test – pressure increase FL":

F_FL	F_FR	F_RL	F_RR
1350	170	115	105

c) Measured brake forces (N) of each wheel at the test step „ESP® test – pressure decrease FL":

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F_FL	F_FR	F_RL	F_RR
350	175	120	115

Judgement of each test step for the wheel FL:

a) Freewheeling at the wheel FL before the ESP® test: 150 ≤ 250 **= ok**

b) ESP® test „pressure increase FL“: 1200 ≥ 800 **= ok**

c) ESP® test „pressure decrease FL“: 200 ≤ 350 **= ok**

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4.3 Alternative: Dynamic test on a 4 roller test bench with 4 independent rolls (high speed)

4.3.1 Wheel speed sensor quality tests (during acceleration of the vehicle – rotation direction: forward)

Wheel speed sensor test (signal quality test)			
Test speed $V_{_Test}$:	ECU data to be evaluated:	Proposal for evaluation:	Test time:
1. test step: Signal quality test			
$V_{_min} \leq V_{_Test} \leq V_{_max}$	Wheel FL ΔV_max xx km/h Wheel FR ΔV_max xx km/h Wheel RL ΔV_max xx km/h Wheel RR ΔV_max xx km/h	$\Delta V_{FL_max} \leq \Delta V_{WSS_test_accelerated_vehicle}$ $\Delta V_{FR_max} \leq \Delta V_{WSS_test_accelerated_vehicle}$ $\Delta V_{RL_max} \leq \Delta V_{WSS_test_accelerated_vehicle}$ $\Delta V_{RR_max} \leq \Delta V_{WSS_test_accelerated_vehicle}$	$t_{_WSS_signal_quality_accelerated_vehicle}$
2. test step (optional): WSSi additional information „rotation direction“			
$V_{_min} \leq V_{_Test} \leq V_{_max}$	Rotation direction FL Rotation direction FR Rotation direction RL Rotation direction RR	$RD_{FL} = RD_{_demand}$ $RD_{FR} = RD_{_demand}$ $RD_{RL} = RD_{_demand}$ $RD_{RR} = RD_{_demand}$	$t_{_WSS_rotation\ direction}$
3. test step (optional): WSSi additional information „mounting position”			
$V_{_min} \leq V_{_Test} \leq V_{_max}$	Mounting position FL Mounting position FR Mounting position RL Mounting position RR	$MP_{FL} = MP_{_demand}$ $MP_{FR} = MP_{_demand}$ $MP_{RL} = MP_{_demand}$ $MP_{RR} = MP_{_demand}$	$t_{_WSS_mounting_position_accelerated_vehicle}$

<u>Remarks:</u>

<u>1. test step: signal quality test:</u>

The signal quality test is executed during the acceleration phase before the ESP® or ABS test. As soon as the **lower speed limit $V_{_min}$** is reached, the tester starts a routine within the ECU software for the **test time$_{_WSS_signal_quality_accelerated_vehicle}$** (test time depends on the test speed and wheel circumference). The routine measures and compares with the **sampling time $t_{_sampling}$** the wheel speeds of all wheels within the selected test time or till the **stop speed V_{stop}** is reached. The routine stores the maximal calculated **difference speed per wheel (ΔV_{XX_max}/wheel$_{xx}$)** and outputs (have to be requested by the tester) these data after the selected test time or the stop speed is reached. The tester evaluates these data and compares them with the defined **maximal difference speed $\Delta V_{WSS_test_accelerated_vehicle}$**. The maximal allowed value “maximal difference speed” depends on the deceleration respectively acceleration, WSS, tooth wheel and ECU tolerances. This value must be defined by the customer. **For more details see chapter 4.3.1.3**.

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Details about the used diagnostic function can be read in the chapter 5.4.

2. & 3. test step (optional): WSSi rotation direction and mounting position test:

These additional and optional test steps are only required for vehicles with intelligent wheel speed sensors (e.g. Bosch WSS11i). These tests are directly executed after the signal quality tests. To ensure that the WSS weren't wrongly mounted the tester reads and checks the information "rotation direction" and "mounting position". These data have to be read with the corresponding diagnostic function during one revolution of the wheel. At first the information "rotation direction of all wheels" within the **test time $t_{_WSS_rotation_direction}$** is read and compared with the expected value **rotation direction demand RD_{demand}**. During the following **test time $t_{_WSS_mounintg_position}$** (time depends on the test speed and wheel circumference) the information "mounting position of all wheels" are read 3 times, at the beginning, in the middle and at the end of the test time. The read ECU data are compared with the expected **mounting position demand MP_{demand}**. The requirement to read this data only 3 times is sufficient, because sporadic occurred fault information is stored in the ECU for the **storing time t_{St}**. During this time the stored data can be read via the corresponding diagnostic service. **For more details about the evolution please refer the chapter 4.2.1.5 and 4.2.1.6.**

Parameter:

Kind of parameter:	Parameter name:	Unit:	Physical value:
Time constant in the Software	Sampling time $t_{_sampling}$	Milliseconds (ms)	5
	Storing time t_{St}		3000
Time	Test time $t_{_WSS_signal_quality_accelerated_vehicle}$		tbd
	Test time $t_{_WSS_rotation\ direction}$		≤ 100
	Test time $t_{_WSS_mounting_position_accelerated_vehicle}$		tbd
Condition	Lower test speed $V_{_min}$	Kilometre per hour (Km/h)	≥ 3
	Upper test speed $V_{_max}$		≤ 15
Evaluation parameter	Maximal difference speed $\Delta V_{WSS_test_accelerated_vehicle}$		tbd
	Rotation direction RD_{demand}	---	forward
	Mounting position MP_{demand}		tbd

Notes to the given parameters:

The test times depends on the start speed and the used tire sizes. These times are to be defined and specified by the customer.
For the definition of the test times $t_{_WSS_signal_quality_accelerated_vehicle}$ and $t_{_WSS_mounitng_position_accelerated_vehicle}$ following formula is valid:

test time = wheel circumference / start speed

Example:

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Wheel circumference: 2 Meter

Start speed of the WSS-Signal quality test: 5 km/h (1,39 m/s)

Test time= 1,44 seconds (2 m / 1,39 m/s)

→ recommended test time with some safety: 1,8 seconds

Start speed mounting position: 12 km/h (3,33 m/s)

Test time = 0,60 seconds (2 m / 3,33 m/s)

→ recommended test time with some safety: 0,75 seconds

To ensure that all important wheel speed information (wheel speed signals respectively mounting position (air gap) information – only vehicles with WSS11i) are available, the **test speed $V_{_Test}$** might not below the **lower test speed $V_{_min}$** respectively not exceed the **upper test speed $V_{_max}$**.

Details to all 3 test steps and evaluations can be seen in the graphics on the following pages.

<u>The execution of the wheel speed sensor tests have to ensure the following functions and assembly processes:</u>

1) To ensure the signal quality regarding missing tooth respectively single division error
2) Correct installation (evaluation of the additional WSSi information “rotation direction”)
3) To ensure the correct air gap respectively magnetic field intensity (evaluation of the additional WSSi information “mounting position”)

No WSS failure detected ⇒ **Continue**

WSS failure detected ⇒ **Failure message, abort, print out fault message and send vehicle to repair area**

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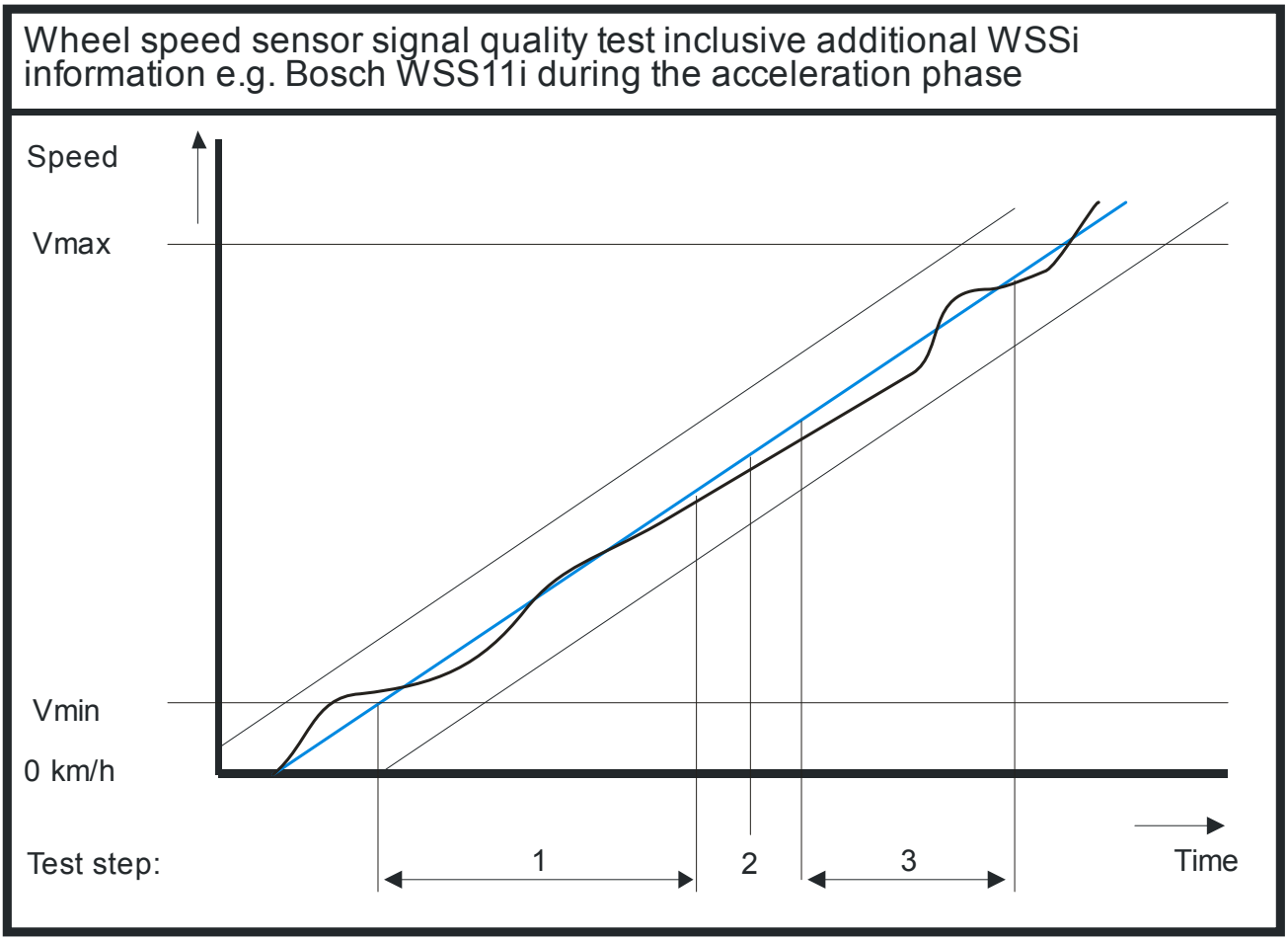
4.3.1.1 Graphical presentation of the entire wheel speed senor tests inclusive the additional WSSi information e.g. Bosch WSS11i during the acceleration phase



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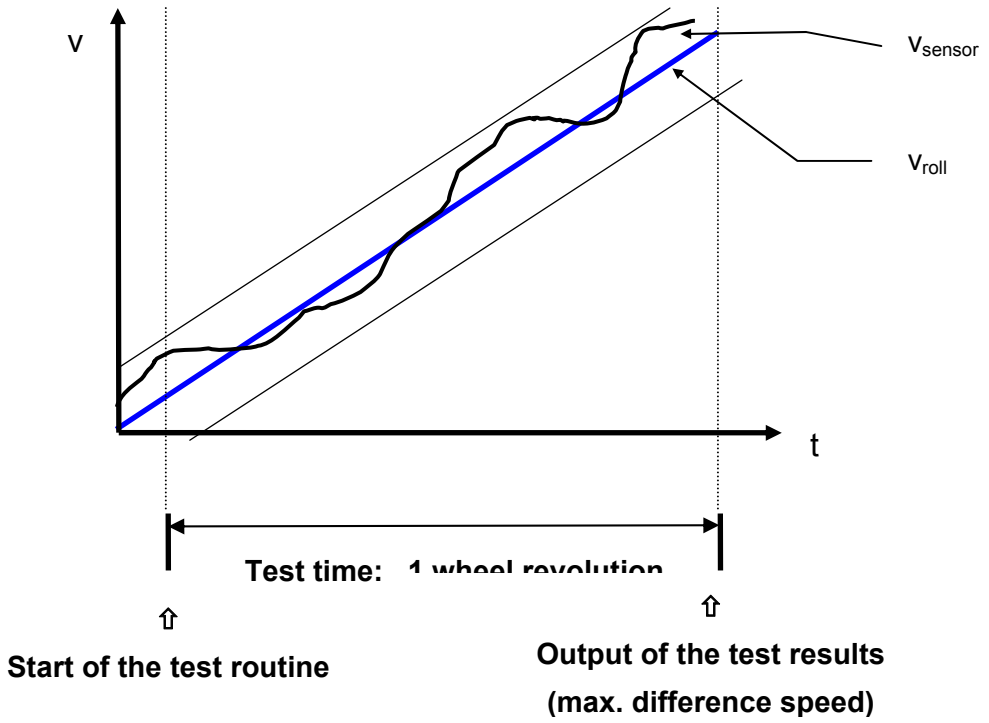
4.3.1.2 Evaluation diagrams to the WSS signal quality test (test step 1) during the acceleration phase

Wheel speed sensor test (signal quality test) during acceleration

Test diagram (exemplary at one wheel):



Remarks:

After the routine is started by tester, the ECU starts to sample cyclic all wheel speeds and calculates the difference speed values as long as one of the stop criteria is reached.
As long as the calculated difference speed is negative or zero the routine will output “zero”. This indicates a good WSS signal (normal or good case).
In case of the calculated difference speed is positive, the routine starts to summarize these positive difference speed values as long as there is again a negative or zero value. The result of the summarised value will be compared with the initial (zero) or stored maximal difference speed value. If the actual calculated difference speed value is larger than the stored or initial value, this value will be stored as new maximal calculated difference speed. At the end of the routine the stored values of all wheels are outputted as routine results. A stored value > zero + ECU and WSS tolerances indicates a abnormally deceleration (e.g. caused by a bad WSS signal --> bad case).
The tester has to compare these data (maximal calculated difference speed per wheel) with the defined limits. These limits have to include all ECU, WSS and tooth wheel tolerances. In addition all fluctuations during a normal acceleration of the vehicle have to be considered.

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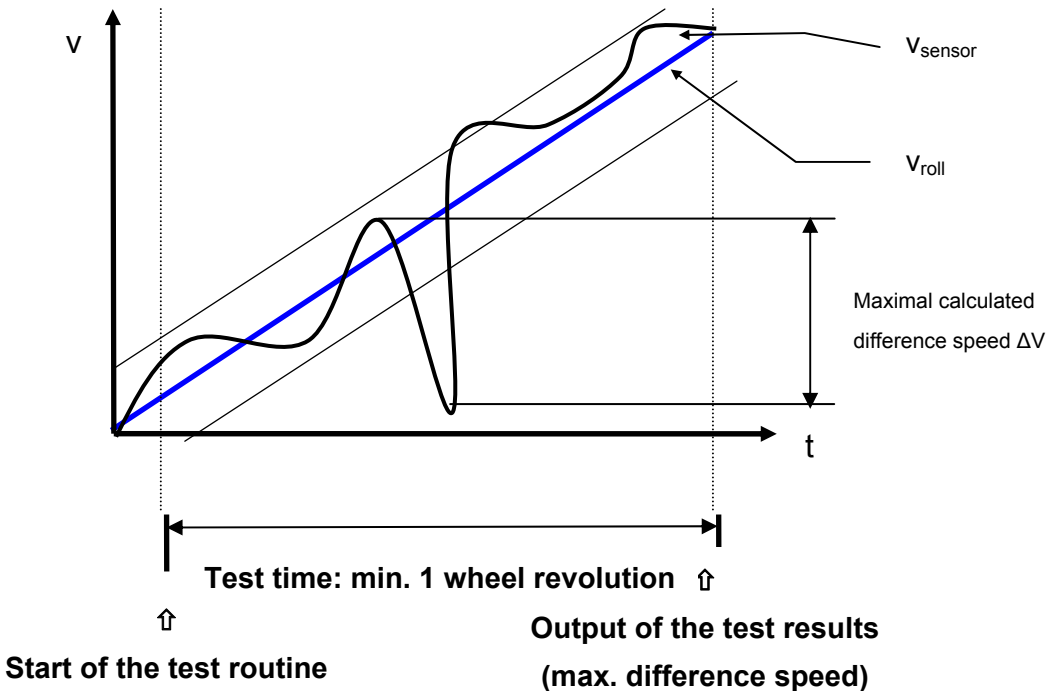
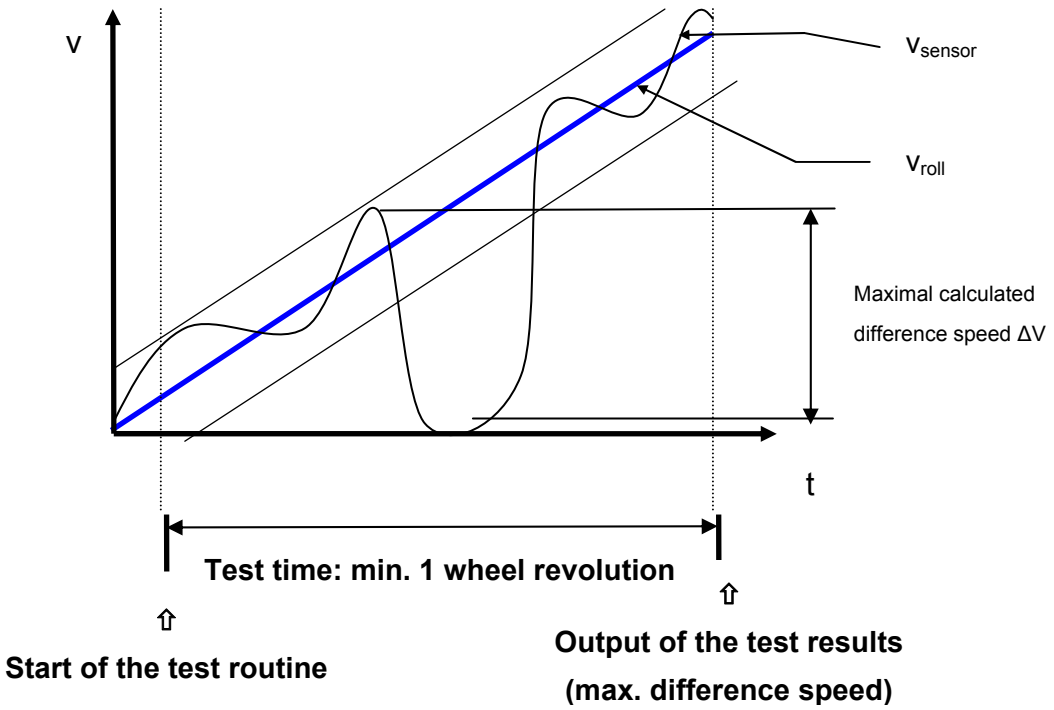
Failure simulation:

Failure simulation: faulty tooth wheel e.g. missing tooth



Failure simulation: Air gap to big or eccentricity



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4.3.1.3 Test step 1: Exemplary definition of the maximal allowed difference speed

Measurement results of vehicle:	ΔV_{FL_max}	ΔV_{FR_max}	ΔV_{RL_max}	ΔV_{RR_max}
1	0,05625	0	0,05625	0,05625
2	0,11250	0,05625	0,11250	0
3	0,05625	0,05625	0,05625	0,05625
4	0	0	0,11250	0,11250
5	0	0,11250	0	0,05625
Average (AV)	**0,05344**			
Standard deviation (S)	**0,04270**			
Maximal difference speed $\Delta V_{WSS_test_accelerated_vehicle}$ (AV + 5*S)	**0,26695**			

Note:

At a „good“ vehicles (no tooth wheel faults, etc.) and a equable acceleration (best case) the ECU will output 0 km/h (initial value) as measured maximal difference speed for each wheel. Caused by the fact that in reality the speed fluctuates on the roller test bench and different normal tolerances (e.g. ECU ± 2 bit, etc.) influences the outputted maximal difference speed per wheel, the maximal allowed tolerance “maximal speed difference $\Delta V_{WSS_test_accelerated_vehicle}$” is > 0 km/h. The right value (tolerance) has to be defined by a statistic evaluation. The definition of this value is up the customer responsibility.

4.3.1.4 Evaluation proposal for the WSS test (signal quality regarding single pitch deviation, eccentricity, etc.):

Comparison of the maximal measured difference speed (ΔVxx_{max}, etc.) with the maximal allowed difference speed $\Delta V_{WSS_test_accelerated_vehicle}$ (calculated from the measured average value (AV) plus 5* standard deviation (S), values are defined by a statistic evaluation):

$\Delta V_{FLmax} \leq \Delta V_{WSS_test_accelerated_vehicle}$
$\Delta V_{FRmax} \leq \Delta V_{WSS_test_accelerated_vehicle}$
$\Delta V_{RLmax} \leq \Delta V_{WSS_test_accelerated_vehicle}$
$\Delta V_{RRmax} \leq \Delta V_{WSS_test_accelerated_vehicle}$

Example:

Measured maximal difference speeds (km/h) of each wheel:

ΔVFLmax	ΔVFR_{max}	ΔVRL_{max}	ΔVRR_{max}
0,05625	0	0,05625	3,60

Judgement:

ΔV_{FLmax} (0,05625 ≤ 0,26695) = ok
ΔV_{FRmax} (0 ≤ 0,26695) = ok
ΔV_{RLmax} (0,05625 ≤ 0,26695) = ok
ΔV_{RRmax} (3,60 ≤ 0,26695) = nok

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4.3.2 Base brake test and dynamic ABS test (function test „brake pipe correlation, pressure increase and release“)

Base brake and ABS test (function test „brake pipe correlation, pressure increase and release“)					
Operator:	Test rig:	To be evaluated brake forces:		Demanded brake forces:	Measure sample:
→ Don’t apply Brake pedal!	Measuring of the free-wheeling decelerations	Wheel FL: Wheel FR: Wheel RL: Wheel RR:	Δ xx km/h Δ xx km/h Δ xx km/h Δ xx km/h	$\Delta V_{FL} \leq \Delta V_{freewheeling_max}$ $\Delta V_{FR} \leq \Delta V_{freewheeling_max}$ $\Delta V_{RR} \leq \Delta V_{freewheeling_max}$ $\Delta V_{RL} \leq \Delta V_{freewheeling_max}$	ΔV1 (V1.1-V1.2)
→ Apply the brake pedal till the ex-pected decelera-tion is reached!	Pressure increase at all wheels Measuring of all decel-erations	Wheel FL: Wheel FR: Wheel RL: Wheel RR:	Δ xx km/h Δ xx km/h Δ xx km/h Δ xx km/h	$\Delta V_{FL} \geq \Delta V_{FA_brakes}$ $\Delta V_{FL} - V_{brakes_tolerance} \leq \Delta V_{FR} \leq \Delta_{FL} + V_{brakes_tolerance}$ $\Delta V_{RL} \geq \Delta V_{RA_brakes}$ $\Delta_{RL} - V_{brakes_tolerance} \leq \Delta V_{RR} \leq \Delta_{RL} + V_{brakes_tolerance}$	ΔV2 (V2.1-V2.2)
→ Keep the brake pedal constant!	Pressure increase FL Measuring of all decel-erations	Wheel FL: Wheel FR: Wheel RL: Wheel RR:	Δ xx km/h Δ xx km/h Δ xx km/h Δ xx km/h	$\Delta V_{FL} \leq (\Delta V_{_unpressurised} + \Delta V_{_freewheeling_FL})$ $\Delta V_{FR} \geq \Delta V_{FA_brakes}$ $\Delta V_{RL} \geq \Delta V_{RA_brakes}$ $\Delta V_{RR} \geq \Delta V_{RA_brakes}$	ΔV3 (V3.1-V3.2)
→ Keep the brake pedal constant!	Pressure decrease FL Pressure increase FR Measuring of all decel-erations	Wheel FL: Wheel FR: Wheel RL: Wheel RR:	Δ xx km/h Δ xx km/h Δ xx km/h Δ xx km/h	$\Delta V_{FL} \geq \Delta V_{FA_pressure_increase_brakes}$ $\Delta V_{FR} \leq (\Delta V_{_unpressurised} + \Delta V_{_freewheeling_FR})$ $\Delta V_{RL} \geq \Delta V_{RA_brakes}$ $\Delta V_{RR} \geq \Delta V_{RA_brakes}$	ΔV4 (V4.1-V4.2)
→ Keep the brake pedal constant!	Pressure decrease FR Pressure increase RL Measuring of all decel-erations	Wheel FL: Wheel FR: Wheel RL: Wheel RR:	Δ xx km/h Δ xx km/h Δ xx km/h Δ xx km/h	$\Delta V_{FL} \geq \Delta V_{FA_pressure_increase_brakes}$ $\Delta V_{FR} \geq \Delta V_{FA_pressure_increase_brakes}$ $\Delta V_{RL} \leq (\Delta V_{_unpressurised} + \Delta V_{_freewheeling_RL})$ $\Delta V_{RR} \geq \Delta V_{RA_brakes}$	ΔV5 (V5.1-V5.2)
→ Keep the brake pedal constant!	Pressure decrease RL Pressure increase RR Measuring of all decel-erations	Wheel FL: Wheel FR: Wheel RL:	Δ xx km/h Δ xx km/h Δ xx km/h	$\Delta V_{FL} \geq \Delta V_{FA_pressure_increase_brakes}$ $\Delta V_{FR} \geq \Delta V_{FA_pressure_increase_brakes}$ $\Delta V_{RL} \geq \Delta V_{RA_pressure_increase_brakes}$	ΔV6 (V6.1-V6.2)

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		Wheel RR:	Δ xx km/h	$\Delta V_{RR} \leq (\Delta V_{_unpressurised} + \Delta V_{freewheeling_RR})$	
→ Keep the brake pedal constant!	Pressure increase RR Measuring of all decel-erations	Wheel FL: Wheel FR: Wheel RL: Wheel RR:	Δ xx km/h Δ xx km/h Δ xx km/h Δ xx km/h	$\Delta V_{FL} \geq \Delta V_{FA_pressure_increase_brakes}$ $\Delta V_{FR} \geq \Delta V_{FA_pressure_increase_brakes}$ $\Delta V_{RL} \geq \Delta V_{RA_pressure_increase_brakes}$ $\Delta V_{RR} \geq \Delta V_{RA_pressure_increase_brakes}$	ΔV7 (V7.1-V7.2)
→ Re-lease the brake pedal!	Measuring of the free-wheeling decelerations	Wheel FL: Wheel FR: Wheel RL: Wheel RR:	Δ xx km/h Δ xx km/h Δ xx km/h Δ xx km/h	$\Delta V_{FL} \leq \Delta V_{freewheeling_max}$ $\Delta V_{FR} \leq \Delta V_{freewheeling_max}$ $\Delta V_{RR} \leq \Delta V_{freewheeling_max}$ $\Delta V_{RL} \leq \Delta V_{freewheeling_max}$	ΔV8 (V8.1-V8.2)

Remarks:

The base brake and ABS tests are executed directly after the WSS tests. As first step during the base brake test the freewheeling decelerations are measured. After the final test speed is reached the rolls are detached at first then the freewheeling decelerations are measured. Afterwards the operator is asked to apply the brake pedal immediately. As soon as the expected decelerations are reached, the operator has to keep the brake pedal constant and shouldn't release it till the deceleration values of the base brake test are measured.
The ABS test is executed immediately after the base brake test. The operator has to keep the brake pedal still constant. The tester activates the corresponding valves and the pump motor step by step for the ABS test. In addition the tester measures the corresponding decelerations at each step and wheel.

The operator has to apply, keep constant and release the brake pedal after he was requested by the tester!

The respective wheel decelerations are calculated by the tester and compared with the fixed limits.

To reduce the entire cycle time of the ABS test, the test cycle respectively corresponding vale activation for each wheel can overlap.

The base brake test has to be executed at all vehicles independent of the installed control system. The test is required to ensure the function and tightness of base brake.
The ABS test can be used as well as for vehicles with an ABS or ESP® system. In case of a 3 channel ABS system is used, the described test procedure for the wheel RL is valid for the total rear axle. The described test step for the wheel RR can be removed, but only in case of a 3 channel system.

The base brake and ABS test have to ensure the following functions and assembly processes:

1) To ensure the function of the base brake system
2) To ensure the correlation of the brake pipes between the ABS / ESP® hydraulic ports and each wheel
3) To ensure the function of the EV and AV valves after brake filling process (e.g. to exclude a impairment of function caused by contamination, etc.)
4) To ensure delivery rate of both pumps

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5) To ensure the correct correlation between the wheels speed sensors and the wheels

ABS test ok	⇒	**Continue**
ABS Test nok	⇒	**Failure message, abort, print out fault message and send vehicle to repair area**

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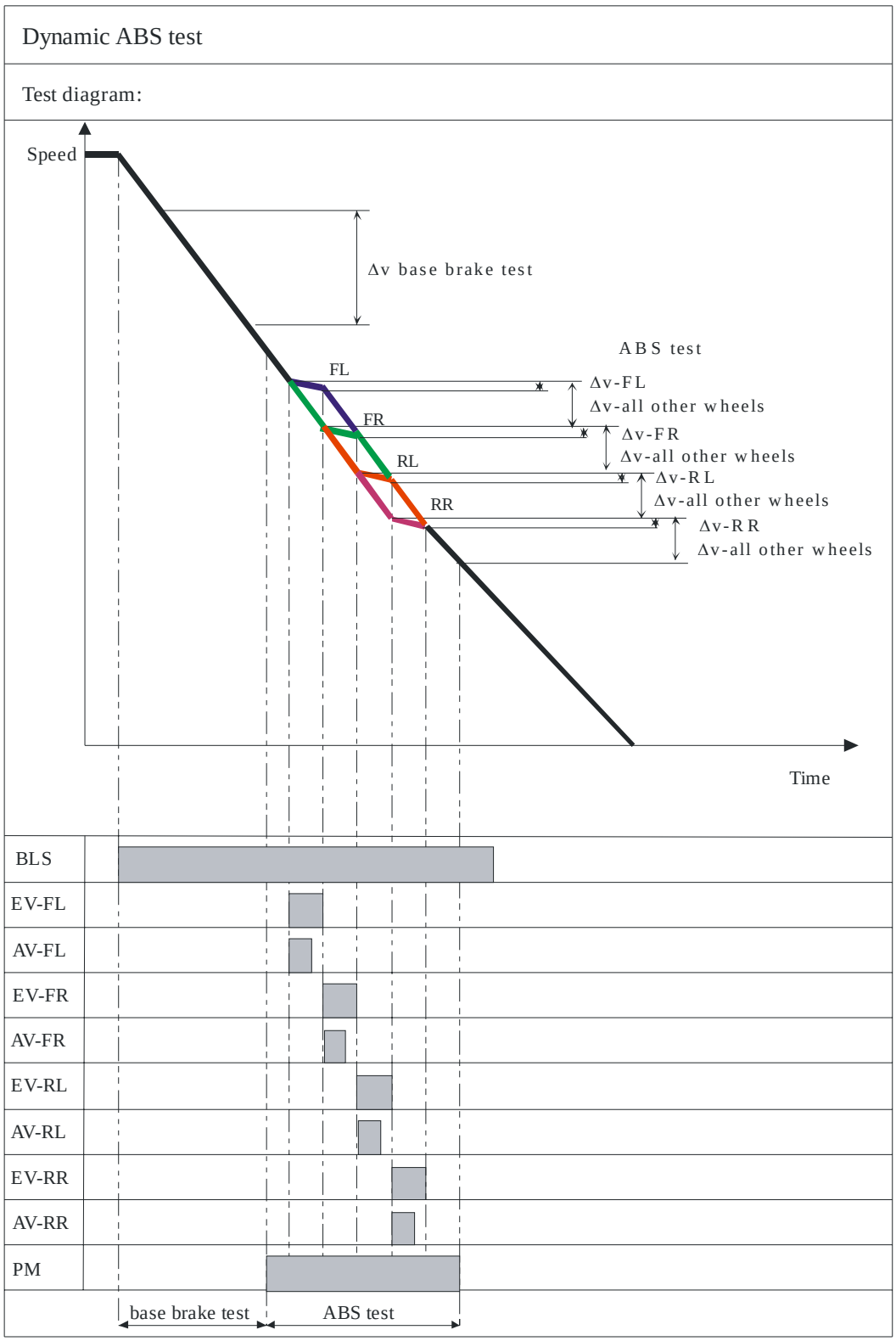
4.3.2.1 Diagram of the dynamic ABS test at all wheels



Remarks:

Please note the maximal allowed valve activation times. More information about these times can be read in the chapter 6.

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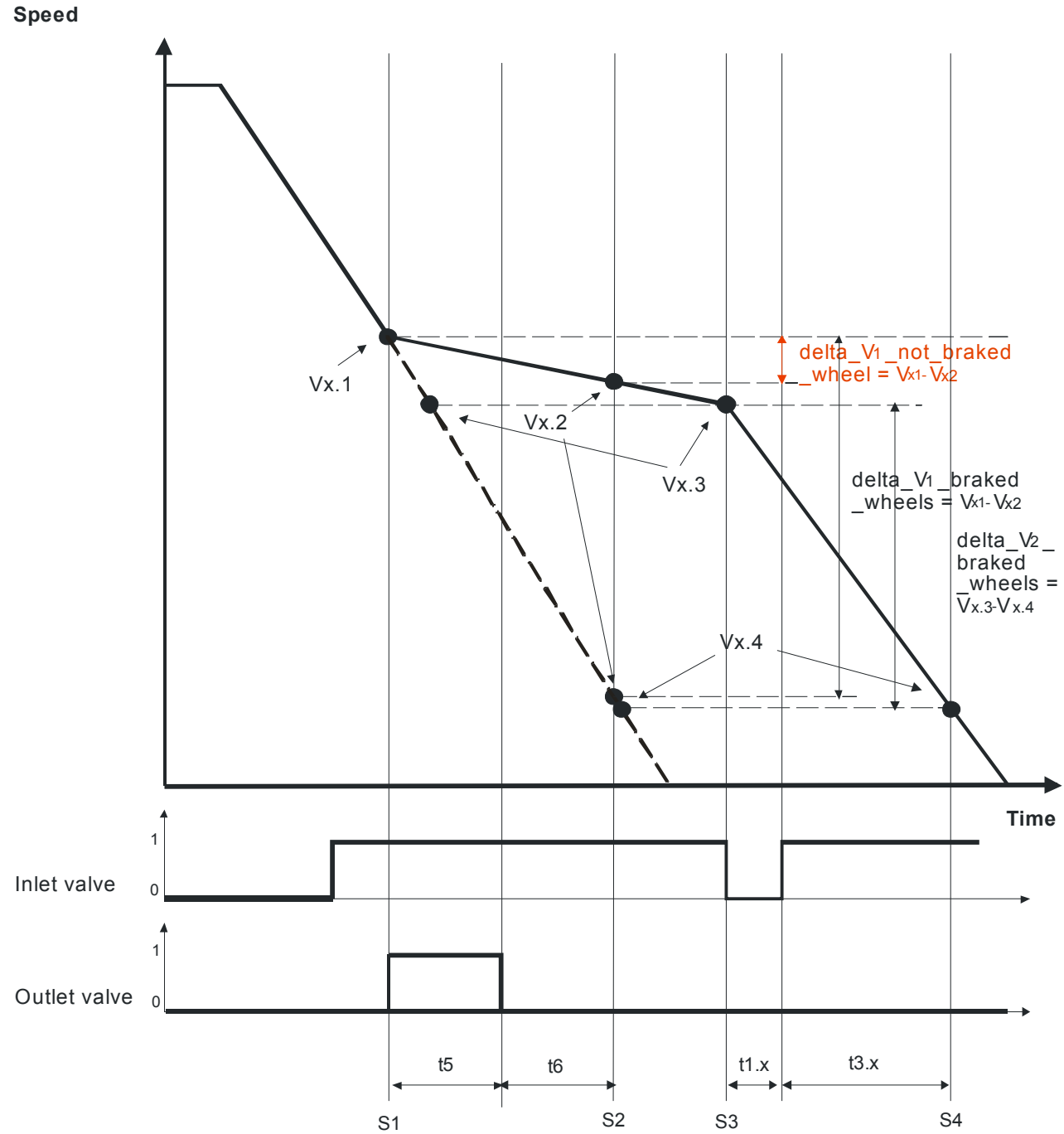
4.3.2.2 Diagram of the dynamic ABS test (exemplary at one wheel)

Dynamic ABS test

Test diagram (exemplary at one whee):



Remarks:

- The diagnostic tester sends the request (S1) to reduce the wheel pressures.

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- The ECU takes the first speed sample (Vx.1) and executes then the requested pressure decrease for the defined time (t5).
- The ECU takes the second speed sample (Vx.2) after the defined stabilisation time (t6) is expired.
- The ECU takes these two speed samples, calculates the differences speed for the pressure decrease cycle (Vx.1-Vx.2 = ΔV1/wheelxx) and outputs the result if the tester request it (S2).
- The diagnostic tester sends the request (S3) to increase the wheel pressure again.
- The ECU takes the third speed sample (Vx.3) and executes then the requested pressure increase for the defined time (t1.x).
- The ECU takes the fourth speed sample (Vx.4) after the defined stabilisation time (t3.x) is expired.
- The ECU takes these two speed samples, calculates the differences speed for the pressure increase cycle (Vx.3-Vx.4 = ΔV2/wheelxx) and outputs the result if the tester request it (S4).

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4.3.2.3 Test sequence for the dynamic ABS test

Start message:	Operator prompt:	Measure sample:	Activation before the time 1:	Time 1:	Activation before the time 2:	Time 2:	Measure sample:
1	Don't apply the brake pedal!	V1.1	---	---	---	---	V1.2
2	Apply the brake pedal till the expected decelerations are reached!	V2.1	---	---	---	---	V2.2
3	Keep the brake pedal constant!	---	Pump motor on	---	---	---	---
4		---	Inlet valves front axle on	---	Inlet valves rear axle on	---	---
5		---	AVFL on	t5	AVFL off	t6	---
6		V3.1	Measurement pressure decrease FL	t7	---	---	V3.2
7		---	EVFL off	t1.3	EVFL on	t.3.3	---
8		---	AVFR on	t5	AVFR off	t6	
9		V4.1	Measurement pressure increase FL & decrease FR	t7	---	---	V4.2
10		---	EVFR off	t1.3	EVFL on	t3.3	---
11		---	AVRL on	t5	AVRL off	t6	---
12		V5.1	Measurement pressure increase FR & decrease RL	t7	---	---	V5.2
13		---	EVRL off	t1.4	EVRL on	t3.4	---
14		---	AVRR on	t5	AVRR off	t6	---
15		V6.1	Measurement pressure increase RL & decrease RR	t7	---	---	V6.6
16		---	EVRR off	t1.4	EVRR on	t3.4	---
17		V7.1	Measurement pressure increase RR	t7	---	---	V7.2
18	Release the brake pedal!	---	Inlet valves front axle off	---	Inlet valves rear axle off	---	---
19		V8.1	---	t8	Pump motor off	---	V8.2

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Remarks:

After the final test speed is reached the rolls are detached by the test rig at first. Then the freewheeling deceleration (ΔV1 = V1.1-V1.2) at all individual wheels are captured (start message 1). These values are subtracted from the afterwards measured wheel decelerations. As next step the operator is prompt to apply the brake pedal till the expected deceleration is reached and measurement for the base brake test (ΔV2 = V2.1-V2.2) is executed (start message 2). Afterwards the operator has to keep the brake pedal constant.
The tester switches on the pump motor (start message 3) and all inlet valves (start message 4). Afterwards the pressure at the wheel FL is released by switching on of the outlet valve for the time t5 (start message 5).
After the stabilisation time t6 is expired the ECU measures the decelerations (ΔV3 = V3.1-V3.2) of all wheels with the dummy message (start message 6) for the time t7 to ensure a correct executed pressure release function at the wheel FL.
As next step the inlet valve FL is immediately opened for the time t1.3 to increase the pressure in the wheel. In parallel the pressure in the wheel FR is released by switching on of the outlet valve for the time t5 (start messages 7 & 8). After this test step respectively after the stabilisation time t3.3 (t6) is expired the decelerations (ΔV4 = V4.1-V4.2) of all wheels are measured again with execution of the dummy message (start message 9) for the time t7. This is required to ensure the pressure increase at the wheel FL and the pressure decrease at the wheel FR.
Afterwards the inlet valve FR is immediately opened for the time t1.3 to increase the pressure in the wheel. In parallel the pressure in the wheel RL is released by switching on of the outlet valve for the time t5 (start messages 10 & 11).
This is ensured by measuring of the wheel decelerations (ΔV5 = V5.1-V5.2) with execution of the dummy message (start message 12) for the time t7 after the stabilisation time t3.3 (t6) is expired.
The test at rear axel is done in the same way (start messages 13 till 17; measurements of wheel decelerations ΔV6 till ΔV7).

At the end of the ABS test the operator is prompt to release the brake pedal. Afterwards all inlet valves are immediately switched off (start message 18). The pump motor is switched off after the time t8 (start message 19). This is required to discharge the accumulator chambers. In parallel the decelerations of all wheels are measured again and then they are compared with the freewheeling forces which were measured at the beginning of the base brake test.

Details to used diagnostic function see chapter 5.6.

Parameter:

See chapter 4.3.3.4.

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4.3.2.4 Proposal of the evaluation of the base brake and dynamic ABS test

a) Evaluation of the freewheeling before the base brake test (exemplary for the wheel FL):

If
ΔV1/wheel_FL ≤ ΔV_freewheeling_max

then ΔV1/wheel_FL = ΔV_freewheeling and execute test step b)

else stop process and output “Freewheeling at the wheel FL nok”

b) Evaluation of the base brake test (exemplary for the front axle):

If
(ΔV2/wheel_FL – ΔV_freewheeling_FL) ≥ ΔV_FA_brakes
and
(ΔV2/wheel_FL – V_brakes_tolerance) ≤ ΔV2/wheel_FR ≤ (ΔV/wheel_FL + V_brakes_tolerance)

then execute test step c)

else stop process and output “Base brake test at the front axle nok”

c) Evaluation ABS test „pressure decrease“ (exemplary for the wheel FL):

If
ΔV3.1/wheel_FL ≤ (ΔV_unpressurised + ΔV_freewheeling_FL)

then execute test step d)

else stop process and output “ABS pressure decrease at the wheel FL nok”

c) Evaluation ABS test „pressure increase“ (exemplary for the wheel FL):

If
(ΔV4.1/wheel_FL – ΔV_freewheeling_FL) ≥ ΔV_FA_pressure_increase_brakes

then execute test step e)

else stop process and output “ABS pressure increase at the wheel FL nok”

e) Evaluation of the freewheeling after the ABS test (exemplary for the wheel FL):

If
ΔV8/wheel_FL ≤ ΔV_freewheeling_max

then output “base brake test and dynamic ABS test ok”;

else output “ Freewheeling at the wheel FL nok ”

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Example:

Evaluation variables*:
ΔV_unpressurised = 0,5 km/h
ΔV_freewheeling_max = 0,8 km/h
ΔV_FA_brakes = 2 km/h
ΔV_FA_pressure_increase_brakes = 1,5 km/h
V_brakes_tolerance = 25%

*All mentioned variables for evaluation are valid for the measure time of 500ms. All values are examples and must be finally defined by the customer!

a) Evaluation of the calculated difference speeds (km/h) of the individual wheels for the test step "freewheeling before the base brake test":

ΔV1_FL	ΔV1_FR	ΔV1_RL	ΔV1_RR
0,5	0,5	0,3	0,3

b) Evaluation of the calculated difference speeds (km/h) of the individual wheels for the test step "base brake test":

ΔV2_FL	ΔV2_FR	ΔV2_RL	ΔV2_RR
2,8	2,9	2,6	2,5

c) Evaluation of the calculated difference speeds (km/h) of the individual wheels for the test step "ABS test – pressure decrease FL":

ΔV3.1_FL	ΔV3.1_FR	ΔV3.1_RL	ΔV3.1_RR
0,9	1,0	0,8	0,8

d) Evaluation of the calculated difference speeds (km/h) of the individual wheels for the test step "ABS test – pressure increase FL":

ΔV4.1_FL	ΔV4.1_FR	ΔV4.1_RL	ΔV4.1_RR
2,3	0,7	0,4	0,8

e) Evaluation of the calculated difference speeds (km/h) of the individual wheels for the test step "freewheeling after ABS test":

ΔV8_FL	ΔV8_FR	ΔV8_RL	ΔV8_RR
0,6	0,5	0,4	0,3

Judgement of each test step for the wheel FL respectively FA:

a) Freewheeling at the wheel FL before the base brake test: 0,5 ≤ 0,8 **= ok**

b) Base brake test front axle: ((2,8-0,5) ≥ 2) && (2,8-0,7) ≤ 2,9 ≤ (2,8+0,7) **= ok**

c) ABS test „pressure decrease FL": 0,9 ≤ (0,5+0,5) **= ok**

d) ABS Test „pressure increase brakes FL": (2,3-0,5) ≥ 1,5 **= ok**

e) Freewheeling at the wheel FL after ABS test: 0,6 ≤ 0,8 **= ok**

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4.3.3 Base brake and dynamic ESP® test (function test „brake pipe correlation, pressure increase and release“)

Base brake and dynamic ESP® test (function test „brake pipe correlation, pressure increase and release“)					
Operator:	Test rig:	To be evaluated brake forces:		Demanded brake forces:	Measure sample:
→ Don’t apply the brake pedal!	Measuring of the freewheeling decelerations	Wheel FL: Wheel FR: Wheel RL: Wheel RR:	Δ xx km/h Δ xx km/h Δ xx km/h Δ xx km/h	$\Delta V_{FL} \leq \Delta V_{freewheeling_max}$ $\Delta V_{FR} \leq \Delta V_{freewheeling_max}$ $\Delta V_{RR} \leq \Delta V_{freewheeling_max}$ $\Delta V_{RL} \leq \Delta V_{freewheeling_max}$	ΔV1 (V1.1-V1.2)
→ Apply the brake pedal till the expected deceleration is reached!	Pressure increase at all wheels Measuring of all decelerations	Wheel FL: Wheel FR: Wheel RL: Wheel RR:	Δ xx km/h Δ xx km/h Δ xx km/h Δ xx km/h	$\Delta V_{FL} \geq \Delta V_{FA_brakes}$ $\Delta V_{FL} - V_{brakes_tolerance} \leq \Delta V_{FR} \leq \Delta_{FL} + V_{brakes_tolerance}$ $\Delta V_{RL} \geq \Delta V_{RA_brakes}$ $\Delta_{RL} - V_{brakes_tolerance} \leq \Delta V_{RR} \leq \Delta_{RL} + V_{brakes_tolerance}$	ΔV2 (V2.1-V2.2)
→ Release the brake pedal!	Measuring of the freewheeling decelerations	Wheel FL: Wheel FR: Wheel RL: Wheel RR:	Δ xx km/h Δ xx km/h Δ xx km/h Δ xx km/h	$\Delta V_{FL} \leq \Delta V_{freewheeling_max}$ $\Delta V_{FR} \leq \Delta V_{freewheeling_max}$ $\Delta V_{RL} \leq \Delta V_{freewheeling_max}$ $\Delta V_{RR} \leq \Delta V_{freewheeling_max}$	ΔV3 (V3.1-V3.1)
→ Don’t apply the brake pedal!	Pressure increase FL Measuring of all decelerations	Wheel FL: Wheel FR: Wheel RL: Wheel RR:	Δ xx km/h Δ xx km/h Δ xx km/h Δ xx km/h	$\Delta V_{FL} > (\Delta V_{FA_pressure_increase} - \Delta V_{freewheeling_FL})$ $\Delta V_{FR} \leq (\Delta V_{unpressurised} + \Delta V_{freewheeling_FR})$ $\Delta V_{RL} \leq (\Delta V_{unpressurised} + \Delta V_{freewheeling_RL})$ $\Delta V_{RR} \leq (\Delta V_{unpressurised} + \Delta V_{freewheeling_RR})$	ΔV4 (V4.1-V4.2) ΔV4.n (V4.n-V4.n+1)
→ Don’t apply the brake pedal!	Pressure decrease FL	---	---	---	---
→ Don’t apply the brake pedal!	Pressure increase FR Measuring of all decelerations	Wheel FL: Wheel FR: Wheel RL: Wheel RR:	Δ xx km/h Δ xx km/h Δ xx km/h	$\Delta V_{FL} \leq (\Delta V_{unpressurised} + \Delta V_{freewheeling_FL})$ $\Delta V_{FR} > (\Delta V_{FA_pressure_increase} - \Delta V_{freewheeling_FR})$ $\Delta V_{RL} \leq (\Delta V_{unpressurised} +$	ΔV5 (V5.1-V5.2) ΔV5.n (V5.n-V5.n+1)

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			Δ xx km/h	$\Delta V_{_freewheeling_RL}$) $\Delta V_{RR} \leq (\Delta V_{_unpressurised} + \Delta V_{_freewheeling_RR})$	
→ Don't apply the brake pedal!	Pressure decrease FR	---	---	---	---
→ Don't apply the brake pedal!	Pressure increase RL Measuring of all decelerations	Wheel FL: Wheel FR: Wheel RL: Wheel RR:	Δ xx km/h Δ xx km/h Δ xx km/h Δ xx km/h	$\Delta V_{FL} \leq (\Delta V_{_unpressurised} + \Delta V_{_freewheeling_FL})$ $\Delta V_{FR} \leq (\Delta V_{_unpressurised} + \Delta V_{_freewheeling_FR})$ $\Delta V_{RL} > (\Delta V_{RA_pressure_increase} - \Delta V_{_freewheeling_RL})$ $\Delta V_{RR} \leq (\Delta V_{_unpressurised} + \Delta V_{_freewheeling_RR})$	ΔV6 (V6.1-V6.1) V6.n (V6.n-V6.n+1)
→ Don't apply the brake pedal!	Pressure decrease RL	---	---	---	---
→ Don't apply the brake pedal!	Pressure increase RR Measuring of all decelerations	Wheel FL: Wheel FR: Wheel RL: Wheel RR:	Δ xx km/h Δ xx km/h Δ xx km/h Δ xx km/h	$\Delta V_{FL} \leq (\Delta V_{_unpressurised} + \Delta V_{_freewheeling_FL})$ $\Delta V_{FR} \leq (\Delta V_{_unpressurised} + \Delta V_{_freewheeling_FR})$ $\Delta V_{RL} \leq (\Delta V_{_unpressurised} + \Delta V_{_freewheeling_RL})$ $\Delta V_{RR} > (\Delta V_{RA_pressure_increase} - \Delta V_{_freewheeling_RR})$	ΔV7 (V7.1-V7.2) ΔV7.n (V7.n-V7.n+1)
→ Don't apply the brake pedal!	Pressure decrease RR	---	---	---	---
→ Don't apply the brake pedal!	Measuring of the freewheeling decelerations	Wheel FL: Wheel FR: Wheel RL: Wheel RR:	Δ xx km/h Δ xx km/h Δ xx km/h Δ xx km/h	$\Delta V_{FL} \leq (\Delta V_{_unpressurised} + \Delta V_{_freewheeling_FL})$ $\Delta V_{FR} \leq (\Delta V_{_unpressurised} + \Delta V_{_freewheeling_FR})$ $\Delta V_{RL} \leq (\Delta V_{_unpressurised} + \Delta V_{_freewheeling_RL})$ $\Delta V_{RR} \leq (\Delta V_{_unpressurised} + \Delta V_{_freewheeling_RR})$	ΔV8 (V8.1-V8.2)

Remarks:

The base brake and ESP® tests are executed directly after the WSS tests. As first step during the base brake test the freewheeling decelerations are measured. After the final test speed is reached

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the rolls are detached at first then the freewheeling decelerations are measured. Afterwards the operator is asked to apply the brake pedal immediately. As soon as the expected decelerations are reached, the operator has to keep the brake pedal constant and shouldn't release it till the deceleration values of the base brake test are measured. Afterwards the operator has to relapse the brake pedal before the ESP® tests are started.
The ESP® test is executed immediately after the base brake test. The operator has to keep the brake pedal released. The tester activates the corresponding valves and the pump motor step by step for the ESP® test. In addition the tester measures the corresponding decelerations at each step and wheel.

The operator might not apply the brake pedal the complete ESP® test!

The respective wheel decelerations are calculated by the tester and compared with the fixed limits.

To reduce the entire cycle time of the ESP® test, the test cycle respectively corresponding vale activation for each wheel can overlap.

The base brake test has to be executed at all vehicles independent of the installed control system. The test is required to ensure the function and tightness of base brake.

The base brake and ESP® test have to ensure the following functions and assembly processes:

1) To ensure the function of the base brake system
2) To ensure the correlation of the brake pipes between the ESP® hydraulic ports and each wheel
3) To ensure the function of the EV and AV valves after brake filling process (e.g. to exclude a impairment of function caused by contamination, etc.)
4) To ensure delivery rate of both pumps respectively the active ESP® pressure increase function at all wheels
5) To ensure the correct correlation between the wheels speed sensors and the wheels

ESP® test ok ⇒ **Continue**

ESP®Test nok ⇒ **Failure message, abort, print out fault message and send vehicle to repair area**

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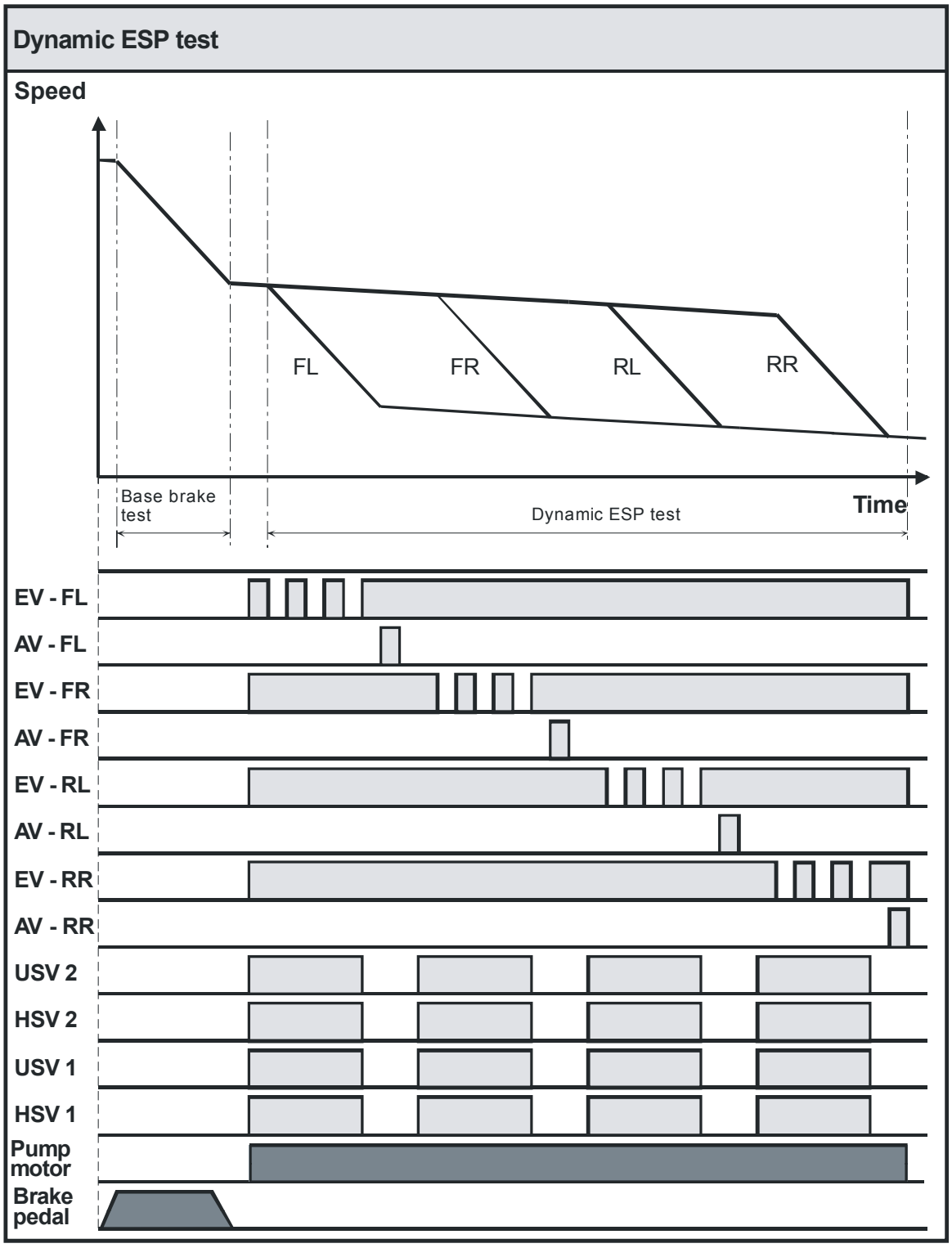
4.3.3.1 Diagram base brake and dynamic ESP® test (independent of the brake split distribution)



Remarks:

Please note the maximal allowed valve activation times. More information about these times can be read in the chapter 6.

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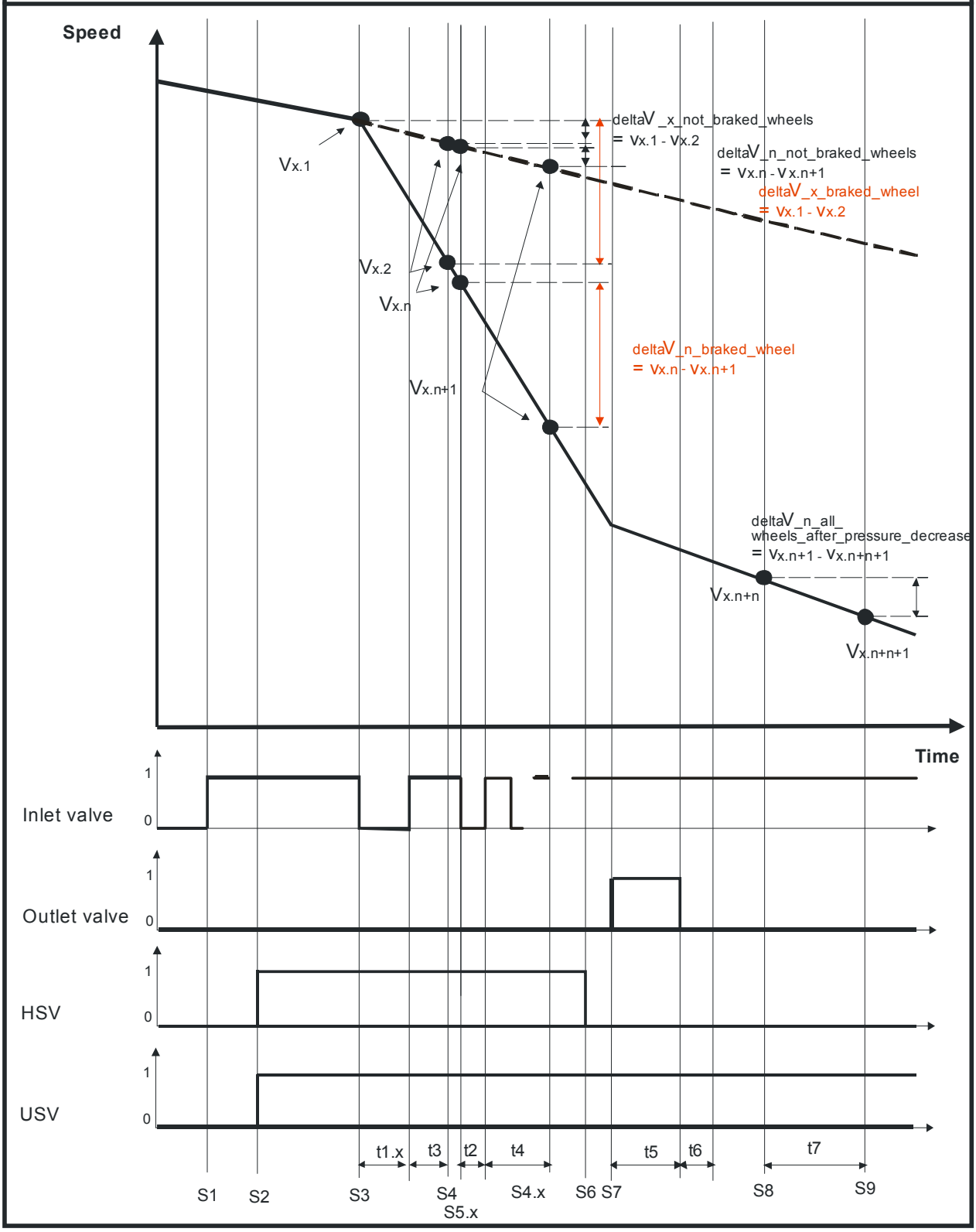
4.3.3.2 Diagram to the dynamic ESP® test exemplary at one wheel

Dynamic ESP test

Test diagram (exemplary at one wheel):



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Remarks:

- The diagnostic tester sends the request (S1) to close the inlet valve. Then it sends a second request (S2) to close the pilot valve (USV) and to open the priming (HSV) valve.
- As next step the tester sends a further request (S3). The ECU takes the first speed sample (Vx.1) and then it executes the first pressure increase pulse for the defined time (t1.x) to reduce the air gap between the brake disk and pads.
- The ECU takes the second speed sample (Vx.2) after the defined stabilisation time (t3) is expired.
- The ECU takes these two speed samples, calculates the differences speed for the pressure increase cycle (Vx.1-Vx.2 = ΔV_braked_wheel/wheel_xx & ΔV_not_braked_wheels/wheel_xx) and outputs the result if the tester request it (S4).
- Depending on the expected deceleration the tester sends a number of requests within a loop (S5.x) to increase the pressure (n pressure increase cycles with the time t2 and the stabilisation time t4). The ECU takes these individual speed samples, calculates the differences speed for the loop of pressure increase cycles (Vx.n-Vx.n+1) and outputs the result if the tester request it (S4.x). The loop is executed as long as the expected deceleration is reached.
- As next step the tester sends the request (S6) to close the priming valve (HSV). Afterwards the tester executes a pressure decrease for the defined time t5 and waits then the stabilisation time t6 (S7).
- The ECU takes before and after the measure time (t7) further speed samples (Vx.n+n; Vx.n+n+1).
- The ECU uses these speed samples and calculates the differences speed for the pressure decrease cycle (Vx2.1-Vx.2.2 = ΔV_all_wheels_after_pressure_decrease/wheel_xx) and outputs the result if the tester request it (S9).

To guarantee the comparability between the individual calculated speed differences, the measure times between the first longer pressure increase cycle, the following shorter increase cycles (loop of pulses) and the pressure decrease cycle should always be identically --> (t1+t3) = (t2+t4) = (t5+t6) = t7.
To save test time, the pressure decrease function can be ensured by evaluation of the speed difference at the next tested wheel.

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4.3.3.3 Test sequence for the dynamic ESP® test (independent of the brake split distribution)

Start message:	Operator prompt:	Measure sample:	Activation before the time 1:	Time 1:	Activation before the time 2:	Time 2:	Measure sample:
1	Don't apply the brake pedal!	V1.1	---	t7	---	---	V1.2
2	Apply the brake pedal till the expected decelerations are reached!	V2.1	---	t7	---	---	V2.2
3	Release the brake pedal!	V3.1	---	t7	---	---	V3.1
4	Don't apply the brake pedal!	---	Pump motor on	---	---	---	---
5		---	Inlet valves front axle on	---	Inlet valves rear axle on	---	---
6		---	USV1 and USV2 on	---	---	---	---
7		---	HSV1 and HSV2 on	---	---	---	---
8		V4.1	EVFL off	t1.1	EVFL on & measurement pressure increase FL	t3.1	V4.2
---		Loop pressure increase (number x1)					
9.x		V4.n	EVFL off	t2	EVFL on & measurement pressure increase FL	t4	V4.n+1
10		---	HSV1 and HSV2 off	---	---	---	---
11		---	AVFL on	t5	AVFL off	t6	---
12		---	HSV1 and HSV2 an	---	---	---	---
13		V5.1	EVFR off	t1.1	EVFR on & measurement pressure release FL & increase FR	t3.1	V5.2
---		Loop pressure increase (number x1)					
14.x		V5.n	EVFR off	t2	EVFR on	t4	V5.n+1
15		---	HSV1 and HSV2 off	---	---	---	---

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16		---	AVFR on	t5	AVFR off	t6	---
17		---	HSV1 and HSV2 an	---	---	---	---
18		V6.1	EVRL off	t1.2	EVRL on & measurement pressure increase RL	t3.2	V6.2
---		Loop pressure increase (number x1)					
19.x		V6.n	EVRL off	t2	EVRL on & measurement pressure increase RL	t4	V6.n+1
20		---	HSV1 and HSV2 off	---	---	---	---
21		---	AVRL on	t5	AVRL off	t6	---
22		---	HSV1 and HSV2 an	---	---	---	---
23		V7.1	EVRR off	t1.2	EVRR on & measurement pressure release RL & increase RR	t3.2	V7.2
---		Loop pressure increase (number x1)					
24.x		V7.n	EVRR off	t2	EVRR on & measurement increase RR	t4	V7.n+1
25		---	HSV1 and HSV2 off	---	---	---	---
26		---	AVRR on	t5	AVRR off	t6	---
27		V8.1	measurement pressure release RL	---	---	t7	V8.2
28		---	USV1 and USV2 off	---	---	---	---
29		---	Inlet valves front axle off	---	Inlet valves rear axle off	---	---
30		---	---	t8	Pump motor off		

Remarks:

After the final test speed is reached the rolls are detached by the test rig at first. Then the freewheeling deceleration (ΔV1 = V1.1-V1.2) at all individual wheels are captured (start message 1). These values are subtracted from the afterwards measured wheel decelerations. As next step the operator is prompt to apply the brake pedal till the expected deceleration is reached and measurement for the base brake test (ΔV2 = V2.1-V2.2) is executed (start message 2). These values are used to judge the base brake function. Afterwards the operator has to release the brake pedal. This is ensured by measuring of the wheel decelerations (ΔV3=V3.1-V3.2) again (start message 3).

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Then the tester switches on the pump motor (start message 4), all inlet valves (start message 5), both USV (start message 6) and both HSV valves (start message 7).
To get a sufficient deceleration at the wheel FL, the tester runs a number of pressure increase pulses. The first pulse should be longer than the following. This is required to reduce the air gap between the brake pads and disk (start message 8 with the pressure increase time FA t1.1 & stabilisation time FA t3.1). Afterwards the tester runs a number of loops with a smaller pressure increase time (start message 9.1 till 9.n; pressure increase time - general t2 & stabilisation time – general t4).
The loop is executed as long as the expected deceleration (ΔV_FA_pressure_increase) or the maximal allowed number (x1) of pressure increase pulses is reached.
The actual speed is measured by the ECU before each pressure increase pulse is executed and after the stabilisation time is expired.
The ECU uses these speed samples and calculates the speed differences for the pressure increase cycle (ΔV4=V4.n-V4.n). After the stop criteria are reached, the tester switches off both HSV valves (start message 10) at first, then it releases the pressure at the wheel FL by switching on of the outlet valve (start message 11, time t5).
To be able to execute the same procedure at the wheel FR both HSV valves have to be switched on again (start message 8). Afterwards the pressure is first increased then released at the wheel FR (start messages 9-12).
The pressure release function is guaranteed by evaluation of the deceleration values after first pressure increase cycle at the wheel FR. This is ensured by evaluation of the wheel decelerations (ΔV5 = V5.1-V5.2).
The test at the wheel FR and rear axle is done in the same way (start messages 12 till 27; measurements of wheel decelerations ΔV5 till ΔV8).
At the end of the ESP® all valves, which are still switch on, are switched off (start messages 28&29).
The pump motor is switched off after the time t8 (start message 30).

Details to used diagnostic function see chapter 5.6 or chapter 5.7.

Important remark:

At ESP® systems pilot valves without pressure limitation are used. To protect these systems against over pressure, the pressure is limited by controlling of the pilot valve current. This current corresponds to the maximal system pressure (fluctuation by the allowed tolerances possible). More details can be read in the chapter 6.

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4.3.3.4 Parameter (base brake, dynamic ABS and ESP® test)

Kind of parameter:	Parameter name:		Unit:	Physical value:
Time	1. pressure increase pulse FA (ESP® test)	t1.1	Milliseconds (ms)	120
	1. pressure increase pulse RA (ESP® test)	t1.2		80
	general pressure increase pulse FA (ABS test)	t1.3		60
	general pressure increase pulse (ABS test)	t1.4		40
	pressure increase loop pulse (general)	t2		20
	stabilisation time FA (1. pressure increase pulse – ESP®)	t3.1		380
	stabilisation time RA (1. pressure increase pulse – ESP®)	t3.2		420
	stabilisation time FA (pressure increase pulse – ABS)	t3.3		440
	stabilisation time RA (pressure increase pulse – ABS)	t3.4		460
	stabilisation time general – after the following pressure increase pulses (loop)	t4		480
	pressure decrease pulse – general	t5		300
	stabilisation time (general) after the pressure decrease pulse	t6		200
	Measurement time - general	t7		500
	Post run time of the pump motor	t8		2000
Number	Number of loops for the pressure increase	x1	---	5
Evaluation parameter	ΔV_FA_pressure_increase		Kilometre per hour (km/h)	tbd
	ΔV_RA_pressure_increase			tbd
	ΔV_unpressurised			tbd
	ΔV_freewheeling_max			tbd
	ΔV_FA_brakes			tbd
	ΔV_FA_pressure_increase_brakes			tbd
	ΔV_RA_brakes			tbd
	ΔV_RA_pressure_increase_brakes			tbd
	V_brakes_tolerance		Percent (%)	25

Remarks:

The mentioned times are proposed times such as the activation times for the pressure increase and release cycles, the stabilisation and measure times. All these times depend on the vehicle (e.g. size of brake, orifices in the hydraulic unit, etc.) respectively the test rig, therefore these times must be confirmed respectively modified during trials at the customer plant.
Also the mentioned evaluation parameter have to be defined by a statistic evaluation with the first tested vehicles .The definition of all these data is up the customer responsibility.

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4.3.3.5 Proposal of the evaluation of the base brake and dynamic ESP® test

a) Evaluation of the freewheeling before the base brake test (exemplary for the wheel FL):

If
ΔV1/wheel_FL ≤ ΔV_freewheeling_max

then ΔV1/wheel_FL = ΔV_freewheeling and execute test step b)

else stop process and output “Freewheeling at the wheel FL nok”

b) Evaluation of the base brake test (exemplary for the front axle):

If
(ΔV2/wheel_FL – ΔV_freewheeling_FL) ≥ ΔV_FA_brakes
and
(ΔV2/wheel_FL – V_brakes_tolerance) ≤ ΔV2/wheel_FR ≤ (ΔV/wheel_FL + V_brakes_tolerance)

then execute test step c)

else stop process and output “Base brake test at the front axle nok”

c) Evaluation of the freewheeling after the base brake test (exemplary for the wheel FL):

If
ΔV3/wheel_FL ≤ ΔV_freewheeling_max

then execute test step d)

else stop process and output “Freewheeling at the wheel FL nok”

d)) Evaluation of the ESP® test „pressure increase“ (exemplary for the wheel FL):

If
(ΔV4.n/wheel_FL – ΔV_freewheeling_FL) ≥ ΔV_FA_pressure_increase

then execute test step e)

else stop process and output “ESP® pressure increase at the wheel FL nok ”

e) Evaluation of the ESP® test „pressure decrease“ (exemplary for the wheel FL):

If
ΔV5.1/wheel_FL ≤ (ΔV_unpressurised + ΔV_freewheeling_FL)

then output “Base brake and dynamic ESP® test ok”;

else output “ESP® pressure decrease at the wheel FL nok”

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Example:

Evaluation variables*:
ΔV_FA_pressure_increase = 2,0 km/h
ΔV_unpressurised = 0,5 km/h
ΔV_freewheeling_max = 0,8 km/h
ΔV_FA_brakes = 2 km/h
V_brakes_tolerance = 25%

*All mentioned variables for evaluation are valid for the measure time of 500ms. All values are examples and must be finally defined by the customer!

a) Evaluation of the calculated difference speeds (km/h) of the individual wheels for the test step "freewheeling before the base brake test":

ΔV1_FL	ΔV1_FR	ΔV1_RL	ΔV1_RR
0,5	0,5	0,3	0,3

b) Evaluation of the calculated difference speeds (km/h) of the individual wheels for the test step "base brake test":

ΔV2_FL	ΔV2_FR	ΔV2_RL	ΔV2_RR
2,8	2,9	2,6	2,5

c) Evaluation of the calculated difference speeds (km/h) of the individual wheels for the test step "freewheeling after the base brake test":

ΔV3_FL	ΔV3_FR	ΔV3_RL	ΔV3_RR
0,6	0,5	0,4	0,3

d) Evaluation of the calculated difference speeds (km/h) of the individual wheels for the test step "ESP® test – pressure increase FL":

ΔV4.n_FL	ΔV4.n_FR	ΔV4.n_RL	ΔV4.n_RR
2,8	0,7	0,4	0,8

e) Evaluation of the calculated difference speeds (km/h) of the individual wheels for the test step "ESP® test – pressure decrease FL":

ΔV5.1_FL	ΔV5.1_FR	ΔV5.1_RL	ΔV5.1_RR
0,9	1,0	0,8	0,8

Judgement of each test step for the wheel FL respectively FA:

a) Freewheeling before the base brake test: 0,5 ≤ 0,8 **= ok**

b) Base brake test front axle: ((2,8-0,5) ≥ 2) && (2,8-0,7) ≤ 2,9 ≤ (2,8+0,7) **= ok**

c) Freewheeling after the base brake test: 0,6 ≤ 0,8 **= ok**

d) ESP® test „pressure increase FL": (2,8-0,5) ≥ 2,0 **= ok**

e) ESP® test „pressure decrease FL": 0,9 ≤ (0,5+0,5) **= ok**

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4.4 Finishing actions & standstill tests

4.4.1 Evaluation of the fault code memory

Evaluation of the fault code memory	
Tester:	Remarks:
Reading and evaluating of the fault code memory.	More details can also be read in the chapter 4.1.10.

Remarks:

At the end of the ABS/ ESP® end of line (EOL) tests the fault code memory is finally checked. In case of fault codes are stored, the vehicle has to send to the rework area and after the root cause of is removed the ABS/ ESP® EOL test has to repeat again.

The evaluation of the fault code memory has following purpose:

1) To ensure that the ABS / ESP® system is free of faults after the EOL tests

ABS/ESP® system free of faults ⇒ **Continue**

ABS/ESP® system not free of faults ⇒ **Abort, print out fault message and send vehicle to repair area**

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4.4.2 Warning lamp test (change to the default diagnostic mode)

Warning lamp test (change to the default diagnostic mode)	
Operator / Tester:	Remarks:
Check and confirm if all ABS / ESP® warning and signal lamps are on!	All ABS/ ESP® signal and warning lamps are activated in the customer specific diagnostic mode!
Change to the default diagnostic mode.	
Check and confirm if all ABS / ESP® warning and signal lamps are off after the time $t_{_signal_warning_lamps_off_}$ is expired!	ABS/ ESP® executes a software reset. After the time $t_{_signal_warning_lamps_off}$ is expired respectively the system is free of faults all signal and warning lamps are switched off.
Ignition/engine off.	
Evaluation of the entire test data & print of a test protocol	

Remarks:

As long as the ECU is within the customer specific diagnostic mode all control functions are deactivated and all ABS / ESP® warning respectively signal lamps are illuminated. This special mode (test mode) can be left by changing into the customer default mode (no hardware or software reset required).

Parameter:

Kind of parameter:	Parameter name:	Unit:	Physical value:
Time	$t_{_signal_warning_lamps_off}$	Milliseconds	≤ 5000

After the EOL tests are finished the tester has to print out the complete test results and a short judgement :ok / nok

The test of the ABS/ ESP® signal and warning lamps has following purpose:

1) To ensure the function of all signal and warning lamps

Signal and warning lamp test ok ⇒ **End of the EOL tests**

Signal and warning lamp test nok ⇒ **Abort, print out fault message and send vehicle to repair area**

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4.5 EOL driving test (only ESP® systems - optional)

The EOL driving test is a optional test (customer specific). This test can be executed as additional test to the previous described tests. This test consists of a standstill and a dynamic driving tests.

EOL driving test	
Standstill tests:	- Pressure sensor test - Yaw rate and lateral signal offset test - Storing of the lateral sensor signal offset
EOL driving tests:	- Sensitivity test of the yaw rate sensor signal - Sensitivity test of the lateral sensor signal - Steering angle sensor signal test

Remarks:

The EOL driving test is activated via diagnostics. Afterwards a defined driving manoeuvre has to be executed. The driver identifies the state of the test by checking of the signal and warning lamps. The individual activation of these lamps shows which test has to be executed as next (e.g. standstill test finished; ECU waits for execution of the dynamic driving test). In addition these lamps show if the tests are finished successfully or if a sensor fault is present.

The EOL driving tests are designed in way, that each individual test can be selected via diagnostics and to do the tests via a very simple manoeuvre.

Details to the EOL driving test can be read in the customer specific documentations.

EOL driving test ok ⇒ **End of the EOL driving test (successfully finished)**

EOL driving test nok ⇒ **Abort, print out fault message and send vehicle to repair area, repeat EOL driving test**

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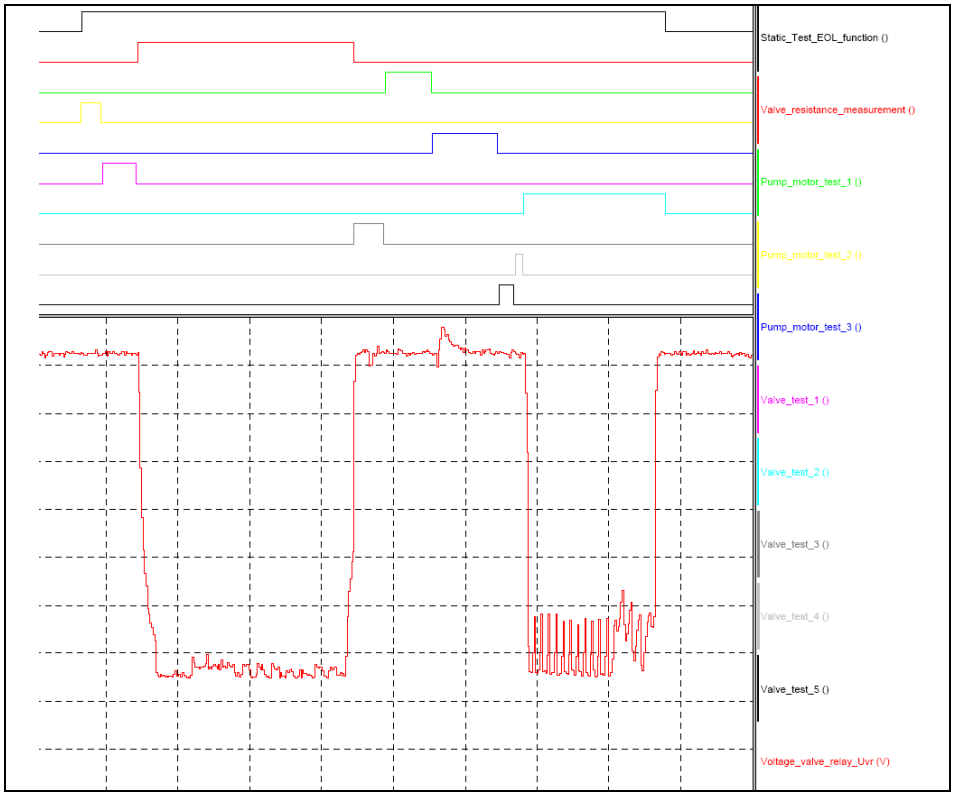
5 Bosch specific EOL diagnostic routines at generation 9

5.1 Diagnostic routine „Static Test“



5.1.1 Start parameter „Static Test“

Start parameter „Static Test“

Remarks:

This routine has no specific start parameter.

5.1.2 Evaluation parameter „Static Test“

Evaluation parameter „Static Test“

Remarks:

This routine has no specific evaluation parameter.

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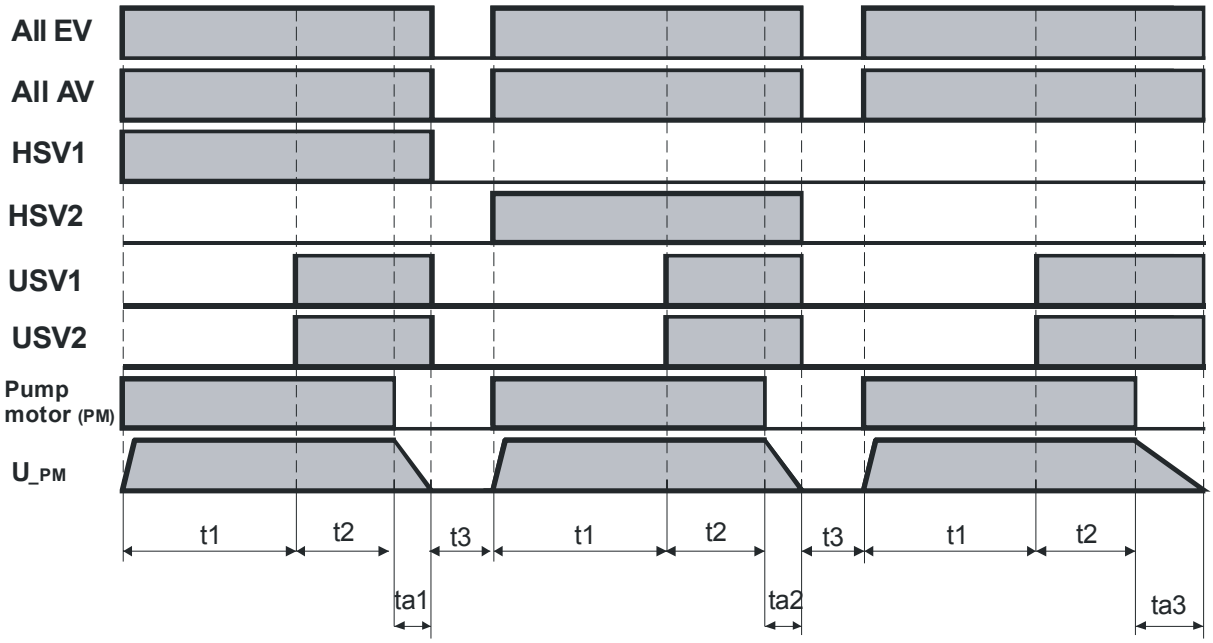
5.2 Diagnostic routine „Pump motor slow-down time test“



5.2.1 Start parameter „Pump motor slow-down time test“

Start parameter „Pump motor slow-down time test“
Delay time t_1 (time till the USV1 & USV2 is switched on)
Delay time t_2 (time till the PM is switched off)

Remarks:

The length (byte or word) of the start parameter and resolution is customer specific.

5.2.2 Evaluation parameter „Pump motor slow-down time test“

Evaluation parameter „Pump motor slow-down time test“
Post run time t_{a1}
Post run time t_{a2}
Post run time t_{a3}

Remarks:

The length (byte or word) of the evaluation parameter and resolution is customer specific.

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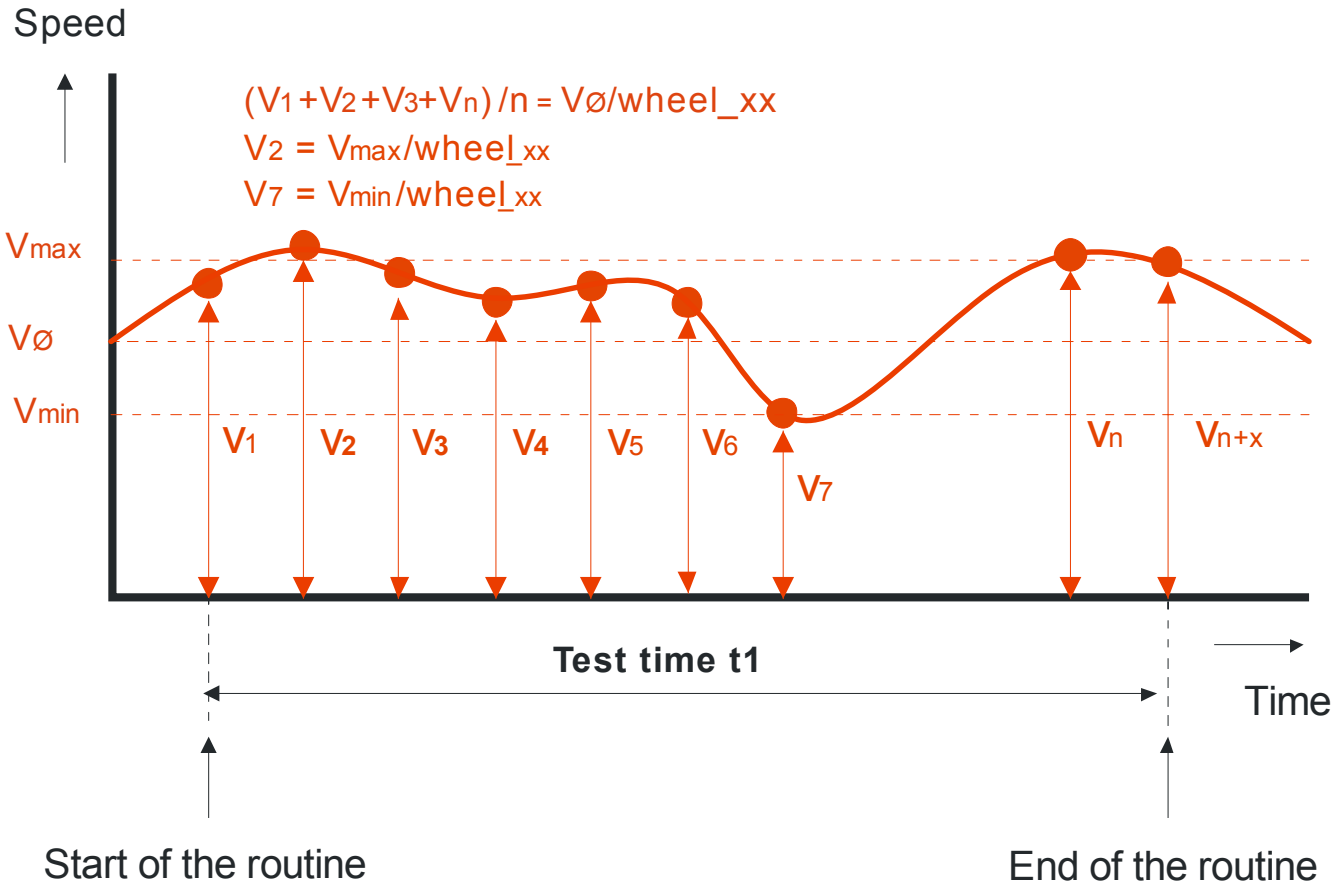
5.3 Diagnostic routine „Wheel Speed Sensor Test“



5.3.1 Start parameter „Wheel Speed Sensor Test“

Start parameter „Wheel Speed Sensor Test“
Test time t_1

Remarks:

The length (byte or word) of the start parameter and resolution is customer specific.

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5.3.2 Evaluation parameter „Wheel Speed Sensor Test“

Evaluation parameter „Wheel Speed Sensor Test“
V_max/wheel_FL
V_min/wheel_FL
V_max/wheel_FR
V_min/wheel_FR
V_max/wheel_RL
V_min/wheel_RL
V_max/wheel_RR
V_min/wheel_RR
*V_Ø/wheel_FL
*V_Ø/wheel_FR
*V_Ø/wheel_RL
*V_Ø/wheel_RR

Remarks:

The length (byte or word) of the evaluation parameter and resolution is customer specific.

* The output of the average speed values can be switched on or off depending on the customer request.

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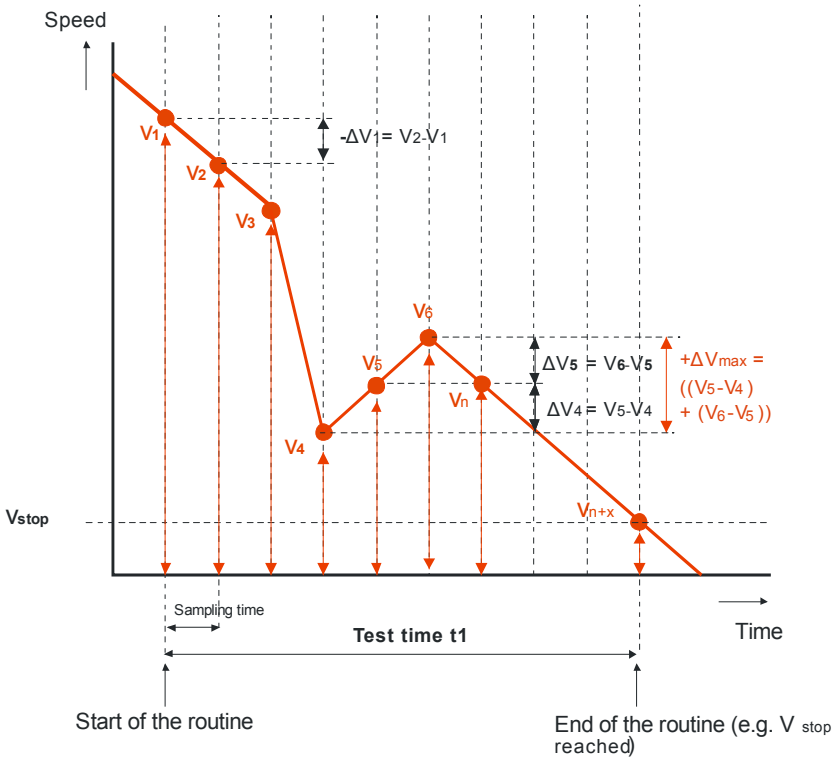
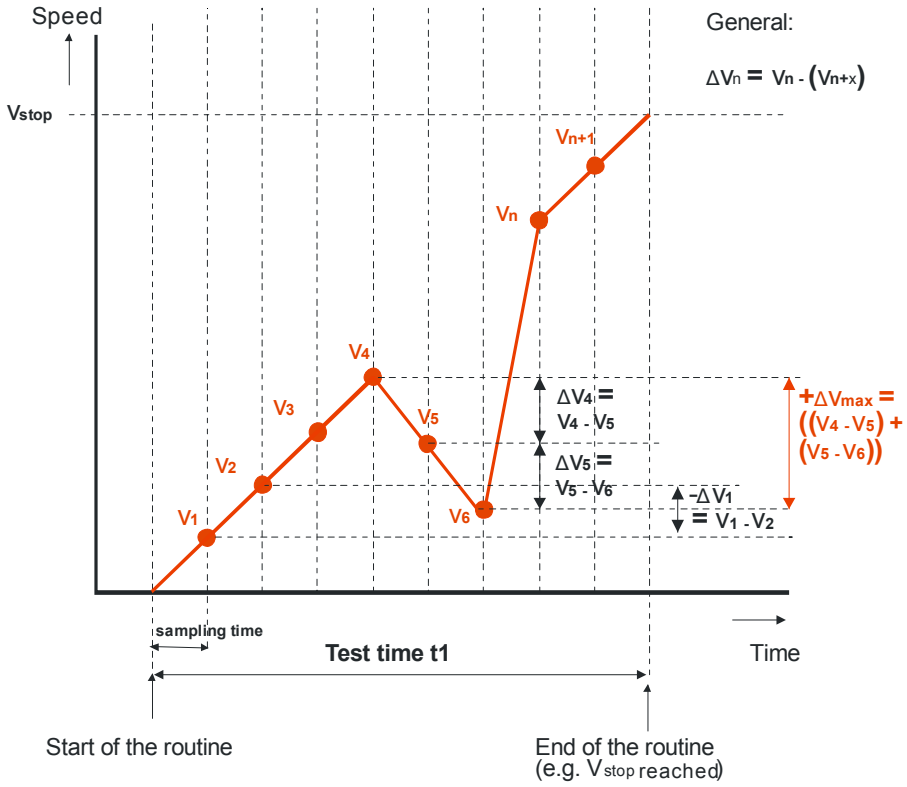
5.4 Diagnostic routine „Wheel Speed Sensor Test – accelerated/decelerated vehicle“

Test phase: during deceleration:



Test phase: during acceleration:



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5.4.1 Start parameter „Wheel Speed Sensor Test – accelerated/decelerated vehicle“

Start parameter „Wheel Speed Sensor Test – accelerated/decelerated vehicle“
Test time t_1
V_{Stop}
Test phase (during acceleration or deceleration)

Remarks:

The test time t_1 for the wheel speed sensor signal quality test has to be adapted on the roller speed. This is required to ensure that one complete rotation of the wheel is tested. The alternate parameter stop speed V_{Stop} has to be defined in the same way. Which of both parameters are reached at first (expired test time t_1 respectively stop speed V_{Stop} reached) stops the test routine. After that the tester can pick up and evaluate the test results. V_{stop} was defined that the routine stops if the last wheel exceeds (acceleration phase) or is below (deceleration phase) the selected stop speed.

The routine evaluation algorithm can be changed by the parameter “test phase”, so that the test can be executed during acceleration or deceleration of the vehicle.

The length (byte or word) of the start parameter and resolution is customer specific.

5.4.2 Evaluation parameter „Wheel Speed Sensor Test – accelerated/decelerated vehicle“

Evaluation parameter „Wheel Speed Sensor Test – accelerated/decelerated vehicle“
ΔV_{max}/Wheel_FL
ΔV_{max}/Wheel_FR
ΔV_{max}/Wheel_RL
ΔV_{max}/Wheel_RR

Remarks:

The length (byte or word) of the evaluation parameter and resolution is customer specific.

The routine provides the sum of all in a row following positive difference speed values of each wheel (e.g. ΔV_{max}/wheel_FL, etc.) after the test time t_1 or the selected stop speed V_{stop} is reached.

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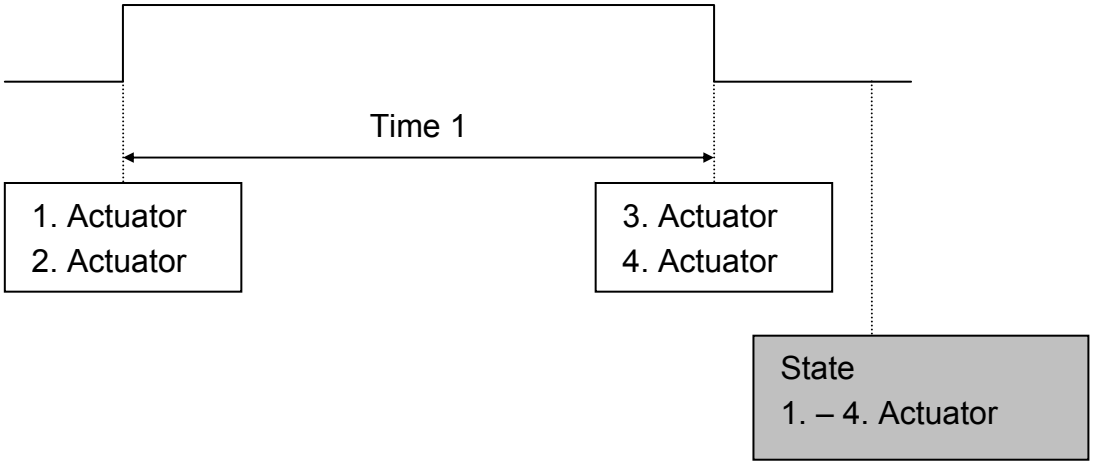
5.5 Diagnostic routine „Actuator Test“



5.5.1 Start parameter „Actuator Test“

Start parameter „Actuator Test“
Actuation value 1. actuator
1. actuator
Actuation value 2. actuator
2. actuator
Time 1
Actuation value 3. actuator
3. actuator
Actuation value 4. actuator
4. actuator

Remarks:

The length (byte or word) of the start parameter and resolution is customer specific.

The number and definition of the required actuators respectively the definition of the actuation value is customer specific.

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5.5.2 Evaluation parameter „Actuator Test“

Evaluation parameter „Actuator Test”
State (on/off) 1. actuator
State (on/off) 2. actuator
State (on/off) 3. actuator
State (on/off) 4. actuator

Remarks:

The length (byte or word) of the evaluation parameter and resolution is customer specific.

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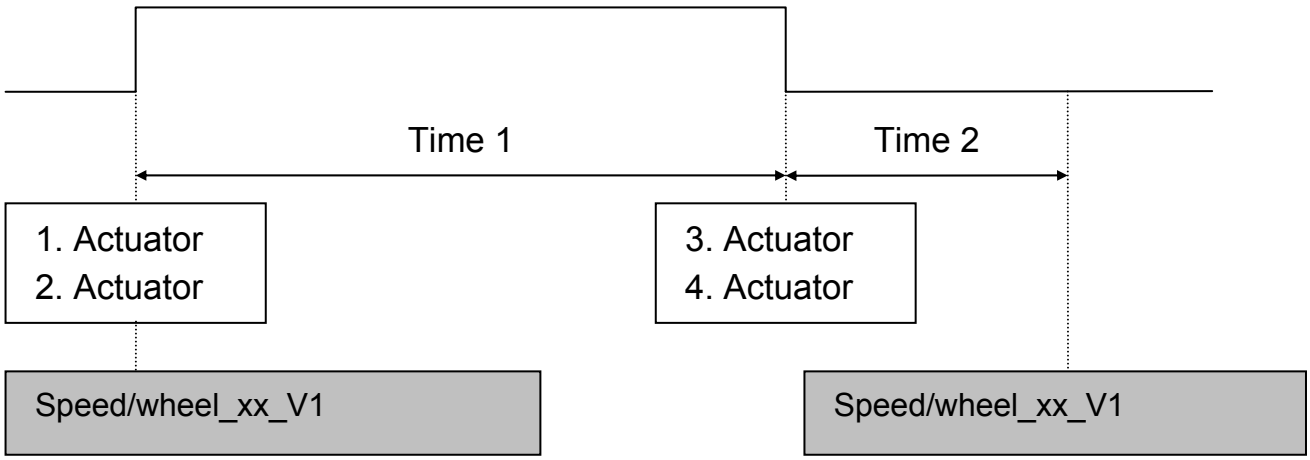
5.6 Diagnostic routine „Dynamic Test“



5.6.1 Start parameter „Dynamic Test “

Start parameter „Dynamic Test“
Actuation value 1. actuator
1. actuator
Actuation value 2. actuator
2. actuator
Time 1
Actuation value 3. actuator
3. actuator
Actuation value 4. actuator
4. actuator
Time 2

Remarks:

The length (byte or word) of the start parameter and resolution is customer specific.

The number and definition of the required actuators respectively the definition of the actuation value is customer specific.

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5.6.2 Evaluation parameter „Dynamic Test“

Evaluation parameter „ Dynamic Test “
Speed_Δ/Wheel_FL (= Speed/wheel_FL_V1 - Speed/wheel_FL_V2)
Speed_Δ/Wheel_FR (= Speed/wheel_FR_V1 - Speed/wheel_FR_V2)
Speed_Δ/Wheel_RL (= Speed/wheel_RL_V1 - Speed/wheel_RL_V2)
Speed_Δ/Wheel_RR (= Speed/wheel_RR_V1 - Speed/wheel_RR_V2)

Remarks:

The length (byte or word) of the evaluation parameter and resolution is customer specific.

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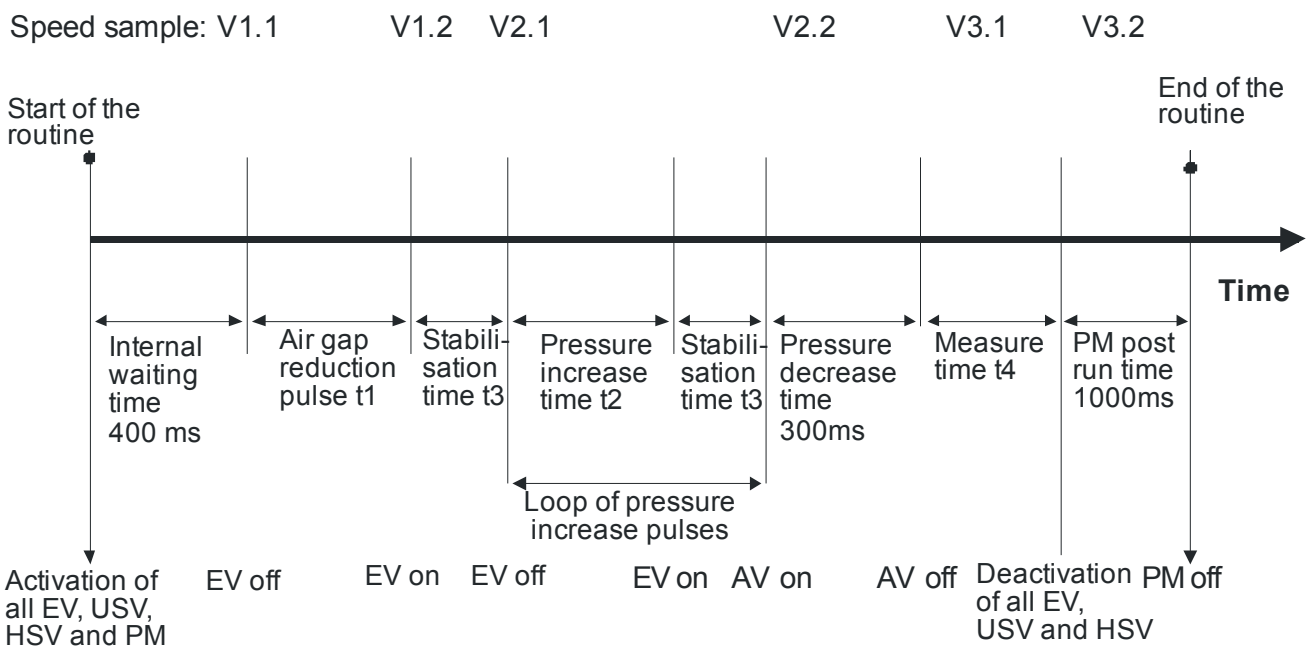
5.7 Diagnostic routine „Extended Dynamic Test“ – only for ESP® systems available



5.7.1 Start parameter „Extended Dynamic Test”

Start parameter „Extended Dynamic Test“
Selected Wheel (FL, FR, RL or RR)
EV actuation time t1 (air gap reduction pulse – to reduce the air ga tween disk and pad)
EV actuation time t2 (loop to increase the pressure)
Number of the max. allowed pressure increase pulses (actuation time t2)
Stabilisation time t3
Demanded speed difference ΔV
Measure time t4 to calculate the ΔV after the pressure release pulse

<u>Remarks:</u>

The length (byte or word) of the start parameter and resolution is customer specific.

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5.7.2 Evaluation parameter „Extended Dynamic Test“

Evaluation parameter „Extended Dynamic Test“
Demanded speed difference ΔV reached? (yes, no, failure)
Number of required pressure increase pulses
ΔV after air gap reduction pulse
ΔV after the 1. pressure increase pulse of the pressure increase loop
ΔV after the n. pressure increase pulse of the pressure increase loop
ΔV after the last pressure increase pulse of the pressure increase loop
ΔV after the pressure decrease pulse

Remarks:

The length (byte or word) of the evaluation parameter and resolution is customer specific.

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6 Pump motor and valve protection measures respectively notes to the percentaged activation

	ABS	ESP®
Valve coil:	Diagnostic mode	
EV 7,3 Ω	If U_{VR} > 9V, than lowering of the duty cycle to U_{eff} = 9V. If the valve activation value is > 0%; than the ECU starts the up/down counter and counts with a ratio up/down = 2/1. The total time is limited to 120s. For each EV valve a separate counter is used.	
AV 7,3 Ω	If the valve activation value is > 0%; than the ECU starts the up/down counter and counts with a ratio up/down = 2/1. The total time is limited to 30s. For each AV valve a separate counter is used.	
HSV 3,3 Ω	---	20ms with 100% duty cycle, than lowering of the duty cycle to U_{eff} = 3,5V and cyclic all 200ms for 20ms with 100%
USV 7,3 Ω	---	Max. duty cycle is limited to max. I = 0,73 A (~ 110 ± 20 bar at active pressure increase)
USV 4,3 Ω		tbd
PM	---	If the USV1 & HSV1 or/and USV2 & HSV2 and the PM are switched on, than the ECU starts the PM up/down counter and counts with a ratio up/down = 2/1. After the PM counter has reached its limitation of 30s the PM activation is reset.

Remarks:

The described valve and PM protection measures are always switched on. They cannot be disabled. These measures are required to prevent predamages at the valve coils caused by overheating. In addition the current limitation of the USV coils protect the system against predamages by overpressure. In case of the allowed activation times are exceeded, the valve activations are immediately reset.

Following activation values can be selected by the tester:

Valve coil:	Kind of output stage activation / valve:	Kind of allowed activation via tester:
EV	duty cycling of the voltage / regulator valve	percentaged in 1% steps between 0-100%
AV	duty cycling of the voltage / switching valve	digital: 0 % = off; 100% = on
USV	current controlled / regulator valve	percentaged in 1% steps between 0-100%
HSV	duty cycling of the voltage / switching valve	digital: 0 % = off; 100% = on

Remarks:

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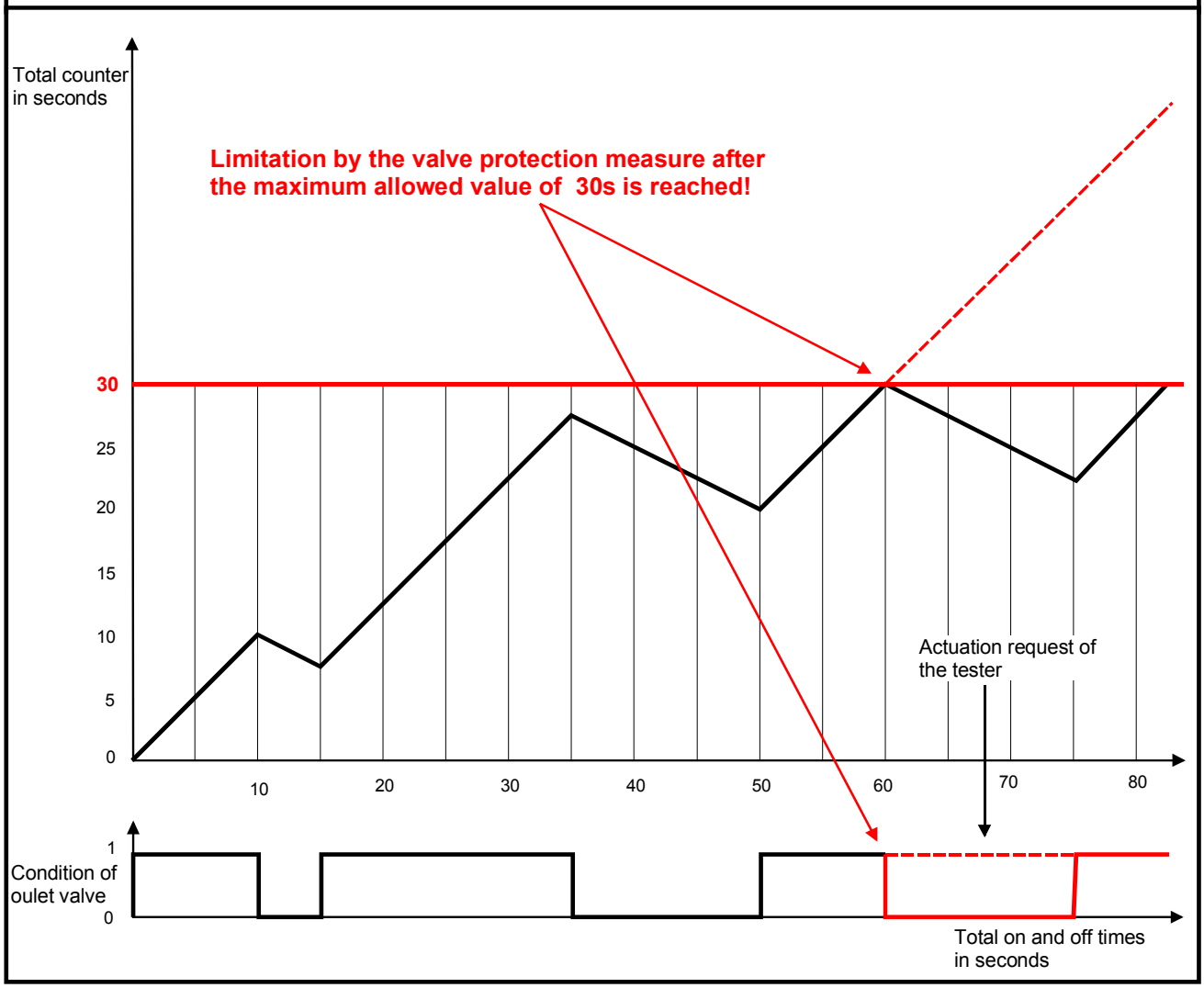
The AV valves are switching valves. To guarantee a correct function (opening or closing of the valve), the valves have been switched on only with 100% and off with 0% duty cycle.
The HSV valves have always been switched on with 100% duty cycle to guarantee, that they have been completely opened also in case of the brake pedal is applied. After that the duty cycle can be reduced to a lower level. This HSV switching on and reduction profile is realised with the valve protection measures therefore the tester does not need to reduce the duty cycle after the HSV valves were switched on with 100%.
For the voltage controlled EV and current controlled USV valves following notes should be noticed:
Depending on the selected percentaged activation value the brake pressure can be hold respectively controlled. A 100% activation value corresponds the maximal brake pressure which is allowed by the valve protection measure. This is especially valid for the USV valves.

Graphical presentation of a up/down counter (AV coil):

Gen 9: Valve protection measure: Up/down counter with a ratio of = 2/1

Example: Outlet valve with a maximum limitation to 30 seconds



Remarks.

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The forced limitation by the valve protection measure is active as long as valve coils and the ECU are sufficient protected against predamages caused by overheating. Only after an appropriate cooling time the valves can be activated again.
To guarantee a sufficient cooling time also after restart of the system (e.g. ignition cycle) the provided coil resistance values are evaluated by the valve protection measure. The resistance the coils are measured after each system power up and then cyclic all 20 seconds. These values are used to initialize the up and down counter value for the EV and AV coils.

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7 Test rig information

1. Dynamometer design:
The ABS / ESP® tests are designed to be performed either on a 2-axle brake dynamometer (low speed) or a roller test bench with 4 independent rolls (high speed).
To allow individual wheel test cycles the rolls of each wheel should be independently driven by the test rig drives.

2. Diagnostic Information:
The diagnostic computer should be as close as possible near the vehicle to avoid communication problems. The timing (time gaps between messages and bytes within messages) should be set close to the minimum possible limits in order to reduce the test time.
To be able to execute high speed tests the communication has to be started below the customer specific diagnostic speed limit. In addition this speed limitation function has to be switched off with the corresponding customer specific diagnostic function.

3. Process bytes:
The final definition of the process byte data must be synchronized between the EOL facility and the evac/fill resepectively repair bleed equipment supplier or the customer.

4. Editor menus / vehicle types:
ECU Identification codes may change during production due to updates of software revisions.
It is important to set up the software program to modify identification numbers easily by using an editor menu.
The timing of the dynamic valve test must be found during test trials.
Timing of valve actuation cycles, such as actuation times and stabilization times should be included in an editor menu.
Upper and lower limit values for evaluation should also be included in an editor menu for simple adaptation or modifications.
Due to the various vehicle types and configurations, different actuation times for the valves cycles of the dynamic ABS / ESP® tests might be required. These data should be also available in separate parameter file within the test equipment software and also easily be modified by using an editor menu.
The selection of the vehicle type should be done by either a bar code reading or an operator selection when driving into the test rig.

5. Test protocol:
In case of an ok test result, a print out (”test passed”) is made.
In case of a nok test result, a failure protocol must be printed out.
This protocol must include all test data of the test step that failed in order to allow a better failure analysis.
An entire test protocol of all test results must be available on request.

6. Vehicle conditions:
The battery must be charged properly in order to allow a sufficient power supply to the system.
The engine should be running in order to apply a sufficient power supply.

7. Electrical circuit diagrams:
It is the user’s responsibility to make sure they are using the latest available electrical circuit diagram!

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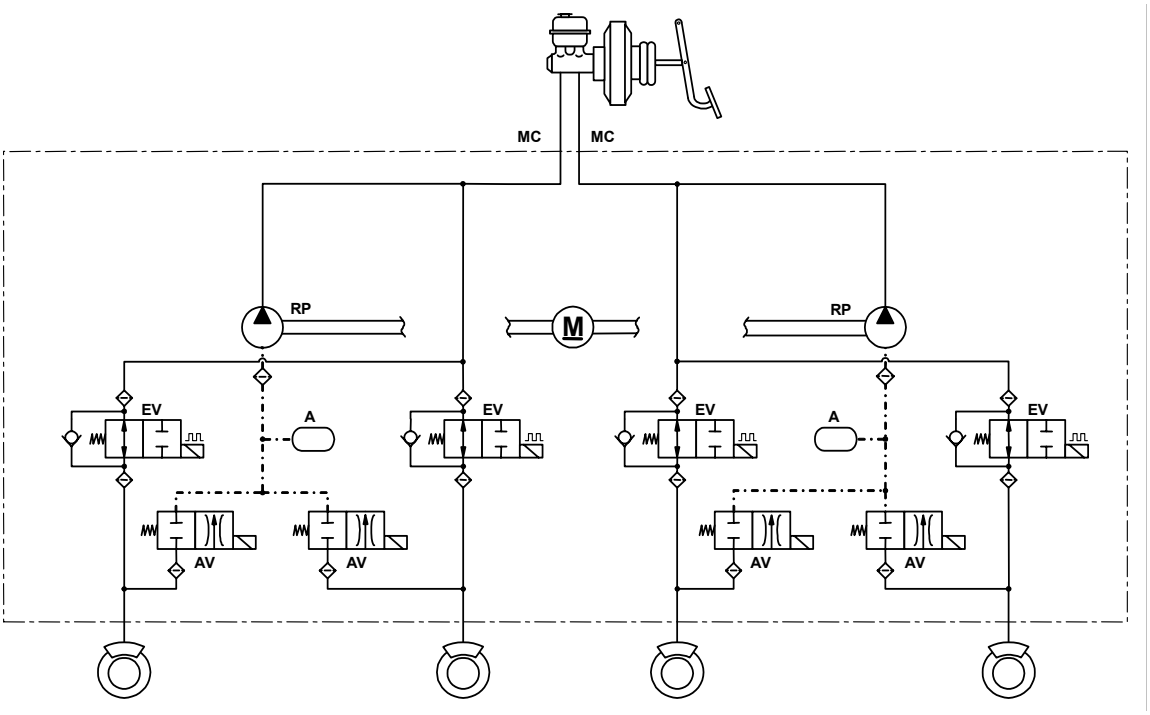
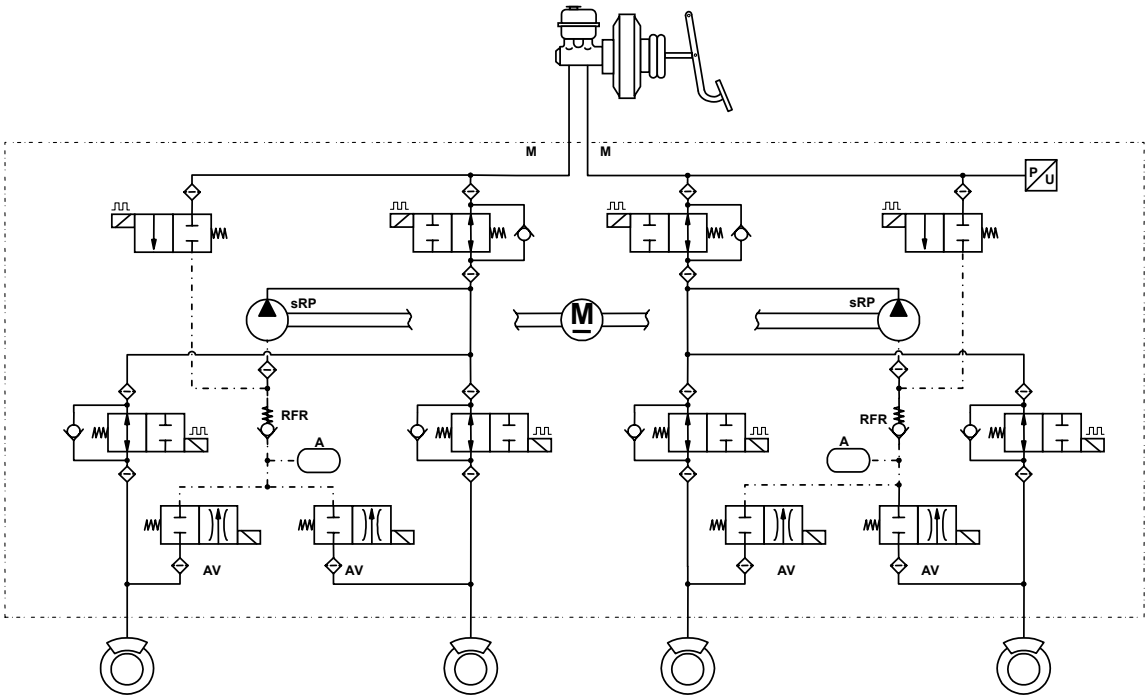
8 Attachments

8.1 Hydraulic circuit diagrams

8.1.1 ABS 9 – independent of the brake split distribution



8.1.2 ESP® 9- independent of the brake split distribution



The dashed line within the ABS respectively ESP® Hydraulic shows the area of the secondary and priming circuits

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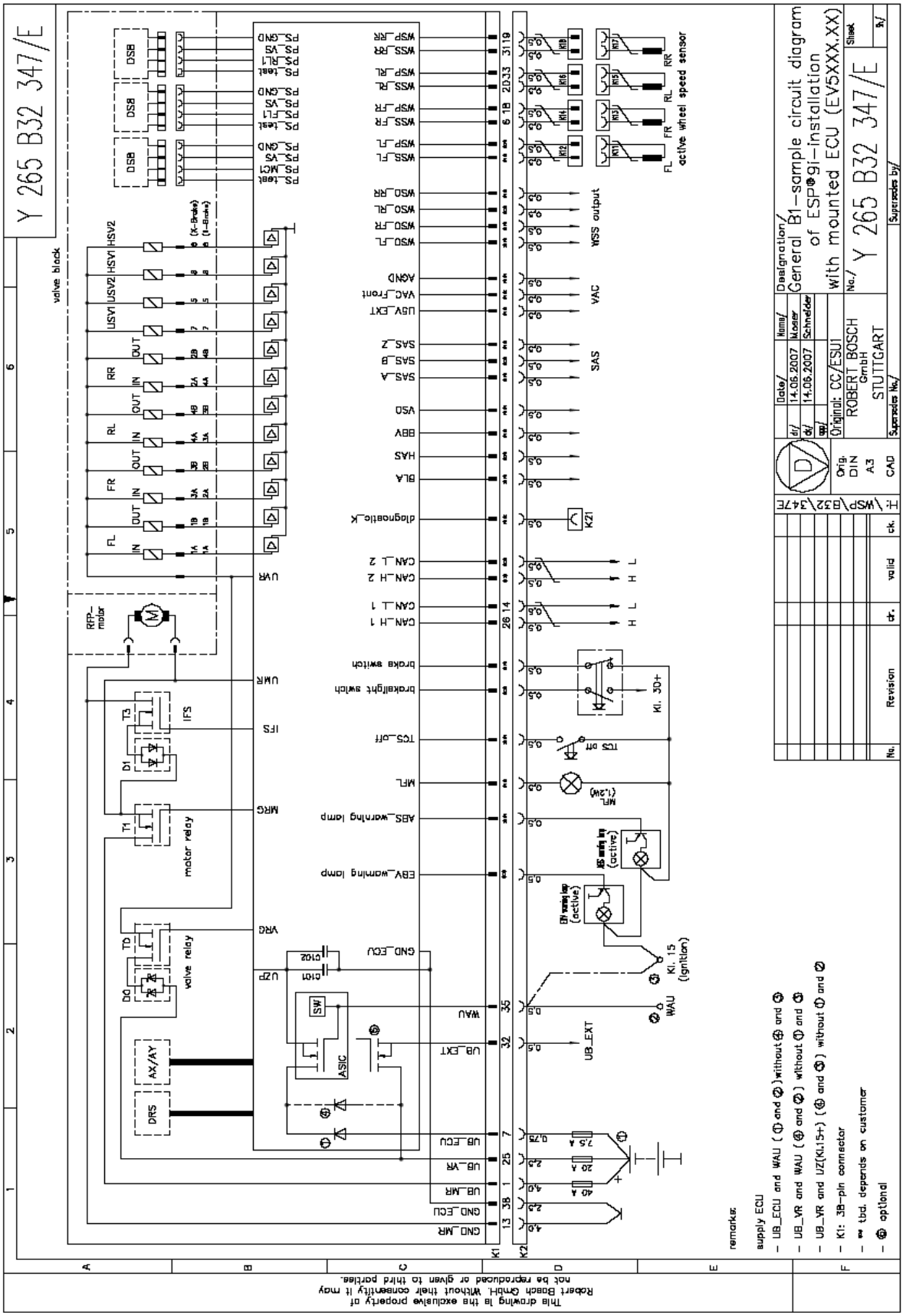
8.2 Electrical circuit diagram ESP® 9i



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